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10 Drone/UAS Competitor Reports

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Research complete. Here is the report.

AeroDrone (Ukraine) — Company Research Report

Compiled 24 September 2026. All figures as reported by cited sources; this is a private Ukrainian LLC with no audited public filings, so financial data is fragmentary and partly contradictory.

0. DISAMBIGUATION — READ FIRST

This report covers AeroDrone / **ТОВ «Аеродрон»** (Aerodron LLC), a Kyiv, Ukraine-based manufacturer of heavy-lift, long-range unmanned aircraft systems (UAS), founded by Yuriy Pederiy and co-owned by Dmytro Shymkiv. Its two flagship platforms are the D-80 Discovery and the E-300 Enterprise. Official site: <https://aerodrone.ua/> (contact: info@aerodrone.ua).

It is NOT any of the following, all of which pollute search results for this name:

- AeroDrone Robotics Pvt Ltd (India) — unrelated Indian robotics/drone services firm. No connection.
- aerodrone.tech — a parked domain currently listed for sale via GoDaddy (<https://aerodrone.tech/> redirects to a GoDaddy "for sale" page). This is NOT the Ukrainian company's site; the correct domain is aerodrone.ua.
- AeroMotors — a *separate* Ukrainian company (drone engines) that received Swedish investment from Front Ventures. CB Insights maintains a distinct profile at <https://www.cbinsights.com/company/aeromotors/> which surfaces on "AeroDrone Ukraine" searches. Different entity.
- Aeros — the Ukrainian light-aircraft/hang-glider manufacturer (<https://en.wikipedia.org/wiki/Aeros>). Related only in that AeroDrone's E-300 Enterprise is reportedly built on the Aeros SkyRanger airframe; Aeros is a supplier/airframe origin, not the same company.
- Escadrone, Aerorozvidka, Ukrspecsystems, Airlogix — other Ukrainian drone makers with superficially similar names or adjacent coverage.
- "Aerodrone" as a generic precision-farming listing — Crunchbase (<https://www.crunchbase.com/organization/aerodrone>) describes "Aerodrone" as "a precision farming company that protects crop from any kind of weeds, insects and diseases." This IS the same Ukrainian company, but the profile is badly stale (pre-2022 agricultural framing only).

Critical data-quality warning on aggregators. The business-intelligence databases contain mutually contradictory records that should not be trusted:

- CB Insights (<https://www.cbinsights.com/company/aerodrone>) lists founding year 2013 and HQ at Franzstr. 10, Munich, 80802, Germany — almost certainly a data error, possibly conflated with founder Pederiy's years at Microsoft's Munich office. The same profile's news feed simultaneously describes a "Kiev-based AeroDrone enterprise," contradicting its own HQ field.
- PitchBook (<https://pitchbook.com/profiles/company/99799-75>) lists founding year 2012, Kyiv, category "Other Hardware."
- Tracxn (<https://www.tracxn.com/d/companies/aerodrone>) lists total funding of \$65.1K across one round.
- Ukrainian press variously reports founding in 2015, 2018, or company registration in 2007 and 2019 (two different legal entities).

None of these reconcile. The Ukrainian corporate-registry evidence (Section 1) is the most reliable.

One further naming note: the founder's surname is **Педерій** / Pederiy in the large majority of sources (Forbes Ukraine, Focus, Militarnyi). One outlet, dev.ua, renders it «**Передрій**» / Peredriy (<https://dev.ua/news/ukrainskyi-aerodrone-ta-pidpriyemets-dmytro-shymkiv-otrymaly-dozvil-na-vykorystannia-droniv-v-zsu-s-hcho-tse-za-bpla>). Treat "Peredriy" as a typo for the same person.

1. COMPANY OVERVIEW

1.1 Identity and positioning

AeroDrone describes itself as "a Ukrainian company specializing in the design, development, and manufacturing of unmanned aircraft systems," built around three stated attributes: heavy payload, long range, high endurance (<https://aerodrone.ua/>, <https://drone.ua/en/blogs/news/aerodrone-at-the-ukrainia-n-pavilion-commercial-uav-expo-2025-in-las-vegas>).

The company's own site lists five target markets:

1. Precision agriculture
2. Oil, gas and power infrastructure inspection
3. Logistics and emergency delivery
4. Aerial photography
5. Military

Its differentiating technical approach is converting manned light aircraft into unmanned platforms rather than scaling up a multirotor or building a clean-sheet UAV. Co-owner Dmytro Shymkiv described the process to Ekonomichna Pravda as taking a two-seat aircraft and making multiple modifications while preserving the original aerodynamic characteristics, adding custom electronics and control systems, with programming done largely in-house and partially outsourced (as excerpted at <https://texty.org.ua/fragments/111989/yak-pererobyty-dvomisnyj-litak-na-bezpilotnyk-dosvid-aerodrone/>). The conversion technology is explicitly stated to be proprietary to AeroDrone.

This places AeroDrone in an unusual niche: while most of Ukraine's ~500 wartime drone manufacturers build small FPV or tactical ISR quadcopters, AeroDrone builds aircraft with 6.3 m to 8.5 m wingspans and payloads measured in hundreds of kilograms — closer to a light general-aviation aircraft than to a conventional drone.

1.2 Founding and corporate history

The founding date is genuinely disputed across sources:

- 2007 — «А е р о д р о н - С е р в і с» LLC (Aerodron-Service) registered. Per YouControl data cited by Forbes Ukraine and reproduced at <https://borgexpert.com/news/dmytro-shymkiv-kupyv-50-vyrobnyka-droniv-aerodrone>.
- 2015 — The team began work with three types of small aircraft described as "flying laboratories"; dev.ua states the startup was "founded in 2015" (<https://dev.ua/news/ukrainskyi-aerodrone-ta-pidpriemets-dmytro-shymkiv-otrymaly-dozvil-na-vykorystannia-droniv-v-zsu-shcho-tse-za-bpla>). HackYourMom also gives 2015.
- March 2016 — Pederiy leaves Microsoft specifically to launch a drone-technology startup (Forbes Ukraine: <https://forbes.ua/innovations/dmitro-shimkiv-z-kiivstaru-pridbav-polovinu-aktsiy-ukrainskogo-virobnika-droniv-aerodrone-tse-persha-publichna-investitsiya-v-ukrainskiy-miltech-yak-vona-vlashtovana-26012023-11147>).
- 2017 — The DR-60 agricultural drone is developed.
- 2018 — The date Forbes Ukraine and Militarnyi use for the company's launch (<https://militarnyi.com/en/news/the-ministry-of-defense-of-ukraine-awards-aerodrone-a-uav-contract/>).
- 2019 — «А е р о д р о н» LLC (Aerodron) registered. This is the specific legal entity in which Shymkiv later acquired 50%.

The most coherent reading: an R&D effort beginning around 2015–2016 under an older 2007 service company, formalized into a new operating LLC in 2019, with 2018 being when the product became commercially real. Forbes' "2018" and dev.ua's "2015" are describing different milestones of the same story.

1.3 Business trajectory

Phase 1 — Agricultural (2016–Feb 2022). The company sold crop-spraying UAS for applying fertilizers and plant-protection products, positioning the DR-60 as a challenger to conventional ground sprayers and manned crop-dusters. Pederiy told Forbes Ukraine that units were shipped to the US and Canada in 2019, with negotiations reported in 2020 with customers in Brazil, India and Australia. (These export claims are asserted by the company and have not been independently documented; see Section 7.4.)

Phase 2 — Military pivot (2022–present). Following the February 2022 full-scale invasion, the company redirected its airframes to defense. Reuters, which visited the plant in late February 2023, described it as sitting on "an unassuming industrial estate in northern Ukraine"

(<https://www.reuters.com/world/europe/inside-ukraines-scramble-game-changer-drone-fleet-2023-03-24/>, mirrored at <https://mnews.world/en/news/inside-ukraines-scramble-for-game-changer-drone-fleet>).

Phase 3 — Internationalization (2023–present). A Thales UK memorandum of understanding in December 2023, appearance at Commercial UAV Expo in Las Vegas in August 2025, and a stated plan to build an R&D, manufacturing and testing facility in Nevada, USA.

1.4 Headquarters, facilities and size

- HQ / registered base: Kyiv, Ukraine. The armedconflicts.com technical entry lists the manufacturer as "AeroDrone, Kyiv, Ukraine" (<https://www.armedconflicts.com/AeroDrone-E-300-Enterprise-t262580>). Reuters places the production facility on an industrial estate in northern Ukraine.
- Munich listing (CB Insights): disregard — see disambiguation.
- Planned US facility: Nevada (R&D, manufacturing, testing), per DroneUA, August 2025.
- Poland: In March 2023 the company was reported to be in talks with a foreign company about starting production lines in Poland (<https://gagadget.com/en/uav/222223-aerodrone-in-ukraine-starts-mass-production-of-e-300-enterprise-and-d-80-discovery-drones-with-a-range-of-up-to-3100/>). No subsequent confirmation that this materialized.
- Headcount: Never publicly disclosed in any source located. Given stated production of "only a few units per month," the operation is small — plausibly low tens of staff.
- Production rate: Shymkiv stated in March 2024 that output is only a handful of aircraft per month. The 2020 Ukrainian Startup Fund grant was explicitly intended to scale to 4–6 aircraft per month, which suggests that range is roughly the design ceiling of the current facility.

1.5 Mission

The company has no published formal mission statement. The operative mission, as expressed by its principals, has two strands:

1. Commercial: replace manned agricultural aviation and manned light-aircraft logistics with autonomous equivalents — safer, cheaper, and able to operate without prepared runways.
2. Defense/humanitarian: remove humans from high-risk resupply and reconnaissance flights. Shymkiv repeatedly framed the E-300 by reference to the manned helicopter resupply missions into besieged Mariupol in 2022, stating "У мене знайомий загинув в одному з цих гелікоптерів" — "A friend of mine died in one of those helicopters" (<https://texty.org.ua/fragment/s/111989/yak-pererobyty-dvomisnyj-litak-na-bezpiлотnyk-dosvid-aerodrone/>).

Pederiy's own stated motivation, per Forbes Ukraine, combines his father's career as a drone-building engineer at the Antonov design bureau with the 2017 death of a friend, a pilot, in an An-2 crop-duster crash.

2. FINANCIALS

AeroDrone is a private Ukrainian limited-liability company. There are no public financial filings, no audited accounts, and no disclosed valuation. What exists is Ukrainian state-registry data (surfaced via YouControl), a small number of press-sourced figures, and unit-price disclosures.

2.1 Registry data (via YouControl, cited by Forbes Ukraine and BorgExpert)

«А е р о д р о н» LLC (Aerodron, registered 2019) — the entity Shymkiv bought into:

- Registered charter capital: 5,000 hryvnia (roughly \$130–180 at prevailing rates) — a nominal, minimum-style capitalization
- Net loss for 2020: 100 hryvnia (i.e., effectively a dormant shell in that year). Note: the hostile UKR LEAKS piece cites "500 hryvnia loss" for 2020 instead; the Forbes/BorgExpert figure of 100 UAH is better sourced.

«А е р о д р о н - С е р в і с» LLC (Aerodron-Service, registered 2007):

- 2020 revenue: 5,500 hryvnia (approximately \$200)
- 2020 liabilities: over 430,000 hryvnia

Source: <https://borgexpert.com/news/dmytro-shymkiv-kupyv-50-vyrobnyka-droniv-aerodrone> and <https://orbes.ua/innovations/dmitro-shimkiv-z-kiivstaru-pridbav-polovinu-aktsiy-ukrainskogo-virobnika-droniv-aerodrone-tse-persha-publichna-investitsiya-v-ukrainskiy-miltech-yak-vona-vlashtovana-26012023-11147>

These figures make clear that as of 2020 the business had essentially no commercial revenue. R&D; was being funded from the founder's own capital and grant money.

2.2 Cumulative turnover

An anonymous source told Forbes Ukraine (January 2023) that the company's total historical turnover was approximately \$200,000. Given the registry figures above, this is credible for the pre-war period and implies the DR-60 sold in very low single-digit unit numbers, if at all.

2.3 Post-contract financial position

The decisive financial event was the March 2023 Ministry of Defence contract. Its value and unit quantity were never disclosed — Militarnyi noted explicitly that "the number of drones ordered from AeroDrone is not to be disclosed."

A rough floor can be inferred from unit economics: at \$250,000–\$450,000 per system, even a modest order would represent one to two orders of magnitude more revenue than the company's entire prior history.

However, the company was not profitable as of March 2024. Asked directly whether he had earned anything from the investment, Shymkiv answered: "Ні, ніч о г о , т і л ь к и в и т р а ч а ю . Б а р а т о ." — "No, nothing, only spending. A lot."

He also disclosed a severe working-capital problem:

- Production sat idle for roughly two and a half months after the 2023 contract ended, with no replacement contract secured as of the interview.
- Ukrainian law restricts most Ministry of Defence contracts to single calendar years, preventing multi-year production planning.
- Six months of "capability surveys" by the ministry produced no contracts.
- No accessible credit for working capital; state banks discouraging registration.
- The state does not cover software licensing costs in contracts.

Source: <https://epravda.com.ua/publications/2024/03/13/711076/> (direct fetch blocked; content via <https://texty.org.ua/fragments/111989/yak-pererobyty-dvomisnyj-litak-na-bezpilotnyk-dosvid-aerodrone/> and <https://focus.ua/uk/digital/633195-bezpilotniki-aerodrone-za-sotni-tisyach-dolariv-yaki-zavdannya-zsu-vikonuyut-na-fronti-video>)

By 2025 this had not resolved. Shymkiv told the Kyiv Independent: "While waiting for contracts, we are burning through our working capital" (<https://kyivindependent.com/global-acquisition-migration-or-death-ukraines-drone-companies-look-for-opportunities-abroad-amid-defense-spending-shortfall/>).

2.4 Cost structure

The single most important structural fact about AeroDrone's economics, per Shymkiv:

"70% в а р т о с т і – і м п о р т" — "70% of the cost is imports."

Ukraine no longer produces aviation-grade aluminium or aviation fabric, and produces no small internal-combustion aircraft engines. All engines are bought abroad. Shymkiv expressed doubt that domestic manufacturers such as Motor Sich would enter small piston-engine production, suggesting electric motors were the more realistic domestic path. This makes the company's margins hostage to FX rates, customs treatment of dual-use imports, and export-control friction.

2.5 Unit pricing (the clearest available revenue signal)

Product / configuration	Price	Source
DR-60 agricultural drone	\$75,000	Forbes Ukraine (pre-war listing)

Product / configuration	Price	Source
E-300 Enterprise, bare airframe (no payload)	\$300,000	Shymkiv, March 2024
D-80 Discovery complex — two aircraft plus high-end cameras and communications	\$450,000	Shymkiv, March 2024
General range, "depending on modifications"	\$250,000–\$450,000	Gagadget / dev.ua, March 2023
General range, "depending on model and configuration"	\$150,000–\$450,000	Reuters, March 2023

Note the important clarification Shymkiv made: the widely-quoted \$450,000 figure is for a two-aircraft D-80 system with expensive optics and comms, not for a single airframe. Press repetition of "\$450,000 per drone" is a misreading.

Optional add-ons include an anti-jamming system to counter Russian electronic warfare (Reuters).

2.6 Valuation

No valuation has ever been publicly disclosed. Neither party disclosed the price of the September 2022 50% acquisition — Forbes Ukraine states plainly: "С у м у і н в е с т и ц і ї в і н н е р о з г о л о ш у є" ("He does not disclose the investment amount"). PitchBook maintains a profile page titled to include valuation data (<https://pitchbook.com/profiles/company/99799-75>) but the underlying figures are behind a paywall and the profile's basic facts (founding year 2012) are already demonstrably wrong.

For sector context, Forbes Ukraine benchmarked the deal against Portuguese dual-use drone maker Tekever, which raised \$23 million from Arab investors in February 2022 — implying the AeroDrone stake was far smaller.

As of mid-2026, Shymkiv was publicly arguing that Ukrainian defense-tech firms "with real revenues, sustainable production" are undervalued (per <https://thedefender.media/>), consistent with a founder who has not achieved a mark-up event.

2.7 Financial filings summary

- No SEC, Companies House, or equivalent filings — the company is not listed anywhere.
- Ukrainian state registry (ЄДР) records exist for both LLCs; accessible via YouControl (<https://youcontrol.com.ua/>), Clarity Project (<https://clarity-project.info/>), and Opendatabot (<https://opendatabot.ua/>). All three blocked automated retrieval during this research (HTTP 403). The EDRPOU registration codes for both entities were not obtainable from open press coverage and would need to be pulled manually from those registries.
- Ukrainian defense contracts are exempted from Prozorro open-tender publication during martial law, so contract values are not retrievable from the public procurement portal.

3. FUNDING HISTORY

AeroDrone has never raised an institutional venture round. Its capital history is: founder money, one government grant, one private 50% share purchase, and thereafter defense contract revenue.

Round-by-round

1. Founder capital — c. 2016–2019 — \$200,000 Yuriy Pederiy invested \$200,000 of his own money into product development. (Forbes Ukraine)
2. Ukrainian Startup Fund grant — 2020 — \$50,000 A \$50,000 non-dilutive grant from the Ukrainian Startup Fund (USF), a Ukrainian government-backed grant programme (<https://usf.com.ua/>). The stated purpose was scaling manufacturing to 4–6 aircraft per month. USF does not publish a per-grantee register on its public site; the fund reports aggregate figures of 764 startup teams supported and \$22.7M+ deployed.
3. Dmytro Shymkiv — 50% acquisition — August/September 2022 — amount undisclosed
 - Date discrepancy: YouControl registry records show the change of ownership in September 2022 (Forbes Ukraine, Focus.ua, BorgExpert). Shymkiv's own LinkedIn profile (<https://www.linkedin.com/in/dmitryshymkiv>) states "In August 2022 Dmytro became an investor at Ukrainian startup AeroDrone." August is likely the deal date, September the registry filing date.

- Structure: acquisition of 50% of the shares of «А е р о д р о н» LLC. Pederiy retains the other 50%.
- Amount / valuation: not disclosed by either party.
- Significance: widely described as the first publicly disclosed private investment into a Ukrainian military-technology startup. Forbes Ukraine ran it as a cover story on that basis.
- Shymkiv's own characterization of his decision was famously terse: «**М а к с и м а л ь н о з а й ш о в**» — roughly, "It completely got me" / "I'm all the way in."
- He framed his contribution as commercial rather than technical: the scaling of a business requires "д о с в і д і р о з у м і н н я , я к п е р е т в о р и т и т е х н о л о г і ю н а б і з н е с" — experience and understanding of how to turn technology into a business.

4. Mark M. Erjavec — unconfirmed, pre-2022 Forbes Ukraine reported, citing US outlet UAS Magazine, that Mark M. Erjavec — founder of the US-based Agro Fund and Aero Fleet One — may have been an earlier investor in AeroDrone. Erjavec did not respond to Forbes' request for comment and the stake was never confirmed. Treat as unverified.

5. Defense contract revenue — March 2023 onward Not funding in the equity sense, but the company's primary capital source since 2023. Value undisclosed.

What is NOT true

- Tracxn's "\$65.1K raised" — this appears to be a partial capture of the USF grant or a conversion artifact; it does not reflect the \$200K founder capital or the undisclosed Shymkiv purchase.
- CB Insights' "Angel round on Jan 27, 2023" — the date corresponds to the *publication date of the Forbes Ukraine article* about the Shymkiv deal (26 January 2023), not to the deal itself, which closed in August/September 2022. Classic aggregator scraping artifact.

Notable absence

Despite Ukraine's defense-tech funding boom (Brave1 cluster, Diia City tax regime, foreign VC entry), no evidence was found of AeroDrone raising any round after 2022. This is consistent with Shymkiv's public statements about burning working capital and with the Kyiv Independent's framing of the sector's funding squeeze.

4. LEADERSHIP AND PEOPLE

4.1 Yuriy Pederiy (Ю р і й П е д е р і й) — Founder, 50% owner

- Education: Graduate of the National Aviation University (Kyiv).
- Career:
 - 2000–2006 (approx. 6 years): Kyivstar, Ukraine's largest mobile operator.
 - c. 2006–2016 (approx. 10 years): Microsoft. Responsible for a major department in the Kyiv office; transferred to Microsoft's Munich office in 2011.
 - March 2016: Resigned from Microsoft to launch a drone-technology startup.
- Family/motivation: His father was a drone-building engineer at the Antonov design bureau. Pederiy also cited the 2017 death of a friend, a pilot, in an An-2 crop-duster crash as a driver for pursuing unmanned agricultural aviation.
- Role today: Retains 50% of Aerodron LLC. Reuters described the company as "run by Dmytro Shymkiv and Yuriy Pederiy." He is the technical/product principal; Shymkiv is the public-facing and commercial principal.
- Media presence: Low. Pederiy gave agricultural-press interviews in 2020 (AgroPortal) but has been largely absent from the record since the military pivot, with Shymkiv handling nearly all public communication.
- Name variants: Pederiy / П е д е р і й (correct); Peredriy / П е р е д р і й (dev.ua typo); "Yuri Pederiy," "Yuriy Pederii" in transliteration.

4.2 Dmytro Shymkiv (Д м и т р о Ш и м к і в) — Co-owner (50%), Investor/Partner

By far the highest-profile figure associated with the company, and one of the best-known technology executives in Ukraine.

- Microsoft Ukraine: Rose to become General Manager / Country Manager of Microsoft Ukraine. He and Pederiy met at Microsoft's Kyiv offices.
- Government: Served as Deputy Head of the Presidential Administration of Ukraine under President Petro Poroshenko, with responsibility for administrative and social reform. (Noted by Focus.ua and by the UKR LEAKS piece.)
- Darnitsa Group: Executive Chairman of the Board of Directors of Darnitsa, one of Ukraine's largest pharmaceutical manufacturers.
- Kyivstar: Member of the supervisory board.
- AeroDrone: Acquired 50% in August/September 2022. His own framing varies by venue — he is described in Ukrainian press as "partner" and "co-owner," while the Commercial UAV Expo speaker listing describes him as "Co-Founder and an Investor in AeroDrone" (<https://www.expouav.com/speaker/dmytro-shymkiv/>). "Co-founder" is a post-hoc characterization; he was not present at founding.
- LinkedIn: <https://www.linkedin.com/in/dmitryshymkiv>
- Other deals: Was involved in Kyivstar's 2022 acquisition of the medtech startup Helsi.

Selected Shymkiv quotes (useful for retrieval testing):

On the December 2022 certification (Facebook, via Ukrinform):

"Команда AeroDrone пройшла сертифікацію та отримала дозвіл на експлуатацію в ЗСУ наших двох типів безпілотних авіаційних комплексів" — "The AeroDrone team passed certification and received permission to operate our two types of unmanned aviation complexes in the Armed Forces of Ukraine." He added that the next step would be "підконтрольна експлуатація та серійне виробництво" — "controlled operation and serial production," and credited the team of Oleksandr Liiev at the Ministry of Defence with having achieved "a revolution in processes." (<https://www.ukrinform.ua/rubric-ato/3630551-aerodrone-otrimav-dozvil-na-ekspluataciu-v-zsu-dvoh-novih-bezpilotnikiv.html>, <https://interfax.com.ua/news/telecom/877161.html>)

On the March 2023 contract:

"We made a deal with the Ministry of Defense. Now we have serial production." (<https://military.com/en/news/the-ministry-of-defense-of-ukraine-awards-aerodrone-a-uav-contract/>)

On the defense industry:

"Тепер розумію, чому до великої війни ніхто в 'тверезому стані' не заходив у ВПК" — "Now I understand why before the big war nobody in a sober state of mind went into the military-industrial complex." (Headline of his March 2024 Ekonomichna Pravda interview.)

On operational security, declining to describe the E-300's weapons role:

"більше не скажу, бо вас москалі читають" — "I won't say more, because the Russians are reading you."

On the coming industry shakeout (Kyiv Independent):

"First, we will see the death of small companies, because the government cannot contract them." ...
 "Second, we will see acquisitions or movement of Ukrainian companies internationally. And third, there will be consolidation in the Ukrainian market."

On the E-300's payload bay:

Pointing to where the pilot's cockpit had been — "Now, it'll be the payload." And on performance: "It can carry 200 kg for 1200 km." (Reuters, March 2023)

4.3 Valeriy Otkidach — alleged business partner (UNVERIFIED)

Named by the UKR LEAKS Substack piece as Pederiy's business partner and the owner of Aerodron-Servis and of a second drone company, VSSA. This claim appears in no mainstream source and comes from an

outlet with a declared adversarial orientation (see Section 7.4). Flagged for completeness only; should not be treated as established.

4.4 Other personnel

No CTO, COO, CFO, chief engineer, or board members have been publicly identified in any source located. The company does not publish a team page. DroneUA's coverage of its Commercial UAV Expo 2025 appearance named no AeroDrone representatives.

5. PRODUCTS

AeroDrone's product family is small — essentially one agricultural airframe and two military-certified derivatives — but each is unusually large for a "drone."

5.1 DR-60 / Discovery DR-60 (agricultural, 2017)

The company's original product and the genetic ancestor of the military line.

- Developed: 2017
- Concept: Pederiy described it as multi-purpose with "a magic six" — up to 60 kg payload, 6 hours flight time, up to 600 km of automatic flight
- Price: \$75,000
- Cruising speed: 100 km/h
- Take-off weight: 170 kg
- Fuel tank: 30 litres
- Range, visual line of sight: up to 3 km
- Range, beyond visual line of sight: over 30 km
- Dimensions: 6.3 m wingspan × 3.6 m length
- Role: aerial application of fertilizers and plant-protection products; pitched as a challenger to both ground sprayers and manned crop-dusters
- Claimed exports: units to the US and Canada in 2019; talks with Brazil, India, Australia in 2020 (company claim, unverified)

Sources:

<https://militaryni.com/en/news/ukraine-s-military-to-receive-d-80-discovery-and-e-300-enterprise-uavs/>, <https://agroportal.ua/ua/publishing/lichnyi-vzglyad/ukrainskii-startap-aerodrone-brosaet-vyzov-traditsionnomu-opryskivatelyu/>

5.2 D-80 Discovery (certified for AFU, December 2022)

The 80 kg-class platform. Originally a survey and ordnance-detection aircraft, adapted for military ISR.

Payload and performance:

- Payload: up to 80 kg (DroneUA specifies this figure is *including fuel*)
- Maximum speed: 120 km/h
- Cruising speed: 100 km/h
- Total/gross weight: approximately 200 kg
- Service ceiling: up to 2 km
- Range: 100–800 km (some sources give "up to 800 km depending on configuration and fuel system")
- Endurance: up to 23 hours airborne (Shymkiv, describing the platform family)
- Wingspan: ~6.3 m (from the DR-60 lineage)

Sensors and mission equipment:

- Research sensor and magnetometer in the original survey configuration
- Designed for aerial surveying of large territories where ground survey is impractical
- Original/adapted military role: mine and unexploded-ordnance detection, both on land and underwater
- Later reported as a multi-role platform for reconnaissance, target acquisition, and strike

- Payload capacity supports high-end optics — Shymkiv cited 40× optical zoom cameras, and noted the airframe's stability is what makes that zoom usable
- Capable of carrying heavy radio-technical reconnaissance equipment, including, per Shymkiv, gear "я ко го ще на віть не ма є в З С У" — "that the Armed Forces don't even have yet"

Pricing: A complete complex of two D-80 aircraft with premium cameras and communications systems: \$450,000.

Positioning: Shymkiv explicitly differentiated the D-80 from cheaper Ukrainian tactical ISR drones such as the Leleka-100 and Ukspecsystems Shark on the basis of endurance and payload.

5.3 E-300 Enterprise (certified for AFU, December 2022)

The flagship and the aircraft that has generated the most attention. A converted Aeros SkyRanger light aircraft.

Airframe:

- Base airframe: Aeros SkyRanger (per H I Sutton, <https://www.hisutton.com/Ukraine-OWA-UAVs.html>, and South Front). The former pilot's cockpit volume becomes the payload bay.
- Wingspan: 8.50 m (27 ft 10⁵/₈ in)
- Length: 5.72 m (18 ft 9¹/₄ in)
- Wing area: 14.10 m² (151.77 ft²)
- Maximum take-off weight: 540 kg (1,190 lb)
- Configuration: conventional monoplane, CTOL, fixed wheeled undercarriage
- Powerplant: single piston engine (specific type not published; sourced abroad)

(Technical table: <https://www.armedconflicts.com/AeroDrone-E-300-Enterprise-t262580>)

Performance:

- Maximum speed: 150 km/h (93 mph)
- Cruise speed: 135 km/h (84 mph)
- Service ceiling: 5,000 m (16,404 ft)
- Baseline maximum range: 675 km (419 mi) in standard configuration
- Extended range: up to 3,000–3,100 km with an auxiliary fuel tank of up to 350 litres
- Endurance: up to 23 hours (platform family figure)
- Payload: up to 300 kg (DroneUA notes this figure is *inclusive of fuel* — an important caveat)

The payload/range tradeoff. This is the single most misreported aspect of the E-300. The headline figures of "300 kg payload" and "3,100 km range" are mutually exclusive. Militarnyi's analysis states that when the 350-litre auxiliary tank is fitted to reach 3,100 km, usable payload falls to roughly 20–30 kg. Shymkiv's own intermediate data point — "It can carry 200 kg for 1,200 km" (Reuters) — is probably the most useful single characterization of the aircraft's real operating envelope.

Features:

- Fully automated mission execution from takeoff to landing, with real-time manual override and control
- No runway required — can launch and land from any flat surface
- Emergency parachute recovery system allowing slow descent to the ground
- Optical navigation under development — the aircraft orients itself by comparing live camera footage against satellite imagery, providing GNSS-independent navigation in jammed environments
- Optional anti-jamming system to counter Russian EW

Original civil role: Cargo and courier aircraft. armedconflicts.com records the original design purpose as "the transport of mail"; Militarnyi describes it as marketed "as a courier drone for delivering parcels at a speed of 150 km/h."

Military roles, in order of emergence:

1. Cargo resupply to frontline positions — the originally advertised military use, framed by Shymkiv against the Mariupol helicopter resupply losses.

2. Heavy ISR platform — carrying radio-technical reconnaissance equipment beyond what smaller drones can lift.
3. Weapons platform — Shymkiv described the E-300 as adaptable for equipment "як е р а н і ш е н і х т о і н е д у м а в с т а в и т и н а Б П Л А" ("that nobody previously thought of mounting on a UAV"), declining to elaborate. The company was reported in 2024 to be testing a weaponization project intended to support ground-forces aviation.
4. Long-range bomber / one-way attack — the role it has actually become known for. See Section 7. On kamikaze use: Shymkiv argued in March 2024 that the airframe's \$300,000 cost makes expendable one-way use impractical, though technically possible if the military customer requested it. Subsequent events (Section 7.2) show it being used in exactly that manner, and in reusable bombing sorties.

5.4 Product summary table

Model	Payload	Range	Speed	Span	Role	Price
DR-60 Discovery	60 kg	600 km auto	100 km/h cruise	6.3 m	Agricultural spraying	\$75,000
D-80 Discovery	80 kg (incl. fuel)	100–800 km	120 km/h max	~6.3 m	ISR, magnetometry, mine/UXO detection	\$450k per 2-aircraft complex
E-300 Enterprise	300 kg (incl. fuel)	675 km std / 3,100 km extended	150 km/h max, 135 cruise	8.50 m	Cargo, heavy ISR, weapons platform, deep strike	\$300,000 airframe

5.5 Shared technology stack

- Proprietary manned-to-unmanned conversion process preserving original aerodynamics
- In-house electronics, flight-control systems, and flight software (partially outsourced)
- Full automation with manual override
- Parachute recovery
- Optical/visual navigation under development
- Aviation-standard (GOST) inspection and certification of each airframe
- ~70% imported bill of materials

6. PATENTS

6.1 Summary finding

No granted patents could be located that are attributable to AeroDrone, **ТОВ «Аеродрон»**, **ТОВ «Аеродрон-Сервіс»**, or inventor Yuriy Pederiy.

Searches attempted:

- Google Patents assignee and free-text queries for "aerodrone," "Pederiy" (<https://patents.google.com/>) — the service returned HTTP 503 to automated retrieval and could not be queried reliably.
- UAPATENTS.COM (Ukrainian patent database mirror, <https://uapatents.com/>) — the query for "П е д е р і й" returned only the IPC classification index, not inventor results; the query for "а е р о д р о н" was blocked at the network layer.
- Ukrainian state IP office (Ukrpatent / NIPO, <https://sis.nipo.gov.ua>) — requires interactive session, not retrievable.
- No patent references appear in any of the ~25 press articles, company pages, or database profiles reviewed.

This is a negative result under imperfect search conditions, not proof of zero patents. A definitive answer requires manual searches of: (a) Ukrainian state register ІБ /Ukrpatent by applicant name in Cyrillic, (b) Espacenet worldwide by applicant "AERODRONE" and inventor "PEDERII/PEDERIY," (c) WIPO PATENTSCOPE, and (d) the US/EP registers for any filings associated with the planned Nevada entity.

6.2 Why the absence is plausible

Three independent signals suggest AeroDrone's IP strategy is trade-secret-based rather than patent-based:

1. Explicit statement of proprietary know-how. Shymkiv said the conversion technology "є в л а с н о ю р о з р о б к о ю AeroDrone" — is AeroDrone's own development — while giving no indication it is patented. He described the method in only general terms (modifications preserving aerodynamics, custom electronics, in-house programming).
1. Operational security. For a company whose primary customer is a wartime ministry of defence and whose products are actively striking Russian territory, publishing patent specifications would disclose capability. Shymkiv's repeated refusals to describe payload integration ("because the Russians are reading you") are inconsistent with a patent-publication strategy. He separately flagged, as an unresolved concern, the security risk posed by publicly accessible data on Ukrainian defense companies in state registries.
1. Capital constraints. A company with ~\$200K lifetime turnover through 2022 and acute working-capital problems through 2024–2026 is unlikely to have funded an international patent portfolio, which typically costs tens of thousands of dollars per family per jurisdiction.

6.3 Related IP-adjacent assets

- Certification as a moat. The Ukrainian MoD operational-use certification for the D-80 and E-300 (December 2022) functions as a practical barrier to entry. Ukraine's Cabinet of Ministers resolution of September 2022 (effective November 2022) compressed the approval process from roughly one year to three weeks; AeroDrone was among the first beneficiaries, with only 13 drones approved for AFU use out of 50+ applications at the time Forbes reported in January 2023.
- Airframe dependency. The E-300's reliance on the Aeros SkyRanger means a meaningful portion of the underlying aircraft IP is not AeroDrone's.
- Optical navigation. The satellite-image-matching navigation system described in 2024 would be the most patentable asset in the portfolio, if filings exist. None were found.

7. LATEST NEWS AND DEVELOPMENTS

Presented chronologically, with the most consequential recent items in 7.2 and 7.3.

7.1 Foundational milestones (for context)

8–9 December 2022 — Military certification. AeroDrone passed certification and received permission to operate two unmanned aviation complexes — E-300 Enterprise and D-80 Discovery — within the Armed Forces of Ukraine. Announced by Shymkiv on Facebook. He thanked the Ministry of Defence, the Special Operations Forces (SSO) command, and the General Staff. Ukraine had approved seven domestically-produced drone models in the preceding 30 days, with 19 further applications pending.

Sources: <https://www.ukrinform.ua/rubric-ato/3630551-aerodrone-otrimav-dozvil-na-ekspluatatsiu-v-zsu-dvo-h-novih-bezpilotnikiv.html> | <https://interfax.com.ua/news/telecom/877161.html> | <https://military.com/en/news/ukraine-s-military-to-receive-d-80-discovery-and-e-300-enterprise-uavs/> | <https://forbes.ua/news/ukrainskiy-virobnik-bezpilotnikiv-aerodrone-otrimav-dozvil-na-ekspluatatsiyu-v-zsu-ranis-he-vin-vigotovlyav-agrarni-droni-08122022-10357>

26 January 2023 — Forbes Ukraine reveals the Shymkiv deal. Cover treatment of the 50% acquisition as the first public investment in Ukrainian miltech.

7–10 March 2023 — MoD contract and serial production. Contract signed; serial production launched; quantities and value undisclosed; first units to undergo trial use with pre-selected operating units before a formal adoption decision. Reported alongside the \$250,000–\$450,000 price band and the Poland production-line talks. Sources:

<https://military.com/en/news/the-ministry-of-defense-of-ukraine-awards-aerodrone-a-uav-contract/> | [https://gagadget.com/en/uav/222223-aerodrone-in-ukraine-starts-mass-production-of-e-300-enterprise-and-d-80-discovery-drones-with-a-range-of-up-to-3100-/](https://gagadget.com/en/uav/222223-aerodrone-in-ukraine-starts-mass-production-of-e-300-enterprise-and-d-80-discovery-drones-with-a-range-of-up-to-3100/)

24 March 2023 — Reuters plant visit. Major international profile placing AeroDrone among 80+ Ukrainian drone manufacturers working with the government.

<https://www.reuters.com/world/europe/inside-ukraines-scramble-game-changer-drone-fleet-2023-03-24/>

13–14 December 2023 — Thales UK Memorandum of Understanding. During the first UK–Ukraine defence trade mission, led by the UK Department for Business and Trade and Ministry of Defence, Thales UK signed an MOU with AeroDrone. Other UK participants included BAE Systems, Babcock, Thales, AMS Global and KBR; AeroDrone was the Ukrainian counterparty specifically for the Thales agreement. Discussions had begun at DSEI in London. This is the company's most significant Western industrial partnership on record. No subsequent public reporting of what the MOU produced. Source: <https://militaryni.com/en/news/uk-governement-and-uk-defense-industry-conduct-their-first-trade-mission-to-ukraine/>

13 March 2024 — Ekonomichna Pravda interview. The most substantive on-record disclosure of the company's condition: unprofitable, 2.5-month production gap, 70% import dependency, single-year contracting problem, no working capital credit. Also the first mention of the optical navigation project and of weaponization testing. Source: <https://epravda.com.ua/publications/2024/03/13/711076/> | summarized: <https://texty.org.ua/fragments/111989/yak-pererobyty-dvomisnyj-litak-na-bezpilotnyk-dosvid-aerodrone/> | <https://focus.ua/uk/digital/633195-bezpilotniki-aerodrone-za-sotni-tisyach-dolariv-yaki-zavdannya-zsu-vikonuyut-na-fronti-video>

7.2 The E-300 becomes a strategic strike aircraft (2025–2026) — the most significant development

The E-300 has migrated from "cargo drone" to one of Ukraine's most-discussed deep-strike platforms, apparently without the company ever announcing it.

30 January 2025 — E-300s downed over Bryansk carrying aerial bombs. Wreckage recovered in Russia's Bryansk region showed E-300 airframes carrying a Soviet-era FAB-250M-54 aerial bomb paired with a 120 mm mortar shell; a second downed airframe carried an OFAB-100-120. One bomb bore the handwritten inscription "For Mykolaiv." The aircraft were attributed to the 14th Unmanned Aerial Vehicle Regiment of Ukraine's Special Operations Forces. Reported by Defense Express and Militaryni. Source: <https://united24media.com/latest-news/ukrainian-e-300-drones-carrying-fab-250-bombs-spotted-in-strikes-on-russias-bryansk-5483>

This is a notable capability: a reusable, returnable bomber drone delivering standard 250 kg aerial ordnance, as distinct from the one-way Shahed-style attack drones that dominate the theatre. Reporting elsewhere describes "repeated bombing missions using the reusable E-300, built on commercial sport plane airframes."

17 March 2025 — Zelensky's 3,000 km drone announcement. President Zelensky announced after a Staff (Stavka) meeting: "There's good news on long-range drones. Our drone tested at 3,000 kilometres." The model and manufacturer were not identified. Militaryni noted the E-300's 3,100 km figure as a candidate but explicitly stated no confirmed link. Context: a 3,000 km radius would bring the Omsk Refinery — Russia's largest — into range; the longest confirmed prior strike (Olenya airbase, Murmansk) was ~1,800 km. Source: <https://militaryni.com/uk/news/ukrayina-stvoryla-dron-z-dalnistyu-diyi-3000-km/>

1 July 2025 — US special forces interest. Intelligence Online reported that "the E-300 Enterprise drone produced by Ukrainian company AeroDrone" had drawn the interest of US special operations forces following its performance in Ukraine. Source: <https://www.intelligenceonline.com/americas/2025/07/01/e-300-drone-s-success-in-ukraine-draws-interest-of-us-special-forces.110471283-art> (paywalled; headline and lede only)

Reported deep penetration. Forum-sourced material on armedconflicts.com references 2024 imagery purportedly showing the drone over the Shahed production plant in Tatarstan (Alabuga) — a distance consistent with the extended-range configuration. Unverified.

Downed near Moscow. South Front reported Russian air defences downing a "large bomber drone" during an attack on Moscow, identified as the E-300 "developed by AeroDrone from the Aeros SkyRanger light aircraft." Source: <https://southfront.press/russian-air-defenses-downed-large-bomber-drone-during-recent-attack-on-moscow-video/>

2026 — HUR operation. An Aviation Safety Network wikibase entry (<https://aviation-safety.net/wikibase/572595>) documents a 2026 incident involving an E-300 (Aeros

SkyRanger) operated by HUR (Ukraine's Main Directorate of Intelligence), downed by interceptor drones. If accurate, this indicates the type is in service with at least two separate Ukrainian special-operations/intelligence formations (SSO 14th Regiment and HUR).

Analytical caution. H I Sutton's guide to Ukraine's long-range attack drones (updated 25 July 2026) still lists the E-300 and D-80 in an "also of note" section, stating that "currently neither has been confirmed being used in strategic strikes" — a more conservative position than the Bryansk wreckage reporting supports.

<https://www.hisutton.com/Ukraine-OWA-UAVs.html>

7.3 Commercial and international developments (2025–2026)

20 August 2025 — Commercial UAV Expo, Las Vegas. AeroDrone exhibited in the Ukrainian Pavilion, showing heavy UAS with payloads up to 300 kg. Dmytro Shymkiv appeared as a listed speaker

(<https://www.expouav.com/speaker/dmytro-shymkiv/>). Most significantly, the company disclosed a plan to build an R&D, manufacturing and testing facility in Nevada, USA as part of an international growth strategy. Source: <https://drone.ua/en/blogs/news/aerodrone-at-the-ukrainian-pavilion-commercial-uav-expo-2025-i-n-las-vegas> | <https://aeronaut.media/news-en/uavs-drones-news-en/en-ukrainian-drone-tech-at-commercial-uav-expo-2025/>

3 September 2025 — Focus.ua video feature on AeroDrone's frontline missions, reiterating the \$450k/\$300k pricing and the 23-hour endurance figures.

2025–2026 — The funding squeeze and predicted consolidation. The Kyiv Independent's feature "Ukraine's drone companies face 'death, migration, or global acquisition'" positioned AeroDrone as a relative success story — it secured funding when many peers could not — while quoting Shymkiv on burning working capital and forecasting a three-stage shakeout. Source: <https://kyivindependent.com/global-acquisition-migration-or-death-ukraines-drone-companies-seek-opportunities-abroad-amid-defense-spending-shortfall/>

19 December 2025 — AERONAUT.media long-form on the E-300, describing it as one of the heaviest drones in its class. <https://aeronaut.media/articles-en/en-e-300-enterprise-drone-ukraine/> (direct fetch blocked; indexed summary only)

July 2026 — Shymkiv on valuation. Reported as arguing that Ukrainian defense-tech companies "with real revenues, sustainable production" are undervalued (<https://thedefender.media/>). Consistent with a principal positioning for an exit or external raise.

April 2026 — Market position assessment. An industry overview of Ukraine's 2026 drone production lists AeroDrone only among the pre-war small manufacturers ("Ukrspesystems, Byrd, AeroDrone") producing ISR platforms in small quantities, and omits it entirely from its list of key 2026 producers (which names Ukrspesystems, Terminal Autonomy, ISTA Aerospace, Quantum Systems, Saker Scout, Ukroboronprom subsidiaries). Industry context from the same source: Ukraine produces roughly 1–1.5 million drone systems per quarter across all categories; ~200–250 active companies in the Brave1 ecosystem with only 30–40 at meaningful industrial scale. Source:

<https://ukraine-war-analytics.com/technology/ukraine-drone-production-2026.html>

This is an important signal: AeroDrone's aircraft are strategically significant but numerically marginal. It is not a volume producer and does not appear on 2026 "top manufacturer" lists — e.g., it is absent from the Dignitas Fund's top-10 Ukrainian manufacturers for 2025

(<https://dignitas.fund/blog/drone-warfare-top-10-ukrainian-manufacturers-in-2025-dignitas/>).

7.4 Contested and unverified claims — handle with care

Two items circulating in 2025–2026 originate from Russian-aligned sources and should not be treated as factual:

a) UKR LEAKS piece alleging a procurement scheme (10 June 2026). A Substack article titled "'Birds' from the Fields: How AeroDrone Makes Money by Converting Agricultural Drones to the Needs of the AFU" (<https://ukrleakseng.substack.com/p/birds-from-the-fields-how-aerodrone>) characterizes AeroDrone as "a classic example of a failed startup" and insinuates a corruption scheme in UAV procurement. It names Pederiy, Shymkiv, and an otherwise unreported "Valeriy Otkidach."

Assessment: The outlet is self-described as run by Vasily Prozorov, a former officer of the Ukrainian special services who defected and now publishes anti-Kyiv material. The piece offers no contracts, invoices, court records, or audited filings — it rests entirely on recycled open-source financial figures (the same \$200K turnover, \$50K USF grant, and 2020 loss figures reported by Forbes), career-timeline juxtaposition, and the observation that the company's website is a static logo image and its Facebook page has not been updated since December 2022. Its financial claims that can be checked against Forbes are broadly accurate; its interpretive claims of corruption are unsupported. Treat as hostile information operation content with some factually correct underlying data.

Worth noting independently: the observation that aerodrone.ua is effectively a static placeholder and that social media is dormant is verifiable and true. The site contains only a logo, three capability words, five market categories, and an email address — no products, specifications, team, or news. For a company with active US expansion plans, this is unusual, and is most plausibly explained by wartime operational security.

b) Claimed Russian strike on AeroDrone's Kyiv facility (July 2026). CB Insights' news feed surfaces a Sputnik International item dated 11 July 2026 asserting that a Russian strike hit "the Kiev-based AeroDrone enterprise," naming the E-300 Enterprise and D-80 Discovery as its products. A related French-language item (La Nouvelle Tribune, 8 August 2025) covers a Russian counter-strike on a Kyiv drone plant.

Assessment: This originates from Russian state media reporting Russian Ministry of Defence strike claims, which routinely overstate damage and name targets for propaganda effect. No independent or Ukrainian confirmation of a strike on AeroDrone's facility was found. For contrast, a genuinely confirmed and independently reported strike on a Kyiv drone facility in the same period targeted Terminal Autonomy (maker of the AQ-400 Scythe and AQ-100 Bayonet), reported by The Guardian on 31 July 2026 (<https://www.theguardian.com/world/2026/jul/30/us-firm-drone-factory-terminal-autonomy-ukraine-russia>) — a different company. It is possible the AeroDrone claim is a conflation. Treat as unverified.

7.5 Items sometimes wrongly attributed to AeroDrone

- Forbes, "Ukraine's Heavy Lift Drones For Casualty Evacuation" (9 April 2026, David Hambling) — <https://www.forbes.com/sites/davidhambling/2026/04/09/ukraines-heavy-lift-drones-for-casualty-evacuation-and-air-assaults/> — surfaces on AeroDrone searches but concerns the "Vampire" from SkyFall, a heavy-lift multirotor. Not AeroDrone.
- US Army CASEVAC drone programmes (2026) — Flowcopter demonstrations at Sabre Strike 2026, Defense News "last mile" medevac robot solicitations. Unrelated.
- AeroMotors / Front Ventures Swedish investment — a different Ukrainian company (drone engines). Front Ventures' stated plan to invest \$2–4M in defense startups in 2026 (Kyiv Post) does not concern AeroDrone.

8. OVERALL ASSESSMENT AND OPEN QUESTIONS

What AeroDrone is: A small, technically distinctive Ukrainian manufacturer that occupies a genuine capability niche — the conversion of certified light aircraft into large, long-endurance, heavy-payload unmanned systems. It is one of very few companies anywhere fielding an autonomous aircraft that can carry 200 kg for 1,200 km, land on unprepared surfaces, and be reused. Its aircraft have been documented conducting deep bombing sorties into Russian territory with standard aerial ordnance, and have attracted interest from US special operations forces.

What AeroDrone is not: A scaled business. Lifetime turnover through 2022 was roughly \$200,000. It was unprofitable in 2024 and burning working capital into 2025. Production runs to a handful of aircraft per month. It does not appear on 2026 lists of Ukraine's significant drone producers. Its bill of materials is 70% imported. Its website is a placeholder and its social media has been dormant since 2022.

The central tension: strategically valuable product, structurally fragile business — the exact profile Shymkiv described when predicting that Ukrainian drone companies face "death, migration, or global acquisition."

Open questions a follow-up should resolve:

1. The EDRPOU codes and current filed accounts for both LLCs (requires manual YouControl/Clarity Project/Opendatabot queries; all blocked automated access).
2. Whether the December 2023 Thales UK MOU produced any programme, and its current status.
3. Whether the Nevada facility has been incorporated, sited, or funded — and under what US legal entity.
4. The value and quantity of the MoD contracts from 2023 onward, and whether any multi-year contract was ever secured.
5. Whether the E-300 was the aircraft behind Zelensky's March 2025 "3,000 km" claim.
6. Definitive patent search via Ukrpatent/NIPO, Espacenet and WIPO PATENTSCOPE under Cyrillic applicant names.
7. Current ownership — whether the 50/50 Pederiy/Shymkiv split persists in 2026, or whether new capital has entered.
8. Employee headcount and current monthly production rate.

SOURCE INDEX

Company / primary

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- <https://www.facebook.com/aerodroneukraine/> — official Facebook (dormant since Dec 2022)
- <https://www.linkedin.com/in/dmitryshymkiv> — Shymkiv profile
- <https://www.expouav.com/speaker/dmytro-shymkiv/> — Commercial UAV Expo speaker bio

Ukrainian press

- <https://forbes.ua/innovations/dmitro-shimkiv-z-kiivstaru-pridbav-polovinu-aktsiy-ukrainskogo-virobnika-droniv-aerodrone-tse-persha-publichna-investitsiya-v-ukrainskiy-miltech-yak-vona-vlashtovana-26012023-11147>
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- <https://www.cbinsights.com/company/aerodrone>
- <https://pitchbook.com/profiles/company/99799-75>
- <https://www.tracxn.com/d/companies/aerodrone>
- <https://www.crunchbase.com/organization/aerodrone>

Contested / adversarial

- <https://ukrleakseng.substack.com/p/birds-from-the-fields-how-aerodrone> (Russian-aligned, unverified allegations)
- Sputnik International, 11 July 2026, strike claim (unverified)

Research limitations: the session's WebSearch budget was exhausted early, so the latter two-thirds of this research was conducted via direct WebFetch and Brave Search result pages. Forbes.com, Ekonomichna Pravda, AERONAUT.media, Crunchbase, Tracxn, Opendatabot, Clarity Project, aviation-safety.net and Google Patents all blocked automated retrieval (HTTP 403/503); their content was reconstructed from mirrors, syndication and indexed summaries where available. The Ukrainian corporate registries and patent databases could not be queried directly and remain the largest evidence gap.

ARK Electronics, LLC — Research Report

0. DISAMBIGUATION (read first)

This report covers ARK Electronics, LLC of Murray / Salt Lake City, Utah, USA — the open-source-focused designer and manufacturer of PX4/ArduPilot drone flight controllers, GNSS/RTK modules, and robotics electronics, trading as arkelectron.com, founded 2018 by Alex Klimaj.

It is NOT about:

- Ark Electronics, Inc. (Irvine, California) — an electronics manufacturing services (EMS) / contract manufacturer. Unrelated ownership, unrelated products, unrelated market.
- Suzhou Good-Ark Electronics Co., Ltd. (Shenzhen Stock Exchange: 002079) — a Chinese discrete-semiconductor maker (diodes, rectifiers, power devices). Unrelated.
- Ark Invest / ARK ETFs, Ark Robotics, or any similarly named entity.

Distinguishing identifiers for the subject company: domain arkelectron.com (note: no "ics" in the domain), GitHub org github.com/ARK-Electronics, LinkedIn linkedin.com/company/ark-electronics-llc, Utah entity number 10997430-0160, X/Twitter handle @ARK_ElectrOnics.

1. COMPANY OVERVIEW

1.1 Identity and legal form

ARK Electronics, LLC is a Utah limited liability company. Tracxn's registry-derived profile lists the legal entity as "ARK Electronics, LLC." with an incorporation date of September 23, 2018 and a Utah corporate identification number of 10997430-0160 (https://tracxn.com/d/companies/ark-electronics/_9gqHMP198n11qJHrWBz1gWivsWPoYbCao_DzAu0la6c). The company's own About page also gives the founding year as 2018 (<https://arkelectron.com/about/>). Alex Klimaj's LinkedIn record dates his tenure as CEO from September 2018, which corroborates the registry date.

Being an LLC, the company files no public financial statements and is not subject to SEC reporting.

1.2 Headquarters and facilities

The most specific verified address comes from the 47G Utah aerospace-and-defense industry association member directory:

5882 S. 900 E., Suite 102, Murray, Utah 84121 (coordinates 40.642969, -111.866708)
(<https://www.47g.org/members/ark-electronics/>)

Murray is a city inside the Salt Lake City metropolitan area, which is why most secondary databases and press releases simply say "Salt Lake City, UT" — e.g. the SiMa.ai press release datelines the company as "Salt Lake City, UT" (<https://www.prnewswire.com/news-releases/simaaai-and-ark-electronics-partner-to-accelerate-intelligent-autonomous-drones-302865336.html>), as does the Mobilicom release (<https://www.stocktitan.net/news/MOB/mobilicom-and-ark-electronics-partner-to-develop-and-launch-4gnappeno7re.html>). The company's GitHub organization page also lists "Salt Lake City, Utah" (<https://github.com/ARK-Electronics>).

The company completed a facility expansion to support high-volume production; trade press reported it as enabling manufacture of millions of drone and robotic components annually (<https://www.defenseadvancement.com/news/ark-electronics-completes-facility-expansion-to-support-high-volume-drone-production/> — note: this URL returned HTTP 403 to automated fetching, so the detail here is drawn from the indexed summary rather than the full article text). An earlier expansion of the manufacturing network was reported around April 2024 via Evertiq, per the Tracxn news log.

All products are stated to be "designed, manufactured, and tested in the United States" (<https://arkelectron.com/about/>).

1.3 Size

Headcount figures vary by source and date, all pointing to a very small team:

Source	Figure	As of
Tracxn	9 employees	June 30, 2026
Tracxn	10 employees	August 31, 2026
47G member directory	"2-10 Employees" band	current listing
Explorium (data broker)	4 employees	undated / likely stale

(https://tracxn.com/d/companies/ark-electronics/_9gqHMP198n11qJHrWBz1gWivsWPoYbCao_DzAu0la6c ; <https://www.47g.org/members/ark-electronics/>)

Best estimate: roughly 10 people as of mid/late 2026. Dronecode's own 2025 retrospective characterizes ARK as "a small hardware startup out of Salt Lake City, Utah" competing against much larger engineering organizations on open-source contribution volume (<https://dronecode.org/the-2025-year-in-review/>).

Social reach as a rough proxy for market presence: ~1,282 LinkedIn followers, ~4,406 X followers, 178 GitHub org followers.

1.4 Mission and positioning

The homepage tagline is "USA Made Drone and Robotics Electronics," with a hero line of "Mission-Critical Drone Electronics." The stated value proposition is "NDAA-compliant, Blue UAS listed flight controllers, GPS modules, and sensors" for "defense, commercial, and research applications" (<https://arkelectron.com/>).

The About page frames the founding purpose as designing and building American-made electronics for drones and robotics that are "trusted for professional, commercial, and defense applications," emphasizing domestic supply chain, quality control, American manufacturing jobs, and continuous innovation (<https://arkelectron.com/about/>).

Trust badges displayed on the homepage: Made in USA, NDAA Compliant, Blue Framework Listed, Dronecode Foundation Member, ArduPilot Partner, 47G Member.

Strategically, ARK occupies a specific niche: open-standard, open-source, US-manufactured autopilot hardware for the defense-adjacent drone market. Its products are built to published open hardware standards (Pixhawk FMUv6X, Pixhawk Autopilot Bus / PAB) rather than proprietary form factors, which makes them drop-in alternatives to Chinese-manufactured boards (Holybro, CUAV) for buyers who need NDAA Section 848 compliance and Blue UAS Framework listing.

1.5 Relationship to PX4 and the Dronecode Foundation

This relationship is central to the company's identity and is unusually deep for a company of its size.

Membership. ARK Electronics joined the Dronecode Foundation as a Silver Member, announced in a post dated September 22, 2023 (<https://dronecode.org/welcome-ark-electronics-as-a-silver-member/>), with a corresponding company announcement "ARK Electronics Joins Dronecode" on October 1, 2023 (<https://arkelectron.com/news/page/3/>). Dronecode's current member roster confirms Silver tier, alongside peers including CUAV, Holybro, ModalAI, Freely Systems, NXP, SIYI, DroneBlocks, Neros, TII, and Firestorm. Platinum tier is held by Auterion, DeltaQuad, and UVify (<https://dronecode.org/our-members/>).

Code contribution. Dronecode's *2025 Year in Review* places ARK Electronics as the #2 contributing organization to PX4, behind only Auterion:

- Ecosystem totals for 2025: 651 contributors, 99 organizations, 17,979 commits.
- Independent/unaffiliated contributors: 28% of contributions.
- Auterion: 25%. ARK Electronics: 11% (2,945 total contributions, rank #2).
- Note: the same page's badge/stat block shows a "22% of PX4 Contributions" figure for ARK that conflicts with the 11% in the narrative scorecard — treat the exact percentage as uncertain; the #2 rank is consistent.
- Jacob Dahl of ARK broke into the top 10 individual PX4 contributors, credited with authoring the bidirectional DShot feature plus IMU drivers and sensor fixes.
- ARK added four new DroneCAN peripherals to the PX4 ecosystem in 2025, extending PX4 beyond flight controllers into standardized CAN-bus sensor modules.

- Non-code contribution is also noted: Discord support, GitHub issue triage, weekly PX4 developer call participation.

(<https://dronecode.org/the-2025-year-in-review/>)

Upstream hardware support. ARK boards are documented in the official PX4 user guide as "manufacturer supported" targets — see the ARKV6X page (https://docs.px4.io/main/en/flight_controller/ark_v6x) and the ARK FPV page (https://docs.px4.io/main/en/flight_controller/ark_fpv). PX4 explicitly notes it "does not manufacture this (or any) autopilot" and routes hardware/compliance questions to ARK.

Financial sponsorship. The ARK GitHub organization sponsors two individual open-source developers: julianoes (Julian Oes, longtime PX4/MAVSDK maintainer) and facontidavide (Davide Faconti, author of PlotJuggler / BehaviorTree.CPP) (<https://github.com/ARK-Electronics>).

Multi-stack strategy. The About page states ARK contributes to PX4, ArduPilot, and Betaflight to maintain cross-platform compatibility. It is a listed ArduPilot Partner. ARK also maintains forks of the AM32 ESC firmware family (as ARK32, ARK32-bootloader, ark32-configurator) and of PX4's flight_review.

Education. Klimaj collaborated with DroneBlocks, Bill Doyle of Momentum Drones, and Dronecode on the DEXI PX4 Developer Kit, a solder-free STEM education drone (<https://droneblocks.io/px4-unveils-the-droneblocks-dexi-px4-developer-kit-a-collaborative-triumph-in-drone-education/>).

1.6 Certifications and government-market positioning

- NDAA compliant across the product line (relevant to US federal procurement restrictions on Chinese drone components).
- DIU Blue UAS Framework listed — as of the May 2025 Mobilicom release, ARK held 11 autonomy and flight-computer hardware products on the Blue UAS Framework list. By August 14, 2025, the addition of ARK Flow and ARK 4IN1 ESC CONS brought the total to 13 (<https://arkelectron.com/news/page/2/>). An April 28, 2025 post noted that the M.2 LTE additions gave ARK "the most listed products on the framework." Further additions followed (ARK SAM GPS, September 9, 2026), so the current count is likely 15+.
- Hardware described as aligned with DoD Modular Open Systems Architecture goals and built to AUVSI cybersecurity benchmarks (per the Mobilicom release).
- 47G member (Utah aerospace & defense association); Klimaj sits on its Board of Governors and Industry Advisory Board.

2. FINANCIALS

Summary: no reliable public financial data exists. ARK Electronics is a privately held LLC with no disclosure obligations.

Metric	Status
Revenue	Not disclosed. Tracxn explicitly records "no data available." No credible third-party estimate was found.
Valuation	Not disclosed. No priced round, no public comparables event.
Profitability	Unknown.
Public filings	None. No SEC registration, no 10-K/10-Q/S-1, no Reg D Form D filings surfaced. As a Utah LLC, the only public filing is the state business registration (entity 10997430-0160, registered Sept 23, 2018) and annual renewals, which do not contain financials.
Debt / credit facilities	No public record found.
Stock	Not publicly traded. Do not confuse with SZSE:002079 (Suzhou Good-Ark) — see disambiguation.

(https://tracxn.com/d/companies/ark-electronics/_9gqHMP198n11qJHrWBz1gWivsWPoYbCao_DzAu0la6c)

2.1 Indirect revenue inference (analytical, not sourced)

The only way to bound revenue is from observable operating signals, which should be treated as inference rather than fact:

- 58 SKUs currently listed on the webstore, with unit prices from \$20 (adapters) to \$3,587.15 (top-configuration Jetson PAB V3 NDAA bundle). Average selling price for the core flight-controller and GNSS lines sits roughly in the \$150–\$1,200 band.
- ~10 employees, US-based hardware engineering and manufacturing.
- Self-funded facility expansion "to support high-volume drone production" with claimed capacity for "millions of components annually."
- 13-15+ products on the DIU Blue UAS Framework, which is a procurement-gating list — this implies meaningful defense-program channel revenue rather than hobbyist-only sales.

A ~10-person US hardware company with a mid-priced catalog and defense channel access is consistent with low-to-mid single-digit millions USD in annual revenue, but no source confirms this and it should be labeled as an estimate in any downstream use.

2.2 Government contracting

No federal prime-contract award records were retrieved in this research pass (USASpending and SAM.gov both require POST/API-key access not available here). Given Blue UAS Framework listing, ARK's federal exposure is more likely to be indirect — as a component supplier to drone OEMs holding the prime contracts — than as a direct prime awardee. This is a gap worth a dedicated USASpending/SAM.gov lookup.

3. FUNDING HISTORY

Summary: ARK Electronics has raised no disclosed outside capital. It appears to be entirely bootstrapped.

Tracxn classifies the company's stage as "Unfunded" with "Total funding: None raised to date" and "Valuation: not disclosed / no data available." No founders-round, pre-seed, seed, Series A, venture debt, or grant funding is recorded. No investors are listed. No acquisitions made by ARK, and no acquisition of ARK. (https://tracxn.com/d/companies/ark-electronics/_9gqHMP198nl1qJHrWBz1gWivsWPoYbCao_DzAu0la6c)

Data caveat: the Tracxn profile contains contradictory boilerplate — one templated line references "a single undisclosed round" while the substantive structured fields record no funding. The structured fields, plus the absence of any funding announcement across the company's own eight-plus years of news posts, plus the absence of any press coverage of a raise, all point to bootstrapped / no institutional capital as the correct reading.

3.1 Competitive funding context

Tracxn ranks ARK 4th among 68 active competitors in its category, and 6th by total funding — i.e., ARK ranks high on activity/traction while ranking low on capital raised:

Competitor	Founded	Country	Disclosed funding
uAvionix	2014	US	\$10M
Sagetech	1998	US	\$12M
Zerodrag	2022	India	\$1.23M
ARK Electronics	2018	US	\$0 (unfunded)
Starlogic Intelligent Technology	2017	China	\$1.49M
Defendron Technologies	2024	India	\$422K
SpeedyBee	2005	China	Unfunded
Team BlackSheep	2013	China	Unfunded
ModalAI	2021 (listed)	—	Unfunded (per Tracxn: ModalAI has in fact raised outside capital, so this row is unreliable)
UavKart	2024	India	Unfunded

Note the Tracxn competitor table has visible data-quality problems (ModalAI's founding year, country, and funding status are all questionable), so treat the comparison set as indicative only.

3.2 Capital strategy read

The pattern that emerges is growth financed by revenue and by partnerships rather than by equity:

- Facility expansion apparently funded organically.

- Distribution and capability gaps filled through partnerships (SiMa.ai for AI compute, Mobilicom for cybersecurity, Doodle Labs / Silvus / Microhard / DTC for radio integration) rather than through acquisition or heavy R&D; spend.
- A European expansion effort staffed by hire rather than by subsidiary capitalization — Tracxn logs the appointment of Thierry Basse to lead European expansion in June 2023.

Investment-thesis framing found in secondary analysis: full founder control and zero dilution make ARK an attractive target for a strategic acquirer or defense-focused growth-equity buyer, while the absence of disclosed funding and public revenue is a risk factor given that hardware manufacturing is capital intensive.

4. LEADERSHIP AND PEOPLE

4.1 Alex Klimaj — Founder, CEO (and CTO)

The company's founder and chief executive. Several data brokers list him as holding both CEO and CTO titles, consistent with a ~10-person founder-led hardware firm.

Tenure: Founder/CEO since September 2018.

Education: B.S. Electrical Engineering, California Polytechnic State University, San Luis Obispo (Cal Poly SLO), 2010–2014.

Prior career:

- Electrical Engineer at Teal Drones (Utah-based small-UAS maker, later acquired by Red Cat Holdings) — directly relevant to his later founding of a Utah drone-electronics company.
- Electrical Engineer at Code Corporation (barcode/data-capture hardware, Utah).
- Embedded hardware/firmware at Nebula Music Technologies.

PX4 involvement: Began working with PX4 Autopilot in 2017, a year before founding ARK. He is a well-known figure in the PX4 developer community; Dronecode's membership announcement describes him as "recognized within the PX4 developer community for his contributions" and notes he would continue that involvement after ARK's Silver membership.

Community/industry roles:

- Board of Governors, 47G (Utah aerospace & defense association) — since December 2023.
- Industry Advisory Board, 47G.
- Sole listed public member of the ARK GitHub organization (AlexKlimaj).

Selected quotes:

On Dronecode membership (September 2023):

"Working with Dronecode and open source has been a huge benefit to ARK Electronics growth in the drone industry."

The relationship is "not merely a business arrangement but as a pivotal catalyst for innovation," creating "a culture of collaboration and transparency that has elevated our product development," and helping others "get to market quicker with open source technology."

(<https://dronecode.org/welcome-ark-electronics-as-a-silver-member/>)

On the SiMa.ai partnership (August 31, 2026):

"Autonomous intelligence is the next frontier for drones across defense, agriculture, and commercial applications." (<https://www.prnewswire.com/news-releases/simaai-and-ark-electronics-partner-to-accelerate-intelligent-autonomous-drones-302865336.html>)

Public profiles: <https://www.linkedin.com/in/alexklimaj/> | <https://github.com/AlexKlimaj>

4.2 Ike Atkinson — Chief Operating Officer

Identified as COO of ARK Electronics in the Mobilicom partnership press release of May 20, 2025, where he is quoted on the OS3 cybersecurity platform:

OS3 is "the first solution on the market to provide comprehensive security, safety and standard compliance." (<https://www.stocktitan.net/news/MOB/mobilicom-and-ark-electronics-partner-to-develop-and-launch-4gnappeno7re.html>)

This is the only senior non-founder executive named in press materials found.

4.3 Jacob Dahl — Engineer (PX4 core contributor)

Singled out by the Dronecode Foundation's 2025 Year in Review as ARK's standout individual contributor, having reached the top 10 PX4 contributors list. Credited with:

- Authoring the bidirectional DShot feature in PX4 (ESC telemetry over the DShot protocol — a significant upstream capability, not a vendor-specific driver).
- IMU drivers and sensor fixes.
- Community support: Discord answers, GitHub issue triage, weekly developer calls.

(<https://dronecode.org/the-2025-year-in-review/>)

4.4 Thierry Basse — European expansion lead

Appointed June 2023 to lead the company's European expansion, per the Tracxn news log. No further public detail found.

4.5 Organizational notes

- No board of directors or advisory board is publicly listed (consistent with an unfunded, founder-controlled LLC — there is no investor board seat to fill).
- Tracxn's "founders" field is unpopulated (unclaimed profile), but Klimaj is unambiguously the founder per multiple independent sources.
- The company uses a shared alex byline on its news posts, reinforcing that Klimaj personally writes much of the technical marketing.
- Contact points: sales@arkelectron.com, support@arkelectron.com.

5. PRODUCTS

ARK's catalog comprises 58 SKUs across roughly eleven categories, plus airframes documented but not listed in the webstore, plus ARK-OS software. Sources: the full webstore pagination (<https://arkelectron.com/shop/>, /page/2/, /page/3/) and the complete documentation sitemap (<https://docs.arkelectron.com/sitemap.md>).

The architectural through-line: nearly everything is built to two open standards — Pixhawk FMUv6X (the flight-controller electrical/firmware spec) and the Pixhawk Autopilot Bus (PAB) (the connector/form-factor spec that lets a flight-controller module plug into any compliant carrier board). This makes ARK's flight controllers and carrier boards independently mixable with third-party hardware, and vice versa. Peripherals standardize on DroneCAN over CAN bus.

5.1 Flight Controllers

ARKV6X Flight Controller — \$400.00 (flagship) The company's core product, launched January 25, 2023. Built to the FMUv6X and PAB open standards, made in the USA, NDAA compliant, Blue UAS Framework listed (added February 14, 2025). PX4 lists it as a manufacturer-supported target.

Full specs per PX4 documentation (https://docs.px4.io/main/en/flight_controller/ark_v6x):

- MCU: STM32H743IIK6, 480 MHz, 2 MB flash, 1 MB RAM
- IMUs: triple synced — two InvenSense ICM-42688-P + one InvenSense IIM-42652 industrial IMU. The triple-synced arrangement enables data averaging, voting, and filtering (i.e. fault tolerance against a single bad sensor).
- Barometer: Bosch BMP390
- Magnetometer: Bosch BMM150
- Storage: FRAM + MicroSD slot

- 1W integrated heater to hold sensors at temperature in extreme cold (a meaningful accuracy feature — IMU bias drifts hard with temperature)
- Power: 5V input, 500 mA total (300 mA system + 200 mA heater)
- Physical: 5.0 g, $3.6 \times 2.9 \times 0.5$ cm
- 8 UARTs mapped to GPS, TELEM1–3, RC/PX4IO, debug console, secondary GPS. Firmware note: the ARKV6X bootloader enables only UART7 (TELEM1), so serial firmware upload must use that port.
- LED status indicators

Variants and bundles:

- ARKV6X Flight Controller Extended Range — \$450.00 (extended temperature/environmental range part selection)
- ARKV6X Bundle — \$750.00
- ARKV6X Extended Range Bundle — \$800.00
- ARKV6X-RT — referenced in PX4 docs as a related board

ARKV6S Flight Controller — \$340.00 Launched June 2, 2026. Explicitly a "cost-effective, single-IMU variant of the ARKV6X." STM32H743 @ 480 MHz, one industrial-grade IMU, 1W heater, FRAM + MicroSD, PAB form factor, USA-built, NDAA compliant. Ships with PX4 preloaded; ArduPilot supported.

(<https://arkelectron.com/arkv6s-flight-controller-launch/> ; <https://www.unmannedsystemstechnology.com/2026/06/ark-electronics-unveils-arkv6s-ndaa-compliant-flight-controller-for-uav-robotics-applications/>)

The V6S is the strategically interesting SKU: it trades the triple-IMU redundancy for a ~15% lower price than the ARK FPV-class cost tier while retaining the same architecture, aimed at high-volume/attribution platforms where per-unit cost dominates over redundancy.

ARK FPV Flight Controller — \$194.99 Derived from the ARKV6X but repackaged into the 30.5 mm standard FPV mounting pattern. Specs (https://docs.px4.io/main/en/flight_controller/ark_fpv):

- MCU: STM32H743I1K6, 480 MHz, 2 MB flash, 1 MB RAM
- IMU: InvenSense IIM-42653 industrial IMU
- Barometer: Bosch BMP390
- Magnetometer: ST IIS2MDC
- Power: 3–12S battery input, with regulated 12V 2A output for video transmitters and payloads (the distinguishing feature — FPV builds normally need a separate VTX regulator)
- Ships with PX4; optional ArduPilot flashing; Betaflight support added February 2025 (<https://www.unmannedsystemstechnology.com/2025/02/ark-fpv-flight-controller-now-supports-betaflight/>)

The triple-firmware support (PX4 / ArduPilot / Betaflight) on one board is unusual and is a deliberate wedge into the FPV/racing and attribution-drone market that Betaflight dominates.

ARK Pi6X — \$665.00–\$785.00 and ARK Pi6X Flow — \$915.00–\$1,035.00 Integrated flight controller plus Raspberry Pi Compute Module companion computer on one board. The "Flow" variant adds integrated optical flow. Documentation covers block diagram, pinout, CM4 Lite/eMMC flashing guides, Wi-Fi/SSH setup, ARK services, QGroundControl connection, and flight-controller firmware updating (<https://docs.arkelectron.com/sitemap.md>).

5.2 Carrier Boards (PAB)

ARK Pixhawk Autopilot Bus Carrier — \$450.00 The reference PAB carrier: accepts any PAB-compliant flight-controller module (ARKV6X, ARKV6S, or third-party). Documented with pinout, PX4 instructions, 3D models and case files.

ARK Jetson PAB Carrier — \$800.00 Combines an NVIDIA Jetson Orin module with a PAB flight-controller socket on a single board — i.e. autopilot plus AI companion computer, no inter-board cabling. Runs ARK-OS. Documented with full hardware reference: pinout, Mini DisplayPort, block diagram, 3D models and case files, GPIO control, camera overlays, USB port power cycling, flashing guide, micro-USB console.

ARK Jetson PAB V3 — \$800.00 (launched May 26, 2026) Described as "our most advanced Pixhawk Autopilot Bus (PAB) carrier board yet." Key additions over prior generations:

- Integrated 10/100 Ethernet switch providing native, low-latency Ethernet between the Jetson and the flight controller (replacing serial/UART as the high-bandwidth link — this matters for ROS2/DDS workloads)
- External Ethernet ports
- "Swap-optimized form factor"
- Jetson Orin NX / Orin Nano compatible; works with any PAB-compliant FC
- Fully NDAA compliant, USA manufactured

(<https://arkelectron.com/ark-jetson-pab-v3-launch/>)

ARK VOXL2 RTK PAB Carrier — \$1,150.00 Carrier integrating ModalAI's VOXL2 compute platform with RTK GNSS and a PAB flight-controller socket. Documented with pinout, connectors and cables, 3D models, autopilot connections.

5.3 Embedded / Companion Computers

ARK Just A Jetson — \$680.00 A standalone NVIDIA Jetson carrier without the integrated flight-controller socket — for builders who already have an autopilot. Extensive documentation: connect/get-online setup, ARK-OS usage, flight-controller firmware updating, pinout, ports and serial, ICM-42688P IMU guide, INA238 power monitor guide, block diagram, 3D model, GPIO control, camera overlays, USB power cycling, flashing guide, USB-C console.

ARK Just A Pi — \$335.00–\$475.00 (launched August 19, 2026) A budget-friendly, NDAA-compliant Linux single-board computer for robotics, based on the Raspberry Pi CM5 (both Lite + microSD and eMMC variants documented). Getting-started, flashing, Wi-Fi setup, SSH, pinout, block diagram, 3D model. This is the low-cost entry point into ARK's compute line.

Jetson module options. As of September 15, 2026, ARK Jetson bundles offer both Orin Nano and Orin NX in multiple memory sizes, including Orin NX 8GB (<https://arkelectron.com/news/>).

Bundles:

- ARK Jetson PAB NDAA Bundle — \$921.50–\$3,587.15
- ARK Jetson PAB V3 NDAA Bundle — \$921.50–\$3,587.15
- ARK Just A Jetson NDAA Bundle — \$801.50–\$3,167.15
- ARK Jetson PAB V3 with Modalix MLSoC Bundle — \$1,897.20–\$2,764.20
- ARK Just A Jetson with Modalix MLSoC Bundle — \$1,777.20–\$2,344.20

5.4 GPS / GNSS (the broadest product line — 11+ modules)

ARK fields one of the widest GNSS ranges of any autopilot vendor, spanning three silicon suppliers (u-blox, STMicroelectronics, Septentrio) and both simple UART modules and full DroneCAN RTK units.

Entry / UART modules:

- ARK SAM GPS — \$75.00 — u-blox SAM-M10Q, 4 concurrent GNSS constellations, jamming/spoofing detection, ST IIS2MDC industrial magnetometer, Pixhawk-standard UART/I2C 6-pin JST GH, 36 × 36 × 8 mm, 11 g, 15 mA average, USA-made, NDAA compliant. Added to the Blue UAS Framework September 9, 2026 (<https://arkelectron.com/ark-sam-gps-added-to-the-blue-uas-framework/>). PX4/ArduPilot/Betaflight instructions documented.
- ARK SAM GPS MINI — \$75.00 — compact variant.
- ARK DAN GPS — \$145.00 — announced September 4, 2025. PX4/ArduPilot/Betaflight support.
- ARK GPS — \$225.00 — the original module.
- ARK TESEO GPS — \$215.00–\$268.00 — built on the STMicroelectronics TESEO LIV4F receiver. A notable non-u-blox option, useful for supply-chain diversification.

RTK modules:

- ARK RTK GPS — \$615.00–\$670.00 — USA-made, Blue UAS Framework listed (added July 8, 2024), open source, DroneCAN L1/L2 RTK using the u-blox ZED-F9P, with magnetometer, barometer, IMU, buzzer, and safety switch integrated.

- ARK RTK GPS L1 L5 — \$615.00–\$670.00 — L1/L5 band variant, same integrated sensor suite. (L5 improves multipath resistance in urban/canyon environments.)
- ARK X20 RTK GPS — \$695.00–\$780.00 — NDAA-compliant all-band DroneCAN RTK using the newer u-blox X20 platform, with magnetometer, barometer, IMU, buzzer, safety switch. Latest published firmware: version 89-1.16.47e04790, dated 2025-11-17 (<https://docs.arkelectron.com/products/gps/ark-x20-rtk-gps>).
- ARK MOSAIC-X5 RTK GPS — \$1,150.00–\$1,230.00 — built on the Septentrio mosaic-X5 module, the premium option, with PX4 and ArduPilot instructions plus a browser-based Web GUI for receiver configuration.
- ARK G5 RTK GPS — \$745.00–\$835.00
- ARK G5H RTK Heading GPS — \$820.00–\$995.00 — dual-antenna moving-baseline heading solution, which delivers true heading without relying on a magnetometer (critical near large motors, steel structures, or in magnetically hostile environments).
- ARK RTK Base — \$595.00 — the ground-side base station that broadcasts RTK corrections.

Pricing caveat: some aggregator pages report different GNSS prices (e.g. ARK RTK GPS at \$915–\$1,035, X20 at \$320–\$768.50) that appear to be row-misalignment artifacts in the scraped table. The figures above are taken directly from the ARK webstore pagination and should be treated as authoritative as of this research date.

5.5 Sensors

- ARK Flow — \$250.00 — optical flow sensor with integrated distance sensing, DroneCAN. Added to the Blue UAS Framework August 14, 2025. Documented with ArduPilot/PX4 instructions, DroneCAN message definitions, 3D models and case files.
- ARK Flow MR — \$350.00 — medium-range variant: PAA3905 optical flow sensor with a 50 m distance sensor. Announced February 6, 2025.
- ARK DIST SR — \$180.00 — short-range distance sensor / rangefinder.
- ARK DIST MR — \$280.00 — medium-range distance sensor. Announced August 14, 2025. Documented with pinout, firmware, UART/MAVLink interface, ArduPilot/PX4 instructions, 3D models and case files.
- ARK MAG — \$175.00 — standalone external magnetometer module with pinout, ArduPilot/PX4 instructions, 3D models.
- ARK CANnode — \$150.00 — a generic DroneCAN node board, letting integrators put arbitrary sensors/actuators onto the CAN bus. ArduPilot and PX4 instructions documented.

5.6 IMUs (standalone, high-grade)

- ARK SCH16T — \$160.00 and ARK SCH16T-K10 IMU — \$160.00 — based on the Murata SCH16T MEMS gyro/accelerometer, a high-stability automotive/industrial part.
- ARK ADIS16507 — \$1,150.00 — based on the Analog Devices ADIS16507 tactical-grade IMU. This is by a wide margin the highest-priced single component in the catalog and signals a navigation-grade / GPS-denied use case.

Both are documented with PX4 and ArduPilot instructions and 3D models.

5.7 ESCs (Electronic Speed Controllers)

- ARK 4IN1 ESC — \$218.50 — four-in-one FPV-form-factor ESC, NDAA compliant, AM32 firmware, made in the USA. Blue UAS Framework listed (August 14, 2025, as the CONS variant). Documented with pinout, firmware flashing (ArduPilot ESC passthrough, flash ARK32, flash bootloader), ARK32 configuration, PWM calibration, motor spin direction, 3D models.
- ARK 4IN1 ESC CONS — \$238.50 — conformal-coated / ruggedized variant.
- ARK 12S CAN ESC — \$399.00 — single high-voltage (12S) ESC with CAN bus control rather than PWM/DShot, for larger airframes. Documented with pinout, mounting and power connections, firmware, 3D models.

ARK maintains its own fork of the open-source AM32 ESC firmware as ARK32, plus a fork of the AM32 bootloader and configurator.

5.8 Power Modules

- ARK PAB Power Module — \$100.00
- ARK PAB Power Module No Connector — \$100.00
- ARK 12S PAB Power Module — \$100.00
- ARK 12S Payload Power Module — \$195.00

Each documented with PX4 instructions, ArduPilot instructions, and 3D models.

5.9 Radio / Connectivity

- ARK M.2 LTE — \$250.00 — M.2 form-factor LTE modem for BVLOS/cellular command-and-control. Three variants added to the Blue UAS Framework April 28, 2025. Documented with OpenVPN configuration.
- ARK Microhard DDL Carrier — \$320.00–\$768.50 — carrier board for Microhard digital data link radios. 3D model documented.

ARK also publishes radio integration guides for third-party datalinks it does not sell: Doodle Labs Mesh Rider, DTC BluSDR, Mobilicom SkyHopper Pro, Silvus StreamCaster SL5200, plus its own Microhard DDL carrier (<https://docs.arkelectron.com/sitemap.md>). This documentation-as-marketing approach positions ARK as the integration hub for whatever radio a defense customer is mandated to use.

5.10 Accessories and Breakouts

- ARK Primary Avionics Breakout — \$79.99
- ARK Secondary Avionics Breakout — \$79.99
- ARK Pixhawk Payload Bus Breakout — \$89.99
- ARK Servo Expander — \$49.99
- ARK Pixhawk Debug Adapter — \$60.00
- Servo Adapter — \$20.00
- Ethernet Adapter — \$20.00
- Gigabit Ethernet Adapter — \$20.00
- ARK USB Adapter — \$20.00

5.11 Airframes

ARK RIG5 and ARK RIG7 appear in the documentation sitemap under an "Airframes" category but do not appear in the public webstore listing. These are likely reference/development airframes or channel-only products. Worth a follow-up check.

5.12 Software — ARK-OS and open-source tooling

ARK-OS is the company's Linux software stack for its compute products — "software for the ARK Jetson Carrier and ARK Pi6X Flow." Written primarily in Python, MIT licensed, 44 stars / 9 forks on GitHub, last updated September 1, 2026 (<https://github.com/ARK-Electronics>). Documentation covers ARK-OS services and apt package-pinning guidance ("Apt: Hold Back Risky Packages"). Functionality includes autopilot connection brokering (to QGroundControl / Mission Planner), flight-controller firmware updating, camera overlays, GPIO control, and USB port power cycling.

PointPerfect Flex integration (July 15, 2026). ARK-OS gained an NTRIP client for u-blox PointPerfect Flex, enabling centimetre-level RTK positioning on PX4- and ArduPilot-based UAVs without deploying a physical base station — corrections stream from u-blox's network service instead (<https://arkelectron.com/news/>). This is operationally significant: it removes the logistical burden of hauling and surveying an RTK base for each flight, which is a real barrier for survey and inspection operators.

Other open-source repositories (<https://github.com/ARK-Electronics> — 88 repos total):

Repo	Description	Language	Stars
ROS2_PX4_Offboard_Example	Controlling PX4 velocity setpoints via ROS2 teleop	Python	208
ARK-OS	Software for ARK Jetson Carrier and ARK Pi6X Flow	Python	44
isaac_ros-dev	ROS2 workspace for NVIDIA Isaac ROS on Jetson	Shell	29
mavlink-to-lattice	Bridge between MAVLink and Anduril Lattice	Python	28
ark_jetson_kernel	Kernel build scripts for ARK Jetson Carrier	Shell	23

Repo	Description	Language	Stars
logloader	Automatic upload/download of PX4 flight logs	C++	23
ARK32	Fork of AM32 ESC firmware	C	11
ark-meta-simaai	Yocto layer for Jetson carrier on SiMa.ai Modalix SoM	Shell	0
ark_jetson_yocto	Yocto Linux + CUDA BSP for Jetson carriers	Python	0
ark-hardware	(no description — likely hardware design files)	—	0
flight_review	Fork of PX4 flight-log web analyzer	JavaScript	0
ark32-configurator	Fork of am32-configurator	TypeScript	0
ARK32-bootloader	Fork of AM32-bootloader	C	0

The mavlink-to-lattice bridge deserves note: Lattice is Anduril Industries' command-and-control platform. Publishing a MAVLink-to-Lattice bridge (announced August 14, 2025) is an explicit play to make ARK-equipped, PX4-based drones interoperable with the defense C2 stack most likely to be deployed alongside them.

5.13 Knowledge base

ARK publishes an unusually thorough vendor-neutral engineering knowledge base: Microcontrollers, SWD Programming, Serial Communication (UART), CAN Bus, MAVLink, Connectors and Wiring, Power and Grounding, GPS Placement, Communication Buses (I2C, SPI), Recommended Tools, ST-LINK Flashing Guide, DroneCAN GUI Tool Guide, u-blox Firmware Update, PX4 Log Encryption, PX4 Setup, USB Peripherals. Plus ROS2 & PX4 tutorials: teleop example, precision landing, ArUco detection, custom modes hardware demo, custom modes with ROS2/QGC, simulated offboard mode.

6. PATENTS

Summary: no granted patents or published applications assigned to ARK Electronics, LLC, or naming Alex Klimaj as inventor, were located in this research pass. The evidence strongly suggests the company's IP strategy is deliberately not patent-based.

6.1 Search limitations (stated honestly)

Direct patent-database queries were attempted and blocked:

- Google Patents (patents.google.com, assignee and free-text queries, plus the XHR API endpoint) — returned HTTP 503 on every attempt.
- Justia Patents (patents.justia.com) — returned HTTP 403.
- Espacenet (worldwide.espacenet.com) — returned HTTP 403.
- FreePatentsOnline — connection reset.
- USPTO Patent Public Search / PatentsView API — requires POST or an API key, unavailable via the fetch tooling here.
- Web-search budget for this session was exhausted before a patent-specific search could be issued.

Therefore "no patents found" here means "none surfaced across company materials, press releases, trade coverage, and investor-database profiles," not "a negative result from an exhaustive patent-database query." A definitive answer requires a direct USPTO Patent Public Search or PatentsView API query on assignee "ARK Electronics" (Utah) and inventor "Klimaj, Alex."

6.2 Affirmative evidence of a non-patent IP posture

Several independent signals point the same direction:

1. Products are built to open standards ARK does not own. FMUv6X and the Pixhawk Autopilot Bus are Dronecode/Pixhawk-project open hardware specifications. A company whose differentiation rests on *conforming to* someone else's published spec cannot patent the spec, and gains little from patenting around it.
2. ARK publishes its own designs as open source. The webstore describes the ARK RTK GPS as "open source," and ark-hardware is a public repository. Open-publication is antithetical to a patent-first strategy (and,

once published, starts the clock on any filing window).

3. Firmware is derived from GPL/permissive upstream projects. ARK32 derives from AM32; flight controllers run PX4 (BSD-3) and ArduPilot (GPLv3). ARK contributes changes back upstream rather than holding them proprietary.
4. The Tracxn profile — which does report IP activity when it exists — records no patent data.
5. No press release, product page, or datasheet uses "patented," "patent pending," or cites a patent number. For a hardware company, patent marking is both legally advantageous and standard marketing practice; its total absence is informative.

6.3 What ARK's actual defensible moat appears to be

In lieu of patents, the company's competitive protection appears to rest on:

- Regulatory moat: NDAA compliance plus 13-15+ DIU Blue UAS Framework listings, reportedly the most of any component vendor on that list. Framework listing is expensive and slow to obtain and functions as a de facto barrier to entry for the US defense channel — arguably a stronger moat than a patent in this market.
- Domestic manufacturing: US design, manufacture, and test, which competitors offshoring to China structurally cannot replicate.
- Upstream code position: being the #2 contributor to PX4 means ARK's hardware is supported first and best in the firmware everyone else also uses.
- Trademark: the "ARK" and "ARKV6X" marks are presumably protected, though no USPTO trademark registration was verified in this pass (a TSDR lookup would confirm).

6.4 Recommended follow-up

- USPTO Patent Public Search: assignee ARK Electronics AND state UT; inventor Klimaj.
- USPTO TSDR / TESS trademark search for ARK ELECTRONICS, ARKV6X, ARK-OS, ARK32.
- Google Patents at a later time (the 503s appeared to be transient rate limiting rather than a permanent block).

7. LATEST NEWS (reverse chronological, 2023–September 2026)

Primary source for the company-published timeline: <https://arkelectron.com/news/> (three pages).

2026

September 15, 2026 — NVIDIA Jetson Orin NX 8GB now in ARK Jetson bundles. Jetson NDAA bundles expanded to offer Orin Nano and Orin NX options across multiple memory configurations, including the 8GB NX. (<https://arkelectron.com/news/>)

September 9, 2026 — ARK SAM GPS added to the Blue UAS Framework. The \$75 u-blox SAM-M10Q-based GNSS + IIS2MDC magnetometer module (36 × 36 × 8 mm, 11 g, 15 mA) joins the DIU list. ARK notes it is NDAA compliant and will not impair a platform's eligibility for the Blue UAS Cleared List. Notable because it puts a genuinely low-cost, FPV/small-UAS-class GNSS module on the Framework — most Framework-listed GNSS is far more expensive. (<https://arkelectron.com/ark-sam-gps-added-to-the-blue-uas-framework/>)

August 31, 2026 — SiMa.ai partnership. SiMa.ai (San Jose, "Physical AI" silicon company) and ARK announced the ARK-SiMa.ai Modalix MLSoC Bundles: bundled hardware/software kits combining SiMa.ai's Modalix System-on-Module (built on the Modalix MLSoC) and its Palette Neat AI development tooling with ARK's carrier hardware. Marketed as pin/layout-compatible with existing GPU-based edge platforms so drone OEMs can upgrade compute without redesigning the airframe electronics. Available immediately to qualifying manufacturers and integrators through a SiMa.ai portal.

Quotes:

- Durga Peddireddy (SiMa.ai): "winning programs will be built on platforms that pair globally trusted hardware with best-in-class AI compute"

- Alex Klimaj: "Autonomous intelligence is the next frontier for drones across defense, agriculture, and commercial applications"

(<https://www.prnewswire.com/news-releases/simaai-and-ark-electronics-partner-to-accelerate-intelligent-autonomous-drones-302865336.html> ;

<https://arkelectron.com/ark-and-sima-ai-partner-on-intelligent-autonomous-drones/>)

Two resulting SKUs shipped: ARK Jetson PAB V3 with Modalix MLSoC Bundle (\$1,897.20–\$2,764.20) and ARK Just A Jetson with Modalix MLSoC Bundle (\$1,777.20–\$2,344.20), plus a public Yocto layer (ark-meta-simaai).

August 19, 2026 — ARK Just A Pi launch. A budget-friendly, NDAA-compliant Raspberry Pi CM5-based Linux compute board for robotics begins shipping at \$335–\$475. Extends ARK's compute line downward from Jetson pricing.

July 15, 2026 — RTK without a base station: u-blox PointPerfect Flex on ARK-OS. ARK-OS gains an NTRIP client for streaming u-blox PointPerfect Flex corrections, giving PX4 and ArduPilot vehicles centimetre-level RTK without a physical base station.

July 9, 2026 — "How to Brick Your Flight Controller." A technical deep-dive on a flash-memory fault that causes a hard fault on every boot. Notable as an example of the company's engineering-transparency marketing — publishing a failure-mode post-mortem rather than only product announcements.

June 2, 2026 — ARKV6S Flight Controller launch. The single-IMU, lower-cost (\$340) sibling to the ARKV6X ships. STM32H743 @ 480 MHz, industrial IMU, 1W heater, FRAM + MicroSD, PAB form factor, USA-built, NDAA compliant, PX4 preloaded with ArduPilot supported. Covered independently at <https://www.unmannedsystemstechnology.com/2026/06/ark-electronics-unveils-arkv6s-ndaa-compliant-flight-controller-for-uav-robotics-applications/>

May 26, 2026 — ARK Jetson PAB V3 launch. The company's most advanced PAB carrier: Jetson Orin NX/Nano compatible, integrated 10/100 Ethernet switch for native low-latency Jetson↔FC Ethernet, external Ethernet ports, swap-optimized form factor, NDAA compliant, USA manufactured. \$800 standalone; bundles to \$3,587.15.

May 7, 2026 — ARK at XPONENTIAL 2026 (Detroit). Booth presence at the AUVSI autonomy/robotics trade show.

(Undated, 2026) — Facility expansion completed to support high-volume drone production, reportedly enabling manufacture of millions of drone and robotic components annually (<https://www.defenseadvancement.com/news/ark-electronics-completes-facility-expansion-to-support-high-volume-drone-production/>)

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2025

November 19, 2025 — ARK November Update. Roundup covering the new X20 and DAN GPS products and expanded social/community channels.

September 4, 2025 — ARK DAN GPS announcement. New \$145 GNSS module.

August 14, 2025 — MAVLink to Anduril Industries Lattice bridge. Open-source tool streaming MAVLink telemetry into Anduril's Lattice C2 platform (mavlink-to-lattice, 28 stars).

August 14, 2025 — ARK Flow and ARK 4IN1 ESC CONS added to the Blue UAS Framework, bringing ARK's Framework total to 13 products.

August 14, 2025 — ARK DIST announcement. Distance-sensor/rangefinder line release.

May 20, 2025 — Mobilicom (Nasdaq: MOB) strategic partnership. Joint development of affordable, cybersecure technology for small drones and robotics, pairing Mobilicom's OS3 platform (Operational Security, Safety, and Standards compliance — continuous monitoring, detection, alerting, threat prevention) with ARK's US-manufactured, NVIDIA-powered hardware. First co-developed offering targeted for H2 2025; both companies showcased at XPONENTIAL 2025 (billed at 7,500+ attendees). Market sizing cited: mini/small drone sector at \$5.8B (2023), growing at 8.6% CAGR to \$10.4B by 2030. ARK noted as holding 11 autonomy and flight-computer products on the Blue UAS Framework, NDAA compliant, aligned

with DoD Modular Open Systems Architecture, built to AUVSI cybersecurity benchmarks. ARK COO Ike Atkinson quoted. Mobilicom's stock moved -5.08% in the session following the announcement. A second Mobilicom partnership announcement followed in July 2025. (<https://www.stocktitan.net/news/MOB/mobilicom-and-ark-electronics-partner-to-develop-and-launch-4gnappeno7re.html>)

April 28, 2025 — ARK M.2 LTE added to the Blue UAS Framework. Three LTE module variants listed; the post notes this gave ARK the most listed products on the Framework of any vendor.

February 14, 2025 — ARKV6X Flight Controller added to the Blue UAS Framework. The flagship autopilot achieves DIU listing — arguably the single most commercially consequential event in the timeline, since it unlocks the defense procurement channel for ARK's core product.

February 2025 — ARK FPV Flight Controller adds Betaflight support, giving the board three open-source firmware options (PX4, ArduPilot, Betaflight). (<https://www.unmannedsystemstechnology.com/2025/02/ark-fpv-flight-controller-now-supports-betaflight/>)

February 6, 2025 — NDAA Jetson Orin Nano and NX bundles introduced.

February 6, 2025 — ARK SAM GPS announcement.

February 6, 2025 — ARK Flow MR announcement. PAA3905 optical flow with 50 m distance sensing.

(2025, ongoing) — Dronecode ecosystem standing. ARK finishes the year as the #2 contributing organization to PX4 with 2,945 contributions (~11% of the ecosystem), adds four new DroneCAN peripherals to PX4, and sees Jacob Dahl reach the top 10 individual PX4 contributors on the strength of the bidirectional DShot implementation. (<https://dronecode.org/the-2025-year-in-review/>)

2024

July 8, 2024 — ARK RTK GPS added to the Blue UAS Framework.

April 2024 — Manufacturing network expansion reported via Evertiq (per Tracxn news log).

2023

October 1, 2023 — "ARK Electronics Joins Dronecode" (company announcement).

September 22, 2023 — Dronecode Foundation welcomes ARK Electronics as a Silver Member. At the time, the company is described as five years old with hardware designed, manufactured, and tested in the USA. Featured products cited: the "NDAA compliant Pixhawk FMUV6X" and an "NDAA compliant Pixhawk Carrier board for Nvidia Jetson Orin." Klimaj's quotes frame the membership as "not merely a business arrangement but as a pivotal catalyst for innovation." (<https://dronecode.org/welcome-ark-electronics-as-a-silver-member/>)

June 2023 — Thierry Basse appointed to lead European expansion (per Tracxn).

January 25, 2023 — ARKV6X announcement. Launch of the ARKV6X Flight Controller and ARK PAB Carrier, both built to open Pixhawk standards. This is the product that established the company.

8. ANALYTICAL SUMMARY

What ARK Electronics is. A roughly ten-person, founder-controlled, entirely bootstrapped Utah LLC that has, in eight years and with zero outside capital, become the #2 code contributor to PX4 and the vendor with the most products on the DIU Blue UAS Framework. It sells ~58 SKUs spanning autopilots, GNSS/RTK, IMUs, optical flow, rangefinders, ESCs, power, LTE/radio carriers, and Jetson/Raspberry-Pi compute — all designed and built in the USA to open Pixhawk standards.

The strategy, stated plainly. Do not invent a proprietary platform; instead implement the open standard better than anyone, manufacture it domestically, get it onto the DIU list, and become the default component supplier for every US drone OEM that needs to be NDAA compliant. Contribute heavily upstream so the firmware everyone uses supports your hardware first. Fill capability gaps with partnerships (SiMa.ai for AI silicon, Mobilicom for cyber, Anduril-Lattice interop for C2) rather than with capital.

Strengths. Regulatory moat via Blue UAS listings that competitors cannot quickly replicate. Genuine technical credibility in the PX4 community. Zero dilution and full founder control. Breadth of catalog unusual for the headcount. Multi-firmware (PX4/ArduPilot/Betaflight) and multi-silicon (u-blox/ST/Septentrio; NVIDIA/SiMa.ai/Raspberry Pi) hedging.

Risks and open questions. Hardware manufacturing is capital intensive and ARK has no disclosed funding buffer. Key-person concentration in Alex Klimaj is severe. No patent portfolio means no licensing revenue and no defensive IP if a larger player copies the (openly published) designs — the moat is regulatory and reputational, not legal. Revenue is entirely unverified. And as an unfunded, founder-controlled asset sitting on a scarce regulatory position in a hot defense market, ARK is an obvious acquisition target for a defense prime or a growth-equity buyer.

Gaps in this report requiring follow-up. (a) Definitive USPTO patent and trademark queries — blocked here by database 403/503 responses. (b) Federal contract award history via USASpending/SAM.gov — requires POST/API access. (c) Current exact Blue UAS Framework product count — the trajectory is 11 (May 2025) → 13 (Aug 2025) → 15+ (Sept 2026), but the live DIU list should be checked. (d) The ARK RIG5 / RIG7 airframes, documented but not sold publicly. (e) Full text of the facility-expansion coverage (square footage, investment, capacity), blocked by a 403.

Consolidated source list

Company: <https://arkelectron.com/> · <https://arkelectron.com/about/> · <https://arkelectron.com/shop/> · <https://arkelectron.com/news/> · <https://docs.arkelectron.com/sitemap.md> · <https://docs.arkelectron.com/products/gps/ark-x20-rtk-gps> · <https://arkelectron.com/arkv6s-flight-controller-launch/> · <https://arkelectron.com/ark-jetson-pab-v3-launch/> · <https://arkelectron.com/ark-sam-gps-added-to-the-blue-uas-framework/> · <https://github.com/ARK-Electronics> · <https://github.com/AlexKlimaj>

PX4 / Dronecode: https://docs.px4.io/main/en/flight_controller/ark_v6x · https://docs.px4.io/main/en/flight_controller/ark_fpv · <https://dronecode.org/welcome-ark-electronics-as-a-silver-member/> · <https://dronecode.org/the-2025-year-in-review/> · <https://dronecode.org/our-members/> · <https://droneblocks.io/px4-unveils-the-droneblocks-dexi-px4-developer-kit-a-collaborative-triumph-in-drone-education/>

Press / third party: <https://www.prnewswire.com/news-releases/simaai-and-ark-electronics-partner-to-accelerate-intelligent-autonomous-drones-302865336.html> · <https://www.stocktitan.net/news/MOB/mobilicom-and-ark-electronics-partner-to-develop-and-launch-4gnappeno7re.html> · <https://www.unmannedsystemstechnology.com/company/ark-electronics/> · <https://www.unmannedsystemstechnology.com/2026/06/ark-electronics-unveils-arkv6s-ndaa-compliant-flight-controller-for-uav-robotics-applications/> · <https://www.unmannedsystemstechnology.com/2025/02/ark-fpv-flight-controller-now-supports-betaflight/> · <https://www.defenseadvancement.com/news/ark-electronics-completes-facility-expansion-to-support-high-volume-drone-production/>

Databases / directories:

https://tracxn.com/d/companies/ark-electronics/_9gqHMP198nl1qJHrWBz1gWivsWPoYbCao_DzAu0la6c · <https://www.47g.org/members/ark-electronics/> · <https://www.linkedin.com/in/alexklimaj/> · <https://www.linkedin.com/company/ark-electronics-llc> · https://x.com/ARK_Electronics

Skipped: patent-database confirmation (Google Patents/Justia/Espacenet all returned 403/503, web-search budget exhausted) and federal contract history (USASpending needs POST). Add when a USPTO Patent Public Search and a SAM.gov/USASpending lookup can be run directly.

DEFENDRON TECHNOLOGIES — RESEARCH REPORT

Compiled 24 September 2026. All findings sourced from public web material; gaps are flagged explicitly rather than filled in.

0. IDENTIFICATION AND SCOPE NOTE

The name "Defendron Technologies" resolves to exactly one real entity: Defendron Technologies Private Limited, an Indian defence-electronics startup in Pune, Maharashtra, trading as DEFENDRON at <https://www.defendrontech.com>.

Two disambiguation points matter before anything else:

1. This is not a counter-drone (C-UAS) company, despite the name suggesting "defend" + "drone". It builds EO/IR (electro-optical / infrared) ISR payloads that mount onto drones — the sensor turret, not the interceptor. Searches for "Defendron drone defense" surface unrelated counter-UAS firms (Dedrone/Axon, DroneShield, WhiteFox Defense, Drone Defence UK) which have no connection to it.
2. It is frequently confused in search with the similarly-named Australian firm DefendTex (<https://www.defendtex.com/air.html>), which makes the Drone 40 loitering munition. Unrelated company.

Coverage caveat: this is a company roughly 23 months old with essentially no press coverage, no analyst coverage, no filings beyond statutory Indian registry minimums, and no announced funding rounds.

1. COMPANY OVERVIEW

Legal name: DEFENDRON TECHNOLOGIES PRIVATE LIMITED. CIN U30300PN2024PTC235304. Incorporated 28 October 2024. Registered/operating office: Aundh, Pune, Maharashtra 411007 (operating address is inside a coworking space, "11 Cowork"). Status: Active.

Mission (verbatim): "To design and deliver advanced security solutions that ensure reliability, resilience and continuous protection in critical environments." Vision: "To lead the evolution of intelligent security systems through innovation, adaptability and engineering excellence." Marketing tagline: "Precision. Sovereignty. Mission success." Positioning: "ISR Innovation for the Modern Aerial Ecosystem."

Size: no verified headcount; website team page lists 6 named individuals; About page self-describes as "a small team of tinkerers and engineers." Working estimate 6–15 people.

Target customers (stated, unverified): Indian Armed Forces, DRDO, Defence PSUs, UAV OEMs/integrators, loitering-munition developers. No named customer, contract, or order has ever been publicly disclosed.

Competitive set (India): Tracxn lists Defendron among 31 Pune drone startups (DroneAcharya, NavStik Labs). Broader gimbal competitors: Tonbo Imaging, Paras Defence, IdeaForge payload arm.

2. FINANCIALS

Private, unlisted — no public revenue figure or valuation. Registry data: authorised capital ~5.19 lakh, paid-up capital ~1.19 lakh (~USD 1,400) — a nominal founding capitalisation. FY2024-25 revenue in Tracxn's "0–10 Cr" band (widest, least informative band); covers only 5 months of existence (incorporated Oct 2024). No registered secured charges/loans. First financial statements filed for year ended 31 March 2025; first AGM 24 December 2025. No valuation has ever been published; Tracxn tags the company "Unfunded" (likely a data-lag artifact — see Section 3).

3. FUNDING HISTORY

Company's own Investor page (defendrontech.com/investor) names three backers, no amounts or dates disclosed:

1. All In Capital — India pre-seed/seed VC (founded by Kushal Bhagia). Implies a pre-seed/seed cheque in the 1–4 Cr range typical for this fund, but unconfirmed.
2. IIMA Ventures — incubator/early-stage platform at IIM Ahmedabad (formerly CIIE.CO). Could be incubation, accelerator, grant, or small equity.

3. MeitY Startup Hub (MSH) — Government of India programme, typically non-dilutive grant/incubation support.

Third-party contradiction: Tracxn tags the company "Unfunded" — no round in Tracxn's database, and zero funding-round press coverage found (Inc42, Entrackr, YourStory, VCCircle, Business Standard all silent). Most likely explanation: small, unannounced early-stage support never press-released.

Sector context: Indian defence-tech funding reached USD 247M in 2025 (~2x 2024's USD 125M) across 30 rounds, dominated by large rounds like Raphe mPhibr's Series B (General Catalyst). Defendron is far below this radar — a pre-seed micro-cap.

4. LEADERSHIP AND PEOPLE

Board of directors (MCA):

- Omkar Ashok Lad (DIN 10822387) — Director, appointed 28 Oct 2024
- Pratibha Agrawal (DIN 07971758) — Director, appointed 28 Oct 2024 (older DIN block ~2017-18, other directorships unknown, not on team page, role unresolved)

Founder & CEO: Omkar Ashok Lad

- BE (Hons.), BITS Pilani
- Embedded engineer, 10+ years experience, UAV and embedded systems specialist
- Prior: Managing Director at Enarr Unico Infrachem Solutions Pvt. Ltd.; earlier at Unico Engineering Solutions
- Leads vision/technical direction and system architecture (hardware + software)
- LinkedIn: [linkedin.com/in/omkar-lad-b7057228/](https://www.linkedin.com/in/omkar-lad-b7057228/) (not directly fetchable)
- Registered company email of record is his personal Gmail (omkar269@gmail.com)

Team (defendrontech.com/team/), 6 people total including CEO:

- Omkar Lad — CEO & Founder — UAV/embedded systems, system architecture
- Kiran P. — Embedded Firmware (Drive Systems) — motion/stabilisation control code
- Hemanshu Kale — System Software — video pipelines, camera controls, operator interfaces
- Abbaas Zaidi — Mechanical Design — enclosures/structures for field conditions
- Manthan Jogi — Product & BD — roadmap, customer relationships, GTM
- Nikhil Salunkhe — Customer Success & Operations — field deployment, integration, mission readiness

No bios beyond 1-2 sentence descriptions; no prior employers/education for the five non-founder staff.

Company has posted at least one job (UAV Integration Engineer, Pune, 1-3 yrs, now closed).

5. PRODUCTS

Single product category: EO/IR gimballed ISR payloads for UAVs, five current models (160g–1500g). Also a secondary custom-test-rig line for defence-systems validation (not listed as a catalogue product, mentioned only in press).

Note: product line was renumbered — old DTX-250/DTX-500 URLs now 404, mapped to new DTX-350/DTX-750 HDIR naming. Preserved here as a known source-conflict.

Model	Weight	Mech. axes	EO optical zoom	IR resolution	Status
DTX Dual SO	160 g	0 (electronic stabilisation only)	2x	640×480	Available
DTX KZ	220 g	1	2x	640×480 (or 320×240 cost-down)	Out of stock
DTX-350	420 g	2	20x	640×480	Available
DTX-750 HDIR	790 g	2 (assumed)	30x	1280×720 HD	Available
DTX-1500	1500 g	2	40x	1280×720 HD + 4x IR zoom	Available

All models: onboard edge AI target detection/lock/classification; 3-axis stabilisation (2 mechanical + 1 electronic, or all-electronic on the smallest); MAVLink compatible; H.264/H.265 RTP/RTSP over Ethernet; 12-24V power; -20° C to +55° C operating range; IP55 rated (homepage claims IP67 — inconsistent with per-product IP55 specs); onboard recording; digital zoom to 8x; customisable configurations on request.

Stated use cases: border/perimeter surveillance, military reconnaissance, maritime patrol, counter-UAS ops, critical infrastructure protection, disaster response, loitering-munition/interceptor guidance (DTX Dual SO and DTX KZ descriptions explicitly reference "urban neutralisation," "interceptor swarm," and "terminal navigation" — i.e., munition seeker use).

No pricing published for any product.

6. PATENTS

No patents, patent applications, or patent numbers attributable to Defendron Technologies or Omkar Ashok Lad were found in any publicly accessible source. Google Patents assignee query returned an error and remains unverified; Indian Patent Office full-text search was not reachable within the research session. Company makes no "patented" or "patent pending" claims anywhere on its site. Treat as having no verifiable patent portfolio — filings made after ~March 2025 would legitimately still be inside the 18-month pre-grant publication window and could exist unpublished.

7. LATEST NEWS (2024–2026)

- 28 Oct 2024 — Incorporated (RoC Pune), directors Omkar Lad and Pratibha Agrawal appointed day one.
- June 2025 — Featured in Inc42's "30 Startups To Watch" — the company's only substantive press coverage. Confirmed founder background, described payloads "starting at just 280g" with dual EO-IR, 30x zoom, thermal imaging, optional laser range finders (note: current site doesn't list an LRF option), and custom test rigs. Cited market context: global drone payload market projected to reach USD 22.95Bn by 2032; imaging/mapping segment from USD 2.6Bn (2024) to USD 7.2Bn (2029). No funding amount, revenue, or customers named.
- 2025 — Posted and later closed a UAV Integration Engineer job listing.
- FY ending 31 Mar 2025 — First financial statements filed, revenue in 0-10 Cr band.
- 24 Dec 2025 — First AGM held.
- 2025-2026 — Product line renumbered (DTX-250→DTX-350, DTX-500→DTX-750 HDIR); DTX-1500 flagship added; DTX KZ currently out of stock.
- Current (Sept 2026) — Investor page live naming All In Capital, IIMA Ventures, MeitY Startup Hub; no dates given.

Conspicuously absent: no announced funding round, no named customers/contracts/trials, no IDEX/ADITI/TDF grant announcement, no confirmed trade-show presence, no certifications (MIL-STD/DGQA/CEMILAC/ISO) announced, no partnerships, no executive hires, no defence trade-press coverage.

8. SUMMARY ASSESSMENT

Defendron Technologies is a pre-revenue-or-near-pre-revenue, pre-seed-stage Indian defence hardware startup, roughly 23 months old, with a six-person team operating from a Pune coworking space, a five-model EO/IR payload catalogue that has already been renumbered once, three self-reported early-stage/government backers with undisclosed terms, no announced funding, no named customers, no verifiable patents, and exactly one piece of meaningful press coverage (Inc42, June 2025). Real and technically credible on paper, but at the very earliest stage of its public life.

Sources: defendrontech.com (about/team/products/investor/contact + individual product pages), indiafilings.com, thecompanycheck.com, tracxn.com, falcone.biz.com, zoominfo.com, inc42.com, business-standard.com.

SAGETECH AVIONICS — RESEARCH REPORT

Compiled 2026-09-24 from public web sources. All figures as reported by the cited source.

0. CORRECTION TO THE PREMISE

The task described Sagetech Avionics as "based in Sandpoint/Sun Valley Idaho." That is incorrect. Every primary and secondary source checked places the company in the Columbia River Gorge, Washington State:

- Company's own site footer and contact block: 317 W Steuben Street, Bingen, WA 98605 (<https://sagotech.com/about-us/who-we-are/>, <https://sagotech.com/about-us/careers/>)
- USAspending.gov recipient record (UEI T4UNAUXXM3): 317 W STEUBEN ST, BINGEN, WA 98605-9006, congressional district WA-04
- HigherGov awardee profile: Bingen, WA (<https://www.highergov.com/awardee/sagotech-avionics-inc-10280151/>)
- Older press releases datelined White Salmon, WA (the adjacent town, across from Bingen); some databases (CB Insights, ZoomInfo) still list a White Salmon address at 1000 E Jewett Blvd
- Careers page explicitly: *"We're located in Bingen, WA, in the Columbia River Gorge, which is about an hour east of Portland."*

No evidence of any Idaho office, relocation, or satellite facility was found. The likely source of confusion is Quest Aircraft Company / Kodiak Aircraft, an unrelated aircraft manufacturer in Sandpoint, Idaho (acquired by Daher in 2019). Treat the Idaho attribution as a false premise; the rest of this report uses the verified Washington facts.

1. COMPANY OVERVIEW

1.1 Identity and legal structure

Field	Value	Source
Legal name	Sagotech Avionics, Inc.	HigherGov
HQ	317 W Steuben Street, Bingen, WA 98605	sagotech.com
Incorporated	Delaware	HigherGov
Founded (this entity)	October 24, 2019; federally registered Nov 14, 2019	HigherGov
UEI	T4UNAUXXM3	USAspending / HigherGov
CAGE code	8FG94	HigherGov
DUNS (legacy)	117302803	USAspending
Primary NAICS	334511 — Search, Detection, Navigation, Guidance, Aeronautical & Nautical System and Instrument Manufacturing	HigherGov
Secondary NAICS	334515, 336413, 488190, 541715	HigherGov
Small-business status	Small business; HUBZone certified Feb 12, 2021; historically-underutilized-business firm; manufacturer of goods; US-owned	HigherGov / USAspending business_types
Quality cert	ISO-9000 series	HigherGov
GSA Schedule	MAS 47QSWA23D00A5, PoP 9/28/2023 – 9/27/2028	HigherGov / USAspending

Note the confusing "founded" dates across databases: PitchBook lists 1978 (inheriting the predecessor's lineage), CB Insights/others cite 1998 for "miniature surveillance avionics" activity, the Gorge Technology Alliance says 2019 (the carve-out), and HigherGov's federal registration says Oct 24, 2019. All are defensible depending on whether you mean the technology lineage or the current legal entity.

1.2 Origin story and corporate history

The lineage runs through a predecessor, Sagetech Corporation (separate UEI N3GCB5FTDN79, DUNS 030467026), an engineering services firm in the Gorge that supported Boeing's Insitu — doing engine

dynamometer testing, electronics, microelectronics, software and firmware work.

Key milestones:

- ~2004–2006 — Sagatech Corporation wins early federal R&D; contracts including two DARPA contracts (HR0011-04-C-0119, ~\$396.9K, Sept 2004; HR0011-06-C-0008, ~\$397.0K, Oct 2005) and an Army AMCOM contract (W31P4Q-05-C-R174, ~\$399.1K, April 2005). Source: USAspending.gov.
- ~2009 — Boeing/Insitu needed a transponder for small drones that were invisible to radar; existing transponders were too big and heavy. Sagatech Corporation built what it calls the world's first micro transponder — initially a circuit card for the ScanEagle, later productized into a business-card-sized unit. (<https://www.auvsi.org/news/oem-leader-how-sagatech-avionics-builds-layers-of-aviation-safety-for-dual-use-bvlos-capabilities/>, <https://sagatech.com/about-us/who-we-are/>)
- 2012–2019 — XP-series transponders ship broadly; NASA, NOAA, Navy and Army purchase orders appear in USAspending under Sagatech Corporation (e.g., NND12AE16P Mode C transponders 2012; NND14AB72P "Mode transponder with mini ADS-B" 2014; N68335-14-C-0407 Navy SBIR Phase I Topic N142-102, ~\$79.8K, Sept 2014; DOC RA133R15SU0276 NOAA AirCore sampling system transponders 2015; 80NSSC17P1164 NASA "two MX-MP-03 Sagatech ADS-B transponders" 2017; N66001-18-P-0182 "engineering services in support of the ScanEagle UAV transponder system" 2018).
- November 2019 — Columbia River Partners (a Seattle/Portland/San Francisco private investment firm specializing in "corporate divestitures, recapitalizations, carve-outs and other complex situations in the lower middle market") executes a carve-out of the avionics division / acquisition of the transponder IP from Sagatech Corporation, forming Sagatech Avionics, Inc. Tom Furey, who had been COO of Sagatech Corporation for five years, leads the acquisition "with a few like-minded individuals" and becomes CEO. (<https://crgta.org/spotlight/sagatech-avionics-inc>, <https://www.columbiariverpartners.com/>)
- 2020–2021 — Company adds military IFF functionality and develops the MX12B Mode 5 platform; receives DoD AIMS certification for the Mode 5 micro IFF transponder; wins AFWERX contract; wins 2021 AUVSI XCELLENCE Award for Technology for the MX12B; demonstrates Mode 5 Level 2B Out at Bold Quest 2021.
- June 2021 — \$12M investment from Due West Partners.
- February 2023 — FAA certifies the MXS, described as the first FAA-certified small transponder with integrated ADS-B In.
- April 2026 — \$248.5M, 10-year Army IDIQ for the AN/APX-128.

1.3 Mission, vision, values

- Vision: "Safe skies through advancing technology."
- Mission: "To deliver low-SWaP certified avionics solutions that enable safe airspace integration and protect our warfighters."
- Core values (six): Integrity, Quality, Safety, Customer Focused, High Performance, Stewardship. (<https://sagatech.com/about-us/who-we-are/>)

Self-reported track-record metrics repeated across the site: 1.5 million flight hours, 500+ uncrewed aircraft platforms, 40+ countries, 10+ years experience. Company says its products are standard equipment on many Group 2 and Group 3 UAV programs and fly on Class/Group 2, 3, 4 and 5 aircraft.

1.4 Size

Employee counts vary by source and date — the company is small:

- PitchBook: 36 total employees (older snapshot)
- CB Insights / LinkedIn-derived: ~68 employees as of April 2026
- Another 2026 profile: ~69 employees across North America and Asia as of June 2026
- HigherGov: "<500" (SBA size bucket, not a headcount)

Open roles listed on the careers page at time of research: Quality Technician, Supply Chain Manager, Inventory and Logistics Specialist — consistent with a company scaling up manufacturing/production capacity, which matches the APX-128 full-rate production ramp.

Named non-executive staff surfaced in public content: Rudy Johnson (Director, Business Development), Ashley McKinney (Sr. Director of Marketing), Frank Wu (Sr. Embedded Software Engineer), Cole Hunter (IT Director), Megan Furey (LinkedIn, Sagetech Corporation).

1.5 Standards bodies and industry participation

Sagetech personnel sit on the AIMS Configuration Control Board (CCB) and the ASTM F38 UAS Committee, plus "dozens of others." The company is active in AUVSI, the Commercial Drone Alliance, Women and Drones (corporate member), NUAIR, the Drone Logistics Ecosystem, and was selected for NSIN Propel EDI. It publishes commentary on RTCA standards including DO-365B (DAA MOPS), DO-386 (ACAS Xu MOPS), and DO-398a (eVFR OSED).

2. FINANCIALS

Sagetech Avionics is privately held. There is no public revenue disclosure, no SEC filings (no Form D, 10-K, or S-1 surfaced), and no audited statements available. What follows is the hard public data plus third-party estimates.

2.1 Verified federal award data (USAspending.gov, UEI T4UNAUKXMZM3)

Total federal transaction amount to date: \$8,659,602.96 across 18 transactions (per the USAspending recipient endpoint). Obligations by fiscal year:

Fiscal Year	Obligated
FY2021	\$150,000.00
FY2023	\$44,071.20
FY2024	\$1,895,033.48
FY2025	\$139,962.00
FY2026	\$8,659,602.96

The FY2026 jump is the APX-128 production ramp. Note USAspending's recipient "total_transaction_amount" appears to reflect the current-year rollup; the year-by-year series is the more useful signal — federal revenue grew roughly 60x from FY2025 to FY2026.

Individual definitive contracts (prime, Sagetech Avionics Inc.):

Award ID	Amount	Agency	Period	Description
W15P7T26FA009	\$3,602,297.44	Dept. of the Army	2026-06-15 → 2027-06-14	Full rate production delivery order for AN/APX-128 transponders
W15P7T26FA007	\$2,160,892.48	Dept. of the Army	2026-04-30 → 2027-01-29	APX-128 transponders + crypto application KIV-79 services + cyber services
W15P7T26FA004	\$1,523,243.00	Dept. of the Army	2026-04-16 → 2027-04-16	AN/APX-128 with government-provided KIV-79 and engineering services
W15P7T26FA005	\$1,273,183.04	Dept. of the Army	2026-04-16 → 2027-04-16	AN/APX-128 transponders and KIV-79 services
697DCK24C00305	\$998,426.37	FAA	2024-09-27 → 2025-09-26	BAA Call 005 — non-system-specific tech to integrate UAS into the NAS
697DCK24C00369	\$865,457.11	FAA	2024-09-23 → 2025-09-26	BAA Call 005 — same program, second award
N6833525C0258	\$239,949.00	Dept. of the Navy	2025-07-07 → 2026-09-09	(SBIR-style; kick-off briefing materials & FWA cert)
FA864921P0083	\$150,000.00	Dept. of the Air Force	2020-12-17 → 2021-06-17	Low SWaP ACAS-Xu based Detect and Avoid (the AFWERX award)
N0042123P0269	\$44,071.20	Dept. of the Navy	2023-05-11 → 2023-07-06	P/N: 10-0000 (hardware buy)
FA491324P0026	\$31,150.00	Dept. of the Air Force	2024-09-27 → 2024-10-18	IFF transponders and antennas

Indefinite delivery vehicles (IDVs):

Award ID	Agency	Start	Notes
W15P7T26DA001	Dept. of the Army	2026-04-03	Full rate production contract for AN/APX-128 transponders — this is the \$248.5M ceiling, 10-year IDIQ announced in the press release as W15P7T-26-D-A001
47QSWA23D00A5	GSA Federal Acquisition Service	2023-09-28	Federal Supply Schedule (MAS) contract, through 9/27/2028

Predecessor entity (Sagetech Corporation, UEI N3GCB5FTDN79) federal history — useful for lineage but not current revenue: 12 awards totaling roughly \$1.39M, led by W31P4Q05CR174 (\$399,065, Army AMCOM, 2005), HR001106C0008 (\$397,038, DARPA, 2005), HR001104C0119 (\$396,860, DARPA via DCMA, 2004), N6833514C0407 (\$79,820, Navy SBIR Phase I, 2014), N6600118P0182 (\$40,000, Navy ScanEagle transponder engineering services, 2018), and a tail of small NASA/NOAA/Army/Navy hardware buys.

No federal grants were found for Sagetech Avionics (grant award-type query returned zero records). The BIRD Foundation money (see §3) is a binational grant, not a US federal assistance award, which is why it does not appear.

2.2 Third-party revenue and valuation estimates

- No valuation has ever been publicly disclosed. PitchBook and Preqin maintain profiles but gate valuation behind subscription; no post-money figure surfaced for either the 2019 acquisition or the 2021 Due West round.
- Revenue estimate aggregators (ZoomInfo, Growjo, LeadIQ) carry Sagetech profiles but their figures were not retrievable in this session (403-blocked). Treat any such number as a model-derived estimate, not a disclosure.
- Practical revenue framing: with ~68–69 employees, small-hardware ASPs, and \$8.7M of FY2026 federal prime obligations plus substantial non-federal revenue (international resellers in 17 countries, OEM sales to Textron, Insitu, UAV Factory, Primoco, High Eye, Kraus Hamdani, PAE ISR, A-TechSYN, etc.), total revenue is plausibly in the low-to-mid eight figures. This is an inference, not a sourced figure — flag it as such.
- Risk concentration note: one analyst summary surfaced in search flagged that "Army contract concentration is extreme, with APX-128 execution now driving 2026 revenue and credibility — procurement delays or field failures would freeze adoption across helicopters and FMS."

2.3 Ownership

Sagetech Avionics is majority-owned/controlled by Columbia River Partners (acquirer, Nov 2019), with Due West Partners holding a significant minority position following the June 2021 investment. Both firms hold board seats (see §4). Columbia River Partners reports firm-level stats of \$200M+ deployed capital, 7 platform companies, 15 transactions, 1,000+ portfolio employees (<https://www.columbiariverpartners.com/>).

3. FUNDING HISTORY

Sagetech is not a venture-backed startup in the conventional sense — it is a private-equity carve-out with one subsequent growth-equity injection. Total disclosed capital is modest.

3.1 Round-by-round

Date	Event	Amount	Investor(s)	Notes
Nov 2019	Acquisition / carve-out	Undisclosed	Columbia River Partners	CRP acquires the avionics/transponder IP and assets of Sagetech Corporation, creating Sagetech Avionics, Inc. Structured as a corporate divestiture/carve-out. CRP's stated specialty. Tom Furey led the buyer group.

Date	Event	Amount	Investor(s)	Notes
June 29, 2021	Growth investment	\$12,000,000	Due West Partners (lead)	Announced from White Salmon, WA. Terms not disclosed. Use of funds: accelerate product development, specifically UAS Detect and Avoid (DAA) capability and related technologies; expand engineering and go-to-market teams.
Dec 17, 2020	AFWERX / USAF contract (non-dilutive)	\$150,000	U.S. Air Force (AFWERX)	FA8649-21-P-0083, "Low SWaP ACAS-Xu based Detect and Avoid." Frequently miscounted by databases as a "funding round."
Nov 4, 2021	Grant / prize	Undisclosed	BIRD Foundation (US-Israel Binational Industrial R&D; Foundation), with U.S. Dept. of Homeland Security participation	Won jointly with Ciconia Ltd. (Israel) for a collision avoidance system. Tracxn records this as the company's most recent "round."
~2025	HSWERX award (non-dilutive)	Undisclosed	DHS (HSWERX)	Won jointly with MatrixSpace for advanced air detection supporting UAS operations; potential CBP applications.

3.2 Aggregator totals (they disagree)

Source	Total raised	Rounds	Investors
CB Insights	~\$13M	—	—
PitchBook	\$15M	—	—
Tracxn	\$12M	2	3

The discrepancies stem from whether the AFWERX/BIRD non-dilutive money is rolled into "funding." The only confirmed equity event with a public dollar figure is the \$12M Due West round.

3.3 Investor profiles

Due West Partners (DWP) — Seattle-based private investment firm focused on aerospace and defense. Tagline used in the Sagetech release: "Due West is Different"* — patient, long-term capital for established companies pursuing sustained growth. Managing partners quoted: Nick Wellmon ("Proven technologies and customer credibility plus significant market potential...") and Robert Dickinson ("Situational awareness products are required technology for safe integration...").

Columbia River Partners (CRP) — "A private investment firm investing in the 'Picks and Shovels' of technology." Offices in Portland, San Francisco, Seattle; HQ 400 Union St, Suite 400, Seattle, WA 98101. Focus: corporate divestitures, recapitalizations, carve-outs, complex lower-middle-market situations. Other portfolio companies named publicly include TRP Laboratories, TiniFiber, CCS, and Business Solution Partners. CRP publicly promoted the Sagetech APX-128 win on its own news page (<https://www.columbiariverpartners.com/news/sagetech-avionics-awarded-2485m-army-contract-for-apx-128-advanced-iff-production>).

Others listed by databases: Preqin associates EMV Capital with Sagetech; U.S. Department of Homeland Security appears as an "investor" in Tracxn/CB Insights because of the BIRD/HSWERX grants. Green Span Ventures (Prabhu Goel) is represented on the board, implying an investment position.

4. LEADERSHIP AND PEOPLE

Source for this entire section unless noted: <https://sagetech.com/about-us/leadership-team/>

4.1 Executive team

Tom Furey — Chief Executive Officer and Board Member Has led Sagetech since its late-2019 launch, following the carve-out of the avionics division of Sagetech Corporation, where he had served as COO for

five years. He led the acquisition of the critical assets from Sagetech Corporation together with a small group of like-minded individuals. Roughly 20 years of management experience across aerospace, automotive, chemical, and printing industries. Military background: U.S. Navy Naval Flight Officer flying the A-6 Intruder, including service in the Persian Gulf during the first Gulf War. Education: B.S. Mathematics, U.S. Naval Academy; master's degrees in engineering from Cal State and MIT; MBA from MIT (Sloan). Personal: enjoys outdoor activities with his dog. He is the company's principal public voice — appearing at Xponential 2024, on the *Dawn of Drones* podcast, on *Autonomy Global Defense Tech* (June 2026), and in multiple video features.

Matthew (Matt) Hamilton — Chief Technology Officer Oversees technology development for the company's uncrewed aircraft avionics products. With the company since 2015. Previously built his expertise at ACSS (Aviation Communication & Surveillance Systems), the L3Harris/Thales joint venture — a direct pedigree in TCAS/ACAS and surveillance systems. B.S. Electrical Engineering, Embry-Riddle Aeronautical University. Personal: family time and flying his own aircraft. Quoted on the APX-128 award crediting "the deep technical skills of Sagetech's engineering team, our partnership with the Department of War, and the robust collaboration between Sagetech, the Army, and our partners at R Cubed Engineering."

Dan Kinsfogel — Chief Financial Officer Manages accounting, planning, and capital functions. 25+ years of financial leadership. Prior roles: CFO at Dakine; finance positions at SC Johnson, Nike, and Qualcomm. Education: business degree from the University of Arizona; MBA from ASU's W. P. Carey School of Business. Personal: coaches youth soccer.

David Day — Chief Compliance Officer Responsible for ensuring products meet government and industry compliance standards — a critical role given the AIMS/TSO/FCC certification-heavy product line. 30+ years spanning L3Harris-ACSS, Honeywell, and Motorola, covering surveillance engineering and FAA certification. B.S. Electrical Engineering, Arizona State University. Personal: outdoor recreation in the Columbia River Gorge.

Note the pattern: Hamilton and Day both come from ACSS/L3Harris-Thales, the incumbent TCAS/ACAS house — Sagetech essentially recruited surveillance-certification DNA and miniaturized it.

4.2 Board of Directors

Pooneet Goel — Columbia River Partners (Chief Strategy Officer). 20+ years of startup and strategic expertise; earlier advised Fortune 500 executives at Booz & Company. B.S. Computer Science, Stanford; MBA, Kellogg. Personal: home coffee roasting.

Nathan Chandrasekaran — Columbia River Partners (Chief Financial Officer). 15+ years across finance, strategy, and technology, plus private equity and multiple board seats. Background includes Merrill Lynch, inCode Wireless, and Deloitte Consulting. B.A. Neuroscience, Bowdoin; MBA, Kellogg. Stated philosophy: "having impact on Day 1."

Prabhu Goel — Green Span Ventures (Board Member and Advisor). Active private-company investor serving on multiple boards, including renewable energy ventures in Europe and India. Public-company board experience at Synopsys and Cadence Design Systems. Previously led Gateway Design Automation before its 1989 acquisition by Cadence — a notable EDA-industry figure (the "Goel" of the PODEM algorithm lineage).

Nick Wellmon — Due West Partners (Board Member). Founder and manager of Due West Partners, an aerospace/defense-focused investment firm. Previously Director of Finance at Exotic Metals, where he led the company's 2019 sale to Parker Hannifin. Earlier: corporate development and investment banking. Finance degree, Wake Forest.

4.3 Advisors

Douglas Kirkpatrick — Currently CEO of Eridan (5G RF technology). Formerly a venture capitalist and DARPA Chief Scientist, overseeing programs including rapid DNA synthesis. Earlier: R&D; leadership at Fusion Lighting; scientific and VP roles at SAIC. B.S. Physics, College of William & Mary; Ph.D. Physics, MIT (1988). Fellow of the American Physical Society, 30+ journal articles and 60+ patents to his name (his own, not

Sagetechn's).

Prabhu Goel — also listed as Advisor.

4.4 External figures associated with Sagetechn

- Rudy Johnson, Director of Business Development — quoted in the MatrixSpace release.
- Dawn Zoldi, CEO of P3 Tech Consulting, 25-year Air Force veteran — provided the headline industry endorsement of the MX12B: "The ability to make the call on hostile intent under the Rules of Engagement is critical... This is a game-changer."*
- Lori DeMatteis, Chief Revenue Officer, MatrixSpace — partner-side quote.

5. PRODUCTS

Sagetechn organizes its catalog into three go-to-market verticals (Government-Defense, Government-Civil, Commercial) over three product families: transponders, ACAS/DAA solutions, and systems integration services. There is also an interrogator line (MXR) and a newer OEM/military line (ACX-3000M, AN/APX-128).

5.1 MX Series — current-generation transponders

The MX Series is marketed as "transponders and interrogators [delivering] ADS-B In and Out, IFF Mode 5, detect and avoid, omni-directional surveillance, and other advanced functionality." MXS and MX12B share a common hardware platform, which is the line's key commercial argument: OEMs can swap between military and civil configurations without additional integration, software changes, or aircraft changes.

MX12B — Mode 5 IFF Micro Transponder (defense)

(<https://sagetechn.com/transponders/mx12b/>)

Positioning: "the world's first AIMS-certified micro Mode 5 IFF transponder"; the smallest full-function Mode 5 IFF transponder available; the only micro Mode 5 IFF transponder fully compliant with the DoD AIMS Mk XII specification; the first Mode 5 Level 2B Out transponder certified.

Consolidates into one enclosure: ADS-B In/Out, pressure altitude encoder, antenna diversity, Ethernet — eliminating discrete components at integration.

Claimed advantage vs. traditional Mode 5 transponders: 93% smaller, 6x lighter, 5x less expensive.

Spec	Value
Weight	190 g (with KIV)
Dimensions	3.4" H × 2.5" W × 1.0" D
Operating temp	-40 ° C to +71 ° C
Environmental	MIL-STD-810G, RTCA DO-160G, MIL-STD-461F
Military modes	Modes 1, 2, 3, 5 (Levels 1 & 2)
Civil mode	Mode S per RTCA/DO-181
ADS-B In	1090 MHz (ADS-B, ADS-R, TIS-B); growth path to Mode 5 Level 2 broadcast-in
ADS-B Out	RTCA/DO-260B compliant
Interfaces	Ethernet 10BaseT/100BaseT, RS-422, RS-232
Antenna	Configurable diversity or single-antenna
Transmit power	Per AIMS 17-1000
Applications	ISR; Aircraft Groups 2, 3, 4, 5

Supporting documents published: datasheet, FAQ sheet, FCC report, "Why Choose the MX12B?" value-proposition doc, MX12B accessories datasheet. Received a product upgrade announced July 11, 2023. Won the 2021 AUVSI XCELLENCE Award for Technology. Demonstrated Mode 5 Level 2B Out at Bold Quest 2021.

MXS — Mode S Micro Transponder with ADS-B In/Out (civil/commercial)

(<https://sagetechn.com/transponders/mxs-transponder/>)

Positioning: FAA-certified (TSO Authorized), described as the first FAA-certified small transponder with an integrated ADS-B In receiver and "the world's first FAA-certified micro ADS-B receiver." It is the cornerstone of

Sagetech's DAA architecture — its integrated ADS-B In is the critical cooperative-traffic sensor feeding the ACAS X logic.

Spec	Value
Dimensions	3.4" × 2.5" × 1"
Weight	190 g
Modes	A, C, S
ADS-B	1090 MHz In and Out
Antenna	Full diversity
Altitude encoding	to 100,000 ft
I/O	RS-232, RS-422, Ethernet
Certification	FAA TSO
Feature branding	"SafeMission+ Technology"

Key marketing claims: unprecedented micro-SWaP; full output power, no deviations; no additional integration needed; integrates with major autopilots. Documentation includes the *MXS System Description and Installation Manual*, MXS accessories datasheet, and "MXS Micro Transponder Mode" sheet.

MXR — Micro Interrogator

Referenced in partnership coverage rather than on a dedicated product page at time of research. The MXR is an active-surveillance micro interrogator used alongside the transponders. In the Hover, Inc. partnership, the MXR + MXS combined with Hover's embedded computing lets a UAV actively interrogate nearby aircraft, validate ADS-B In tracks, and resolve threats — described as compliant with ACAS-Xu and ACAS-Xr and aimed at UAVs flying above 400 ft BVLOS in the National Airspace System. Sagetech has publicly argued that "transponders and interrogators will serve as core technology enablers for future DAA systems required on autonomous aircraft as Urban Air Mobility integrates into manned airspace."

ACX-3000M

Named in the APX-128 press release as one of the two existing transponder lines the AN/APX-128 extends (alongside MX12B). A military transponder line; no dedicated public product page found.

AN/APX-128 — Advanced IFF Transponder (newest)

The subject of the April 2026 Army award. Fully MOSA (Modular Open Systems Architecture) compliant, includes ADS-B In aligned with anticipated ALERT Act requirements, and is upgradeable to ACAS Xr collision avoidance. Extends the MX12B and ACX-3000M lines. Low SWaP-C replacement for legacy Army IFF hardware. Rapidly developed under a separate SBIR contract with partner R Cubed Engineering. Fields with government-provided KIV-79 crypto appliqué (per the USAspending delivery-order descriptions). Target platforms: Army helicopters, other DoW assets, and Foreign Military Sales.

5.2 XP Series — legacy transponders

The XP family has amassed over 1.5 million flight hours in combat, high-altitude, and other demanding environments, and is "standard equipment in most Group 2 and Group 3 UAV programs." All XP-family units share FCC ID: YT5XP-TR. All are "designed to FAA TSO standards" and "can be seen by TCAS."

XPS — Mode S Legacy Transponder with ADS-B Out

(<https://sagetech.com/transponders/xps-with-ads-b-out/>)

Spec	Value
Modes	3/A, C, S
ADS-B	Out (extended squitter) to RTCA DO-260B
Size	3.5" × 1.8" × 0.7"
Weight	3.53 oz (100 g)
Power	< 1 W standby; 8 W typical; 14 W max
Altitude encoder	integrated, to 85,000 ft per SAE AS392c / SAE AS8003
Performance	RTCA DO-181E
Serial comms	Proprietary protocol RS-232 up to 460.8 kbps; Accord NexNav; NMEA RS-232, adjustable data rate

Spec	Value
FCC	YT5XP-TR

Applications listed: delivery, gliders, inspection, high-altitude aircraft, commercial agriculture, fire agencies, police + border patrol, risk management, first responders. Notable deployment: Primoco UAV Model One 150 endurance record used the XPS; also used in a documented DJI A2 + BeagleBone ATC-visibility integration (XPS-TR).

XPG — Mode S Legacy Transponder, ADS-B Out + integrated GPS

(<https://sagotech.com/transponders/xpg-with-ads-b-out-and-gps/>)

All-in-one with built-in GPS for 24/7 location awareness; Modes A, C, S plus ADS-B Out. Marketed as "the only transponder available in its class that can be seen by TCAS." Operation at full power over full temperature range. *(Caution: the live XPG page's spec block appears to reuse MX12B/Mode 5 copy — listing "just over 1 lb with KIV," 3.4" × 2.5" × 1.0" , -40 ° C to +71 ° C, MIL-STD-810G/DO-160G/MIL-STD-461F, and "Military Modes 1, 2, 3, 5 (Levels 1&2)." This looks like a CMS content error on Sagotech's site; verify against the datasheet before relying on XPG numbers.)*

XPC — Mode C Legacy Transponder (two variants)

XPC commercial/civil *(<https://sagotech.com/transponders/xpc/>)*: "ultra-reliable performance at a cost-effective price" for basic situational awareness on UAS.

Spec	Value
Modes	3/A, C
Size	3.5" × 1.8" × 0.7"
Weight	3.5 oz (100 g)
Power	< 1 W standby; 4 W typical; 11 W max
Altitude encoder	to 85,000 ft per SAE AS392c / SAE AS8003
Performance	RTCA DO-144A
FCC	YT5XP-TR

XPC Mode C Defense *(<https://sagotech.com/transponders/xpc-mode-c-transponder/>)*: same physical/electrical specs, but certified to DoD AIMS Mark XA for guaranteed NATO interoperability. "Used worldwide, for no-fail missions." Tailored for Group 2 and Group 3 programs; ISR applications. Customer named explicitly: Textron. An "XPC Toolkit" is offered alongside the datasheet.

5.3 ACAS Solutions / Detect and Avoid

(<https://sagotech.com/acas-solutions/>)

Architecture: an onboard ACAS DAA computer paired with SafeMission+ Software, fed by the MX-series transponder's integrated ADS-B In (cooperative traffic) and plug-and-play third-party non-cooperative sensors (radar, EO, acoustic). Built on ACAS X, which "uses advanced algorithmic techniques and probabilistic models to provide collision avoidance for all classes of piloted and unpiloted aircraft." Sagotech integrated ACAS X algorithms onto its own DAA computer.

Three variants, segmented by aircraft class and airspace:

Variant	Aircraft class	Envelope	Description
ACAS Xu	Fixed wing	>55 lbs, >400 ft AGL	Certifiable system with an integrated cooperative traffic interrogator; plug-and-play compatible with common non-cooperative sensors; for BVLOS across airspace types
ACAS sXu	Small uncrewed	<55 lbs, <400 ft AGL	Compact, low-SWaP; integrates with autopilots and non-cooperative sensors; dual-band ADS-B receiver for cooperative traffic; supports fixed- and rotary-wing sUAS
ACAS Xr	Rotary wing	>55 lbs, >400 ft AGL	Targets UAM/AAM and crewed rotorcraft; supports in-corridor and non-corridor routes; certified low-SWaP hardware

Marketing collateral includes a "Why Choose ACAS X" flyer. Sagatech provided flight test data to FAA researchers evaluating ACAS Xr logic for crewed and uncrewed rotorcraft. Sagatech has published explainers on the ACAS Xu MOPS (RTCA DO-386) and on DO-365B.

5.4 Systems Integration Services

(<https://sagatech.com/integration-services/>)

Three named service offerings aimed at OEMs, defense departments, civil governments, and commercial operators:

1. Aircraft Integration & Testing — Sagatech's own team performs the integration ("Save Time").
2. Product Success Training — hands-on instruction on Sagatech hardware and software ("Get There Faster").
3. BVLOS Advisory Services — access to Sagatech's partner/vendor network to support BVLOS goals ("Unlock Expertise").

A "Systems Integration Services" datasheet is published.

5.5 Accessories and support products

MX12B accessories, MXS accessories, XPC Toolkit, antennas (the FA491324P0026 Air Force award covers "IFF transponders and antennas"), and published Service Bulletins (<https://sagatech.com/service-bulletins/>).

5.6 Integrations, platforms, and named deployments

Autopilot/flight-control and avionics integrations: ArduPilot (via Kraus & Hamdani Aerospace), Embention, UAV Navigation, Elsieht (connectivity), Iris Automation (non-cooperative detection, integrated into UAV Navigation's flight control system), MatrixSpace (AI radar/non-cooperative sensing), Becker Avionics (European market), Hover, Inc. (interrogator-based DAA).

Aircraft/platform deployments named publicly: Insitu ScanEagle (origin platform), UAV Factory Penguin C (and the broader Penguin family), PAE ISR Resolute Eagle, Primoco UAV Model One 150, High Eye Airboxer, A-TechSYN CGT50 VTOL, American Aerospace AiRanger, Joby Aviation eVTOL, and NASA research aircraft.

5.7 Certifications and regulatory approvals summary

- FAA TSO Authorization — MXS
- DoD AIMS certification — Mode 5 micro IFF (MX12B), Mk XII B compliant, Mode 5 Level 2B Out
- DoD AIMS Mark XA — XPC Defense (NATO interoperability)
- FCC ID YT5XP-TR — XP family
- ANATEL certification (Brazil) — MXS, obtained via partner Advanced Technologies, Security & Defense
- Environmental/EMI: RTCA DO-160G, MIL-STD-810G, MIL-STD-461F
- Performance standards referenced: RTCA DO-181E, DO-181, DO-144A, DO-260B; SAE AS392c, AS8003; AIMS 17-1000

5.8 International reseller network

(<https://sagatech.com/international-resellers/>) — 17 countries:

Australia (The AMS Group; Capital Aircraft Services/CAPAIR) · Brazil (RWK; Advanced Technologies, Security & Defense) · Egypt (Makville Business Consulting Agency) · Hungary (MILTECH) · India (Apollo Micro Systems Limited; Aero SoluStax) · Ireland (A-TechSYN) · Israel (FMS Aerospace) · Italy (Eurolink Systems) · Japan (Cornes Technologies) · Mexico (Descef Sistemas) · Romania (MarcTel) · Singapore (Oriens Group) · South Africa (SSASS Holdings) · South Korea (WI Engineering Co., Ltd.; Pan West; AccessWe) · Spain (Aerogenix; Embention; UAV Navigation) · Turkey (BSM Innovative Technologies)

6. PATENTS

6.1 Methodological caveat — READ THIS

I was unable to retrieve authoritative patent records in this session. Every patent database attempted was blocked at the network layer:

- patents.google.com — HTTP 503 via WebFetch; anti-bot block via direct request
- patents.justia.com — Cloudflare challenge
- api.patentsview.org / search.patentsview.org — 301/empty (legacy API retired; current API requires a key)
- assignment-api.uspto.gov — empty response
- freepatentsonline.com — connection reset
- patentguru.com — HTTP 468
- worldwide.espacenet.com — HTTP 403
- uspto.report — HTTP 403
- fccid.io — HTTP 403
- sbir.gov public API — {"message":"Forbidden"}
- DuckDuckGo HTML/Lite fallbacks — bot challenge

Web search budget for the session was also exhausted (200/200 calls) before patent-specific queries could be run against an indexable source. No patent numbers or titles should be attributed to Sagetech on the basis of this report. What follows is what is defensible from non-patent sources.

6.2 What is known about Sagetech's IP position

The 2019 transaction was explicitly an IP acquisition. The Gorge Technology Alliance profile states Sagetech Avionics "was founded in 2019 through the acquisition of intellectual property from Sagetech Corporation." This means:

- Any patents filed before ~2019 would be assigned to Sagetech Corporation (UEI N3GCB5FTDN79), not Sagetech Avionics, Inc.
- Ownership would have transferred via the carve-out, which would show as an assignment record in the USPTO Patent Assignment Database — the right place to look when a search tool is available.
- Searching only "Sagetech Avionics" as assignee will under-count the portfolio.

Search strings to use when a patent database is reachable:

- Assignee: "Sagetech Corporation", "Sagetech Avionics, Inc.", "Sagetech Avionics Inc", "Sagetech Holdings"
- Inventor: Furey, Hamilton (Matthew), Day (David) — note that Hamilton and Day both came from ACSS/L3Harris and Honeywell, so their earlier patents would be assigned to those firms, not Sagetech
- CPC classes: G01S 13/78 (secondary radar / IFF transponder systems), G01S 13/76, G08G 5/04 (aircraft collision avoidance), G08G 5/0021, H04B 7/185

Likely subject matter, inferred from product claims (these are technology areas, NOT confirmed patents):

- Miniaturization of a full-function Mode A/C/S transponder into a ~100 g / 3.5" × 1.8" × 0.7" package — the "world's first micro transponder" claim
- Integration of Mode 5 IFF (Levels 1 & 2), Mode S, ADS-B In and ADS-B Out, pressure altitude encoder, and antenna diversity into a single 190 g enclosure at full AIMS 17-1000 transmit power "with no deviations"
- The "SafeMission+" software/technology branding, applied to both the MXS transponder and the ACAS DAA computer — a named, likely-trademarked technology block
- Combining a micro interrogator (MXR) with ADS-B In track validation for cooperative DAA, per the Hover, Inc. architecture
- MOSA-compliant, field-upgradeable IFF architecture (APX-128) supporting later insertion of ACAS Xr

Countervailing evidence that the portfolio may be modest: Sagetech's public positioning leans overwhelmingly on certifications (AIMS Mk XII B, AIMS Mk XA, FAA TSO, FCC, ANATEL) and flight-hour track record (1.5M hours) rather than on patents. Its marketing materials, leadership bios, and press releases contain zero patent claims — notable, because the company does loudly claim "world's first" repeatedly. The one place patents are mentioned on the whole site is advisor Douglas Kirkpatrick's bio ("60+ patents"), and those are his personal patents from Fusion Lighting/SAIC/DARPA work, not Sagetech assets. In a

certification-gated market like IFF/ADS-B, the AIMS and TSO certificates are the real moat — they take years and millions to obtain and are far harder for a competitor to replicate than a patent is to design around.

Trademarks are a more likely IP register to yield hits: "Sagotech," "SafeMission+," "MX12B," "MXS" are all candidate marks worth checking in USPTO TESS.

Recommendation for follow-up: run USPTO Patent Public Search (ppubs.uspto.gov) with AN/"Sagotech" and AANM/"Sagotech", plus the USPTO Patent Assignment Search for assignments from Sagotech Corporation to Sagotech Avionics dated late 2019. Also check whether R Cubed Engineering holds co-invented APX-128 IP from the SBIR — SBIR data rights (not patents) are often the operative IP mechanism in Phase III sole-source awards like W15P7T-26-D-A001.

7. LATEST NEWS (2024 – SEPTEMBER 2026)

Sourced from Sagotech's own post sitemap (dated) plus press coverage. Listed newest-first.

2026

September 22, 2026 — OpEd: "The Midair Collision Warning Systems Miss, and How to Close the Gap" (Blog) * (<https://sagotech.com/news-and-events/oped-the-midair-collision-warning-systems-miss-and-how-to-close-the-gap/>)* Prompted by an August 2026 midair collision between two firefighting helicopters in Greece with loss of life. Argues that current collision avoidance tech "does little to protect aircraft flying in close coordination at low altitude." Key data cited: TCAS II and ACAS X only work well when both aircraft are cooperative (working transponder or ADS-B); non-cooperative detection via radar/cameras/acoustics is "much less reliable and only works at short range." Statistics presented: 36 midair collisions since 2020, of which only 11 (just over 30%) had both aircraft broadcasting position; 18 of the 36 occurred in U.S. airspace, and only 7 of those 18 (39%) involved two broadcasting aircraft — against an FAA baseline where >80% of U.S.-registered aircraft carry the equipment and 85–95% of crewed aircraft airborne at any moment are using it. Conclusion: the more aircraft broadcast, the fewer midairs; the gap is aircraft flying in close coordination at low altitude, and closing it "requires a new way for aircraft to talk to each other in real time." This is Sagotech positioning for a low-altitude, close-formation cooperative link — read it as a product-roadmap signal.

June 29, 2026 — CEO discusses ACAS on the *Autonomy Global Defense Tech* podcast (Blog) Tom Furey on ACAS and airspace safety. *(<https://sagotech.com/news-and-events/sagotech-avionics-ceo-discusses-acas-on-autonomy-global-defense-tech-episode/>)*

June 15, 2026 — Full-rate production delivery order, \$3,602,297.44 (USAspending W15P7T26FA009) Largest single APX-128 delivery order to date, PoP through 2027-06-14. Also the "most recent award date" shown on HigherGov.

April 30, 2026 — APX-128 delivery order, \$2,160,892.48 (W15P7T26FA007) — transponders + KIV-79 crypto appliqué services + cyber services.

April 23, 2026 — Sagotech Awarded \$248.5M Army Contract for APX-128 Advanced IFF Production (Press Release, PRNewswire/PRWeb, datelined Bingen, WA) *(<https://sagotech.com/news-and-events/sagotech-avionics-awarded-248-5m-army-contract-for-apx-128-advanced-iff-production/> · <https://www.prweb.com/releases/sagotech-avionics-awarded-248-5m-army-contract-for-apx-128-advanced-iff-production-302751352.html> · <https://www.streetinsider.com/Press+Releases/...> · <https://www.defenseadvancement.com/news/sagotech-avionics-secures-contract-for-advanced-iff-transponder-production/> · <https://www.columbiariverpartners.com/news/sagotech-avionics-awarded-2485m-army-contract-for-apx-128-advanced-iff-production>)*

The single most consequential event in company history. Details:

- SBIR Phase 3 contract W15P7T-26-D-A001, U.S. Army
- 10-year, \$248,500,000 IDIQ; base vehicle recorded in USAspending as W15P7T26DA001, start 2026-04-03
- Scope: integration and deployment of the AN/APX-128 across Army helicopters, other Department of War assets, and Foreign Military Sales

- Technical: MOSA compliant (improving lifecycle and configuration management vs. legacy); ADS-B In aligned with anticipated ALERT Act requirements now before Congress; upgradeable to ACAS Xr
- Lineage: extension of the MX12B and ACX-3000M lines; rapidly developed under a separate SBIR with partner R Cubed Engineering
- Furey: *"The APX-128 is a MOSA-compliant, upgradeable solution which will enhance airspace safety in training missions and protect our warfighters in combat. With a flexible architecture and modular design, this platform can be upgraded and customized for legacy and future aircraft platforms alike."*
- Hamilton: *"The rapid development of the APX-128 was a testament to the deep technical skills of Sagatech's engineering team, our partnership with the Department of War, and the robust collaboration between Sagatech, the Army, and our partners at R Cubed Engineering... The Army's need to quickly replace an obsolete IFF in their aircraft..."*

April 16, 2026 — Two further APX-128 delivery orders, \$1,523,243.00 (W15P7T26FA004) and \$1,273,183.04 (W15P7T26FA005), both PoP through 2027-04-16.

April 2026 — FAA DAA data-collection flights, conducted under the FAA's Broad Agency Announcement program in collaboration with Virginia Tech's Mid-Atlantic Aviation Partnership, with integration partners Sagatech, Collins Aerospace, and uAvionix. This is the flying-out of the two FY2024 FAA BAA Call 005 contracts (697DCK24C00305, \$998,426.37; 697DCK24C00369, \$865,457.11) for "development of non-system-specific technology to integrate UAS into the NAS."

March 13, 2026 — "Safety on the Edge of the New Airspace: Layered Collision Avoidance for BVLOS Drone Operations" (Blog) [\(https://sagatech.com/news-and-events/safety-on-the-edge-of-the-new-airspace-layered-collision-avoidance-for-bvlos-drone-operations/\)](https://sagatech.com/news-and-events/safety-on-the-edge-of-the-new-airspace-layered-collision-avoidance-for-bvlos-drone-operations/)*

2025

July 7, 2025 — Navy contract N6833525C0258, \$239,949.00, PoP through 2026-09-09 (NAVAIR/NAWCAD Patuxent River contracting office).

May 28, 2025 (announced May 20, 2025) — Sagatech Avionics and MatrixSpace Collaborate to Enable Advanced Detect and Avoid Solutions [\(https://sagatech.com/news-and-events/sagatech-avionics-and-matrix-space-collaborate-to-enable-advanced-detect-and-avoid-solutions/\)](https://sagatech.com/news-and-events/sagatech-avionics-and-matrix-space-collaborate-to-enable-advanced-detect-and-avoid-solutions/) * Subhead: "Expanding the compatibility of Sagatech's ACAS X with market-leading noncooperative sensors." Sagatech selected AI-sensing innovator MatrixSpace to support its advanced airspace solutions. The two had already partnered to win an HSWERX award from DHS for advanced air detection supporting UAS operations, with potential applications across DHS Customs and Border Protection. The DHS project sought to identify sUAS with onboard sensors meeting FAA guidelines for safe BVLOS.

- Rudy Johnson, Director of Business Development, Sagatech: *"Our mission is to deliver low-SWaP certified avionics solutions that enable safe, scalable airspace integration. To achieve this requires a sophisticated level of airspace awareness informed by high-performance sensors. MatrixSpace is meeting the challenge, setting new standards in precision detection, making them key to meeting our customers' demanding requirements."*
- Lori DeMatteis, CRO, MatrixSpace: *"As autonomous aviation becomes ubiquitous, the need to integrate it safely into the National Airspace is key. Working with Sagatech Avionics has demonstrated their commitment not only to high customer standards but also reaching for the most advanced, innovative solutions possible to achieve this."*

May 21, 2025 — CAPAIR named Australian reseller (per press coverage; the site's own CAPAIR post is dated July 15, 2024 — the two dates likely reflect an initial announcement and a re-promotion). CAPAIR brings Sagatech's DAA software and IFF transponders to the Australian market.

2024

September 18, 2024 — "How Sagatech Avionics Is Making the Skies Safer: An Inside Look with CEO Tom Furey" (Podcast/video feature)

September 9, 2024 — "Discover Sagatech Avionics' Groundbreaking Collision Avoidance Technology: New Video Explains It All"

September 27, 2024 — Air Force purchase FA491324P0026, \$31,150.00, IFF transponders and antennas.

September 23 & 27, 2024 — Two FAA BAA Call 005 contracts totaling \$1,863,883.48 (697DCK24C00369 and 697DCK24C00305) for non-system-specific UAS-into-NAS integration technology. These two awards are essentially all of FY2024 federal revenue.

July 17, 2024 — Aero SoluStax named official reseller to the India avionics market

July 15, 2024 — Capital Aircraft Services (CAPAIR) named reseller for Australia

May 23, 2024 — American Aerospace AiRanger, equipped with Sagetech technology, receives groundbreaking FAA approval for BVLOS flight *(<https://sagetech.com/news-and-events/american-aerospace-airranger-equipped-with-sagetech-avionics-technology-receives-groundbreaking-faa-approval-for-bvlos-flight/>)* — a significant regulatory proof point for the DAA stack.

May 16, 2024 — "Transforming Airspace Safety: Insights from Sagetech Avionics CEO Tom Furey at Xponential 2024" *(<https://insideunmannedsystems.com/transforming-airspace-safety-insights-from-sagetech-ceo-tom-furey-at-xponential-2024/>)*

April 2, 2024 — Joby and Sagetech Collaborate to Demonstrate Onboard-Piloted Detect and Avoid Capabilities *(<https://sagetech.com/news-and-events/joby-and-sagetech-collaborate-to-demonstrate-onboard-piloted-detect-and-avoid-capabilities/>)* Partnership began in 2022, studying onboard-piloted DAA supporting Joby's proposed enhanced Visual Flight Rules (eVFR) operations — described in RTCA DO-398a OSED as an extension to special VFR. eVFR would provide earlier warnings of impending close encounters plus maneuver guidance even when the pilot cannot see the neighboring aircraft; later implementations would lower visibility and cloud-clearance requirements. The core enabler is the FAA's ACAS Xr; Sagetech's ACAS Xr-based DAA system builds on its TSO-authorized MXS transponder. Joby continues flight and lab evaluation of the prototype. Goal: certify a DAA solution meeting Part 135 eVTOL needs and supporting any aircraft benefiting from eVFR. *(Note: this post carries an April 2, 2024 published date but a May 28, 2025 sitemap lastmod — it was likely re-surfaced/updated in 2025.)*

April 2, 2024 — "We Need to Protect Our Troops Against Drones — Now" (AUVSI opinion piece)

March 2024 — Event calendar: SOF Week, AUVSI Xponential, Industry Day at Fort Novosel (AL), AUVSI Pathfinder Huntsville (AL).

February 23, 2024 — Successful BVLOS flight of Primoco UAV with Sagetech's Mode S micro transponder on board

Slightly earlier, for continuity (2023)

- Nov 2, 2023 — "The MX12B Transponder Challenge: Enhancing Military Operations and Civilian Safety"
- Nov 1, 2023 — "Enhancing Airborne Safety with Sagetech's ACAS X Solutions"; "Get Your Program Off the Ground with Sagetech's MXS Transponder"
- Sept 28, 2023 — GSA MAS schedule 47QSWA23D00A5 awarded
- July 11, 2023 — New upgrade announced to the Mode 5 MX12B IFF transponder
- May 25, 2023 — Sagetech + Elsie integration "takes the industry one step closer to routine BVLOS flight"
- April 11, 2023 — Partner Advanced Technologies, Security & Defense obtains ANATEL certification for the MXS (Brazil)
- March 29, 2023 — Sagetech + A-TechSYN partner to offer the MX product family on the CGT50 VTOL UAV
- Feb 24, 2023 — FAA certifies new Sagetech Mode S micro transponder with integrated ADS-B In/Out (the MXS milestone)
- Jan 17, 2023 — "Sagetech Avionics Takes Lead on a Maturing Detect and Avoid Ecosystem"

8. SYNTHESIS AND ANALYTICAL NOTES

What Sagetech actually is: a ~68-person, PE-backed, HUBZone-certified small manufacturer in rural Washington that occupies a narrow but strategically valuable niche — the smallest certified cooperative-surveillance radios in aviation. Its defensibility is regulatory, not patent-based: AIMS Mk XIIIB,

AIMS Mk XA, FAA TSO, and FCC certifications on a sub-200-gram package that competitors have not matched.

The inflection point is April 2026. Before the APX-128 award, Sagatech's federal prime business was under \$2M/year and lumpy — a \$150K AFWERX award, some \$30–45K hardware buys, and two FAA research contracts. The Army IDIQ changes the company's category: a \$248.5M ceiling over ten years, with \$8.56M already obligated in four delivery orders within three months of award. The sitemap shows the company hiring a Quality Technician, Supply Chain Manager, and Inventory & Logistics Specialist — the signature of a design shop becoming a production house.

Strategic thesis (the "layered" story): Sagatech's three published blog/op-ed pieces in 2026 all argue the same line — cooperative broadcast (transponder/ADS-B) is the reliable layer, non-cooperative sensing is the unreliable layer, and the gap is low-altitude close-coordination flight. The commercial expression of that thesis is: sell the certified transponder (MXS/MX12B/APX-128), make it the cooperative sensor for the ACAS X DAA computer, and plug in third-party non-cooperative sensors (MatrixSpace, Iris Automation) rather than building them. That is a deliberately asset-light, integration-centric architecture.

Dual-use hedge: the MXS/MX12B shared hardware platform lets an OEM flip between civil and military configurations with no aircraft change. That single design decision is why Sagatech can simultaneously serve Joby's Part 135 eVTOL certification path and the U.S. Army's helicopter IFF refresh from one production line.

Key risks visible in the public record:

1. Customer concentration — FY2026 federal revenue is essentially one program (APX-128) with one customer (Army PEO/PM at Aberdeen, contracting office W15P7T).
2. Regulatory dependency — ACAS Xu/Xr/sXu revenue is gated on FAA MOPS maturation (DO-365B, DO-386) and on BVLOS rulemaking; the ALERT Act, cited as justifying APX-128's ADS-B In, was still "currently in Congress" as of April 2026.
3. Supply chain / crypto — APX-128 fielding depends on government-furnished KIV-79 crypto appliqué (explicit in three of four delivery orders), a dependency Sagatech does not control.
4. Scale mismatch — executing a \$248.5M ten-year production program with ~69 people is a real operational stretch; the hiring pattern suggests management knows it.
5. Site data hygiene — the XPG product page appears to display MX12B specs, and the "Active Contracts and Past Performance" page is a live placeholder marked [need content] in its source. Minor, but worth knowing if you are ingesting sagatech.com into a RAG corpus: the site contains at least one internally inconsistent spec block.

9. SOURCE INDEX

Primary (company): <https://sagatech.com/> · [/about-us/](#) · [/about-us/who-we-are/](#) · [/about-us/leadership-team/](#) · [/about-us/careers/](#) · [/transponders/](#) · [/transponders/mx12b/](#) · [/transponders/mx12b/mx12b-faqs/](#) · [/transponders/mxs-transponder/](#) · [/transponders/xps-with-ads-b-out/](#) · [/transponders/xpg-with-ads-b-out-and-gps/](#) · [/transponders/xpc/](#) · [/transponders/xpc-mode-c-transponder/](#) · [/acas-solutions/](#) · [/integration-services/](#) · [/international-resellers/](#) · [/government-solutions/](#) · [/government-solutions/defense-solutions/](#) · [/government-solutions/civil-solutions/](#) · [/government-solutions/active-contracts-and-past-performance/](#) · [/commercial-solutions/](#) · [/service-bulletins/](#) · [/resources/product-sheets/mx-series/](#) · [/news-and-events/](#) (full post index via [/post-sitemap.xml](#)) · [/wp-content/uploads/2021/02/UAS-Transponder-Requirements-and-Guidelines-Sagatech-Avionics-2021-Update.pdf](#)

Government / structured data: <https://api.usaspending.gov> (recipient, spending_by_award, spending_over_time; UELs T4UNAUKXMZM3 and N3GCB5FTDN79) · <https://www.highergov.com/awardee/sagatech-avionics-inc-10280151/>

Press / trade: <https://www.prweb.com/releases/sagatech-avionics-awarded-248-5m-army-contract-for-ap-x-128-advanced-iff-production-302751352.html> · <https://www.streetinsider.com/Press+Releases/Sagatech>

+Avionics+Awarded+\$248.5M+Army+Contract+for+APX-128+Advanced+IFF+Production/26356978.html · <https://www.defenseadvancement.com/news/sagotech-avionics-secures-contract-for-advanced-iff-transponder-production/> · <https://www.defenseadvancement.com/company/sagotech/> · <https://www.prnewswire.com/news/sagotech-avionics-inc./> · <https://www.prnewswire.com/news-releases/sagotech-avionics-receives-afwerx-contract-from-the-us-air-force-301207072.html> · <https://insideunmannedsystems.com/transforming-airspace-safety-insights-from-sagotech-ceo-tom-furey-at-xponential-2024/> · <https://www.auvsi.org/news/oem-leader-how-sagotech-avionics-builds-layers-of-aviation-safety-for-dual-use-bvlos-capabilities/> · <https://verticalmag.com/press-releases/sagotech-avionics-demonstrates-true-detect-and-avoid/> · <https://www.unmannedsystemstechnology.com/company/sagotech-corporation/>

Investors / databases: <https://www.columbiariverpartners.com/> · <https://www.columbiariverpartners.com/news/sagotech-avionics-awarded-2485m-army-contract-for-apx-128-advanced-iff-production> · <https://pitchbook.com/profiles/company/125125-75> · <https://www.cbinsights.com/company/sagotech-avionics> · https://tracxn.com/d/companies/sagotech/_W1xpfbZrUjOJGcs53KmpBnmXTiz0E3YAyhWB35y80ys · <https://www.preqin.com/data/profile/asset/sagotech-avionics-inc-/434543> · <https://www.zoominfo.com/c/sagotech-corp/347884610> · <https://craft.co/sagotech-avionics/locations> · <https://crgta.org/spotlight/sagotech-avionics-inc> · <https://www.linkedin.com/company/sagotech-avionics-inc> · <https://www.linkedin.com/in/tom-furey-584b1/> · https://www.airframer.com/direct_detail.html?company=152932

Known gaps for follow-up: (1) patent portfolio — all databases blocked, needs USPTO Patent Public Search + Assignment Search on both "Sagotech Corporation" and "Sagotech Avionics"; (2) revenue — no disclosure exists, only federal obligations are verifiable; (3) valuation — never disclosed for either 2019 or 2021 transaction; (4) MXR interrogator and ACX-3000M — no public product pages, specs unavailable; (5) exact current headcount — sources range 36 to 69 depending on snapshot date.

SKYNETICS — COMPETITIVE RESEARCH REPORT

Compiled 2026-09-24 from local cached research files plus live web supplementation.

0. DISAMBIGUATION — READ FIRST

"Skynetics" resolves to at least six distinct entities. This report covers exactly one:

SUBJECT: Skynetics Pvt. Ltd. — Bengaluru, Karnataka, India. CIN U26109KA2023PTC174313. Website skynetics.net. Sector: Defense and Space Manufacturing — flight-qualified satellite subsystems, long-range RF transceivers, ground stations.

CORRECTION TO PREMISE: this is not primarily a drone/UAS company. It's a space electronics/satellite subsystems company. UAV/drone is its smallest line (2 of 33 products), positioned as subsystem supply to drone builders, not drone OEM.

EXCLUDED — not this company: Skynetics UK (Cheltenham, landing-gear engineering consultancy); Skynetics Group (Monrovia, Liberia, IT services); Skynetiks Technologies (IT services); Skynetic.store (consumer drone e-commerce); various unrelated "Skynet"-named drone firms (Singapore, Dehradun, Connecticut, Sky-Drones UK); and — critically — the cached Reddit dataset for "skynetics" is 100% Terminator/AI-doom meme contamination with zero signal on the real company; excluded entirely from this report.

1. COMPANY OVERVIEW

Legal name: SKYNETICS PRIVATE LIMITED. CIN U26109KA2023PTC174313. Incorporated 30 May 2023.

Registered address: Sahakaranagar, Bangalore North, Karnataka 560092. HQ Bengaluru.

Founded 2023 by engineers with satellite/space-electronics backgrounds from TU Delft, ETH Zurich, KIT, IISER, and ISRO. Founding story: "born out of the dreams of graduates from the Netherlands and Germany."

Strategic thesis: Indian space-sector import-substitution — "The parts that stayed closed for decades are the parts we build."

Mission: "to develop highly miniaturized and cost-effective embedded solutions... pushing technology beyond limits for extreme scenarios." Homepage tagline: "Space-Grade Technology. Made in India." — "50+ subsystems flown, ISRO-empanelled."

Size: reported figures range 7 (Tracxn) to 9-12 (LinkedIn/Exa) employees, LinkedIn self-declared band 11-50. Realistic estimate: 7-12 people, grew from ~2 to ~9 over the prior year (+350% YoY). LinkedIn followers grew from 3,518 (Jun 2026) to 5,000 (Sep 2026).

Footprint: 3 countries, 3 ground stations (Netherlands, Dubai, Bangalore), India as primary hub. 3 ISRO nanosatellites carrying its hardware, 2 complete satellites delivered.

Certifications/posture: AS9100D in progress (NOT yet certified); SCOMET export-control compliant; DGFT IEC issued; IN-SPACe authorization issued; ISRO-empanelled (claimed). Notable gap: selling flight hardware before AS9100D certification completes.

Flight heritage claimed: INS-1A, INS-1B, INS-1C (ISRO nanosatellites — subsystems), Gaganyaan 2024 (bio-sensor module), CubeSat-1U and CubeSat-3U delivered end-to-end. Aggregate claims: 50+ flight-qualified subsystems at TRL 9, 100% mission success across 50+ deployments, MTBF 12,500 hrs, 4 patents. CAUTION: INS-1A/1B/1C launched 2017-2018, years before the 2023 company existed — this is founder-team heritage carried from prior ISRO/TU Delft work, not company track record.

Named clients/partners: ISRO, DRDO, ADA, URSC, ISTRAC, IN-SPACe (govt/agency); L&T, Bellatrix Aerospace, Kepler Aerospace, Shyena Tech Yantra, Aidin Technologies (industry); IIST, Dayananda Sagar University, Edutech4Space (academic). Also sells to universities for space-environment simulation platforms.

2. FINANCIALS

Pre-revenue-to-early-revenue private company, no public financials filed/accessible beyond one regulatory band.

- Annual revenue: 0–10 crore (~USD \$0–1.2M) as of 31 Mar 2025 (Tracxn) — widest, least informative MCA band; realistically low end given ~7 employees.
- Valuation: not disclosed — no priced round has occurred.
- MCA filings (MGT-7/AOC-4) exist but were not retrievable in this research (Zaubacorp 403, Tofler 404) — biggest open financial gap.
- Board: Tracxn lists 5 active members — 2 founders + 3 others (one named: Vasanthi Ramaswamy Iyengar). A 5-person board at a 7-person unfunded company is atypical, suggesting possible undisclosed investors/advisors holding board seats.
- Tracxn Score: 33/100, ranked 45th of 232 active competitors in its space-subsystem category — lowest score among the top-10 comparison peers shown (DcubeD Germany \$4.8M raised, AAC Clyde Space Sweden public, uAvionix USA acquired, Aitech USA acquired, Ananth Technologies India funded).
- Revenue model inference: subsystem unit sales, complete-satellite delivery, ground-station turnkey builds, custom/NRE engineering (likely dominant near-term line), education sales, DRDO defence programme work.

3. FUNDING HISTORY

No publicly recorded funding rounds — Tracxn states plainly "Skynetics has not raised any funding rounds yet." \$0 total raised, no investors, no rounds, no acquisitions made or received.

Data conflict: same Tracxn page contains a paywalled contradictory line suggesting a funded, undisclosed first round ranking 11th in funding among competitors — likely a generic templated upsell artifact, not reliable. The unfunded reading is more probable, though the atypical 5-person board (see Financials) keeps this unresolved without MCA shareholding filings.

Non-dilutive/quasi-funding signals: IN-SPACe authorization, ISRO empanelment, DRDO Best Innovation Award (via Aeronautical Development Establishment), SCOMET+IEC export-ready posture. India's IN-SPACe seed fund / ISRO tech transfer / DRDO TDF are plausible funding sources, none confirmed specifically.

Funding-readiness read: company shows classic pre-raise behavior in Q3 2026 — trade-show launch (BSX 2026), new flagship product (SKYSATCOM-7985), senior official visits (ISRO Chairman, IN-SPACe Director, DG Defence Space Agency), simultaneous 3-role hiring push. A first institutional round announcement within 6–12 months of Sept 2026 is plausible.

4. LEADERSHIP AND PEOPLE

Founders (per company's own Sept 2026 LinkedIn post, naming 3; Tracxn records only 2):

Dr. Sujay Narayana — Co-Founder, CEO & CTO. PhD, TU Delft space-IoT/embedded systems research background. 3 international IEEE/ACM awards. Verified publication record: 30 papers, 416 citations (Semantic Scholar). Key papers map directly to product lines: "Hummingbird" GPS receiver for small satellites (→ CubeSat GNSS Module); "SOS" satellite health monitoring (→ telemetry subsystems); "SWANS" Space IoT sensor network (→ NTN-01); "Hermes" wind energy harvesting AoA/wind-speed sensing (→ Patent NL2028379B1); "LOCI"/PIR sensor localization (→ Patent NL2024953B1); "SFMAC" high-density LoRaWAN (→ Patent NL2031744B1, LORA-100/LORA-AI); "Heart Watch" ECG synthesis (→ Gaganyaan Bio-Sensor Module). This is an unusually tight research-to-product mapping — the company is essentially a commercialization vehicle for one founder's decade of TU Delft research, with extreme key-person risk.

Yasaswi Reddy Sunkara — Co-Founder & Chief Business Officer. Commercial lead; team page's "BCG · ISB — operators" descriptor likely refers to this role (consulting/MBA background). Detailed bio not retrievable (LinkedIn blocked automated fetch).

Ajay Keshava — Co-Founder. Named in company's own BSX 2026 post and present at the ISRO Chairman walkthrough; not listed as a founder in Tracxn's record. Bio not retrievable.

Core team (self-described via badges, not named individuals — signal of very small team): 3x IEEE/ACM awards, TU Delft moon-rover programme, ISRO satellite subsystems, ETH Zurich, IISER, PhD-level space IoT expertise, BCG/ISB operators.

Advisory board (unnamed): Strategic advisor — "senior DRDO leadership"; Academic advisor — "IEEE Distinguished Lecturers, 350+ publications, lunar engineering"; Industry advisor — "semiconductor C-suite, telecom operators/RF systems."

Board of directors: 5 active members (Tracxn) — 2 founders + Vasanthi Ramaswamy Iyengar + 2 unnamed (paywalled).

Hiring (Sept 2026): 3 open roles — Hardware Design Engineer, FPGA Design Engineer, RF PCB Design Engineer — post drew 174 reactions/56 comments, highest engagement of any captured post. All 3 roles are RF/FPGA/hardware, confirming a pivot toward RF payloads and signals intelligence.

5. PRODUCTS

33 products across 6 lines. Full datasheets gated behind "Request Capability Statement" contact forms; patented-tech briefs available only under NDA.

SKYCOM Series — Long-Range Communication (6 products). Flagship, DRDO Best Innovation Award winner, "indigenous design replacing imported alternatives." Includes SKYCOM-1550 (sub-GHz, 100km, 5W), SKYCOM-4025 (100km, 5W industrial), SKYCOM-7092 (ISM band, 5W), SDR Communication Platform (1MHz-18GHz, >150Mbps full duplex — most technically ambitious item), X-Band Resilient Communication Module (coming soon), Resilient Communication Module (100MHz-6GHz frequency-hopping, drones) — the latter two tied to the pending 4th patent, the clearest anti-jam/contested-spectrum defence differentiator.

Satellite Systems (8 products). Full vertically-integrated 1U CubeSat bus (1U-24U range claimed): 1U CubeSat platform (<1.33kg), SKYSATCOM-1500 (VHF/UHF comms, flight proven), CubeSat Power System (MPPT), CubeSat Battery System (Li-Ion/BMS), CubeSat OnBoard Computer (ARM Cortex-A53, rad-tolerant — a relatively high-performance choice implying edge-AI ambitions), CubeSat Solar Panel (GaAs, 30% efficiency, deployable), CubeSat GNSS Module, 1U CubeSat Structure. Very few 7-person companies can claim a full in-house bus stack.

Ground Stations (4 products). VHF/UHF Ground Station ("highest gain yagi antennas installed in India," ISRO-supported), S-Band Ground Station (2Mbps, 2.4m dish), S/X-Band Dual Ground Station (100Mbps, 3.7m dish), Custom Ground Station. Sites: Netherlands, Dubai, Bangalore. End-to-end mission capability (bus + comms + ground) is a strategically stronger position than pure component vending.

Custom Electronics / Embedded (5 products, NRE line). RF Range Extender (drone-based walkie-talkie relay), Underwater Robotics, Underwater Communication System (acoustic/optical), HAPs Communication System (high-altitude platform), RF GaN Power Amplifier (500W, DC-10GHz — a genuinely hard, high-value defence/EW component). Company deliberately spans seabed to orbit on one RF/signal-processing core competence.

IoT & Sensors (8 products, largest line). NTN-01 (satellite IoT sensor node), SDC-01 (satellite-drone C2, BVLOS — the strongest bridge to the UAS market), TAG-WL (satellite wildlife/bird tracking), LORA-100 (>100km LoRa), GGY-BIO (ISRO Gaganyaan bio-monitoring — marquee credential, human spaceflight hardware), EDG-ML (ultra-low-power edge ML), LORA-AI (AI-powered LoRa sensor), FPGA-AI (FPGA+AI chip board).

UAV & Drone Technologies (2 products — smallest line). Sky Sentinel and Sky Scout — both explicitly "reference stacks"/demonstrators for multi-rotor platforms (RF + edge AI), not saleable aircraft. Confirms Skynetics is a Tier-2 subsystem supplier to drone OEMs, not a drone OEM itself.

Announced/roadmap: SKYSATCOM-7985 (X-band satcom module, launched Sept 2026 at Bengaluru Space Expo, not yet in web catalogue); ELINT/COMINT payloads (shown to ISRO Chairman as "upcoming launches" — major strategic move into signals intelligence); next-gen GNSS modules; next-gen receiver designed to a sensitivity target set by AVM Manu Midha, DG Defence Space Agency.

6. PATENTS

Four patents total — three granted (Netherlands only), one pending.

1. NL2028379B1 — "Method and apparatus for determining angle of attack and wind speed" — Granted, NL. Traces to Narayana's 2021 "Hermes" IEEE RA-L paper. Wind-energy-harvesting self-powered sensor; relevant to UAV flight control / HAPs.
 2. NL2024953B1 — "Systems and methods for infrared sensing" — Granted, NL. Traces to "LOC"/PIR papers (2015, 2020). Privacy-preserving passive-IR multi-person localization; applications in building automation, security, perimeter sensing.
 3. NL2031744B1 — "Channel sensing circuit and apparatus comprising the circuit" — Granted, NL. Traces to LoRa/RF-harvesting MAC papers (2022, 2025). Enabling tech behind LORA-100 and high-density LoRaWAN scaling.
 4. "Beyond Frequency Hopping — A True Resilient RF Communication" — Filing in progress, no number/jurisdiction yet. IP behind the Resilient Communication Module and X-Band Resilient Communication Module. Strategically the most significant patent — aimed at defence procurement, aligned with current RF hiring push and the Defence Space Agency's receiver-sensitivity challenge.
- Portfolio assessment: small (4), young, geographically narrow (Netherlands-only — no US/EP/PCT/Indian grants found, a material weakness given the company's Indian defence/ISRO market and "Made in India" positioning). Research-derived, not product-derived — all granted patents cover TU Delft-era (2020-2022) sensing/channel-access research, not the satellite-bus subsystems that make up most of current sales. Assignment to Skynetics Pvt Ltd (vs. TU Delft or the inventor personally) is unverified since the patents predate the May 2023 incorporation.

7. LATEST NEWS (2026)

Bengaluru Space Expo (BSX) 2026, 7-9 Sept 2026, Hall 1 Booth H7 — the company's coming-out event.

- ISRO Chairman Dr. V. Narayanan visit (~8-9 Sept): all three founders walked him through the roadmap including ELINT/COMINT, GNSS modules, S/X-band comms. Highest-reaction post captured (182 reactions).
- IN-SPACe Director Dr. Vinod Kumar visit (day 3): 91 reactions.
- Day 2 coverage post: 73 reactions.

SKYSATCOM-7985 launch (~mid-Aug 2026): new X-band satcom module, designed/built in Bengaluru, launched at BSX 2026 — pairs with the S/X-band 3.7m ground station for an end-to-end high-rate downlink chain.

Defence Space Agency engagement (~mid-Sept 2026): AVM Manu Midha, DG Defence Space Agency, toured the company's resilient-communications work and set a receiver-sensitivity performance target — arguably the most strategically consequential item after the ISRO Chairman visit, potentially the seed of a shaped defence requirement.

Hiring expansion (~mid-Sept 2026): 3 RF/FPGA/hardware roles posted, 174 reactions/56 comments — highest engagement of any post, confirming the RF/ELINT pivot.

Positioning post (~June 2026): "Once you are in orbit, there is no second attempt... 50+ subsystems... delivered 2 satellites and hold 4 patents... India's space sector has opened to private industry for the first time."

Awards (undated, press page): Best Innovation Award (Aeronautical Development Establishment, DRDO) for SKYCOM line; 3 international IEEE/ACM awards to CEO Dr. Sujay Narayana; recognition as indigenous alternative to imported products.

Notable: zero third-party press coverage found anywhere — all public narrative is first-party (own website/LinkedIn). No trade-press, business-press, or analyst coverage located.

8. SYNTHESIS

Skynetics is a 3-year-old, 7-12-person, unfunded Bengaluru deep-tech hardware company commercializing a TU Delft space-IoT research programme, with genuine (largely founder-carried) ISRO/Gaganyaan flight heritage, a full in-house 1U CubeSat bus, an award-winning long-range RF transceiver line, and three

Netherlands patents. Currently pivoting toward defence RF (resilient/anti-jam comms, ELINT/COMINT) with active senior engagement from ISRO, IN-SPACe, and the Defence Space Agency.

Strengths: rare vertical integration for its size, real flight heritage, DRDO award, near-complete regulatory posture, tight research-to-product mapping, well-timed against India's space-sector liberalization, unusual 3-country footprint.

Weaknesses: unfunded with tiny revenue while announcing an RF/ELINT roadmap that needs capital; extreme key-person risk (one founder's IP); patent estate is Netherlands-only and possibly not corporately assigned; AS9100D not yet certified while selling flight hardware; zero third-party press validation; flight-heritage claims predate the company by 5-6 years; 33 products across 6 lines at ~7-12 headcount implies most SKUs are reference designs/low-volume NRE rather than stocked inventory.

Sources: skynetics.net (about/team/patents/press/careers/products + subpages), linkedin.com/company/skynetics-pvt-ltd, tracxn.com company + Bengaluru space-tech sector pages, Semantic Scholar author search (Sujay Narayana), facebook.com/Skynetics-Pvt-Ltd.

SpeedyBee — Company Research Report

Prepared: 24 September 2026 Research method: Open-web search + direct page fetches. Where sources conflict, I flag it. Several requested categories (audited financials, funding rounds, itemized patents) genuinely do not exist in the public record for this company — I say so explicitly rather than guessing.

0. EXECUTIVE SUMMARY / KEY CORRECTIONS

Three premises in the research brief need correcting before anything else:

- SpeedyBee is not headquartered in Shenzhen. Its operating legal entity is registered in Zhuhai, Guangdong Province, at 珠海市高新区港湾大道田馆路12号厂房四楼A区 (4th Floor, Area A, Building 4, No. 12 Tianguan Road, Gangwan Avenue, Gaoxin/High-Tech District, Zhuhai). Source: 水滴信用 registry record (<https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html>), 爱查ICP registry record (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>), Tracxn (https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIdm-lkDTNdM), Crunchbase (<https://www.crunchbase.com/organization/speedybee>). The Shenzhen association is real but indirect — see the RunCam relationship in §1.3.
- SpeedyBee has raised no venture funding, ever. There is no round, no investor, no valuation, no filing. Tracxn states flatly: "Speedybee has not raised any funding yet." and records zero investments and zero acquisitions by the company. This is not a data gap — it is a private, founder-owned, cash-funded Chinese hardware business.
- The "founded 2005" date circulating in Western databases is wrong. Tracxn (https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIdm-lkDTNdM) lists 2005, which is impossible — the consumer FPV market did not exist in a recognizable form then, and SpeedyBee's first reviewed product dates to 2018. The Chinese business registry gives the authoritative incorporation date of the current entity: 6 January 2021. The brand itself predates that entity by roughly three years (first products mid-2018), which is explained by the RunCam relationship.

1. COMPANY OVERVIEW

1.1 Legal identity and registration

Field	Value	Source
Chinese legal name	珠海市快蜂科技有限公司	水滴信用 (https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html)
English rendering	Zhuhai SpeedyBee Technology Co., Ltd.	ibid.
Brand names	SpeedyBee / Speedy Bee / 快蜂	speedybee.com (https://www.speedybee.com/)
Unified Social Credit Code	91440402MA55TCGE8H	水滴信用 (https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html)
Business registration number	440402000406919	ibid.
Organization code	MA55TCGE8	ibid.
Incorporation date	2021-01-06	ibid.
Business term	2021-01-06 to 无固定期限 (indefinite)	ibid.
Registered capital	RMB 5,000,000 (fully paid in — 实缴资本 500万元)	ibid.
Entity type	有限责任公司(自然人投资或控股) — LLC, natural-person held	ibid.
Status	存续 (active / in operation)	ibid.
Registering authority	珠海市市场监督管理局 (Zhuhai Municipal Administration for Market Regulation)	ibid.
Last approval/amendment date	2026-06-03	ibid.
Registered industry classification	软件和信息技术服务业 (Software & IT services)	爱查ICP (https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html)
Headcount (社保参保人数, social-insurance enrollees)	132	ibid.

Field	Value	Source
Province/city	Guangdong / Zhuhai	ibid.
Registered address	珠海市高新区港湾大道田馆路12号厂房四楼A区	ibid.

Note the industry classification quirk: the company is registered under software and IT services, not hardware manufacturing, despite being a hardware company. This is consistent with its origin story — SpeedyBee began as an *app* company (see §1.2) and the classification was never migrated. Its licensed business scope nonetheless explicitly covers hardware: 电子元器件制造 (electronic component manufacturing), 玩具制造 (toy manufacturing), 智能无人飞行器制造 (intelligent unmanned aerial vehicle manufacturing), plus technical services/development/consulting/transfer, network and information-security software development, computer systems services, electronic product and network equipment sales, and 货物进出口/技术进出口/进出口代理 (goods import-export, technology import-export, import-export agency).

The 132-employee social-insurance figure is the single most reliable size indicator available, since neither Crunchbase nor Tracxn discloses headcount. Treat it as a floor for total staff — contract manufacturing labor and any Shenzhen/Hong Kong affiliated staff would not appear on this entity's Zhuhai social-insurance roll.

The company is actively hiring: the registry aggregator shows 81 open job listings, including an "AI应用工程师" (AI Application Engineer) role posted 2026-08-13 at RMB 10–15k/month via BOSS直聘 (爱查ICP (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>)). That an FPV hardware maker is recruiting AI application engineers in 2026 is a meaningful forward signal about product direction.

1.2 Brand history and origin

SpeedyBee's real history is best reconstructed from the review record, because the company publishes almost nothing about itself. The most complete public chronology is Oscar Liang's archive ([oscarliang.com search](https://oscarliang.com/search) (<https://oscarliang.com/?s=speedybee>)), which spans 10 pages of SpeedyBee coverage from 2018 to 2025.

Phase 0 — 2017/2018, OEM camera era. The earliest SpeedyBee-labelled item reviewed is the "Eachine SpeedyBee SEC FPV Camera (<https://oscarliang.com/page/3/?s=speedybee>)" (2 January 2018), sold under the Eachine banner and positioned as a budget racing camera against RunCam and Foxeer. SpeedyBee at this point was a design house, not a brand with its own retail presence.

Phase 1 — 2018, the Bluetooth insight. The founding product idea was not a drone part at all; it was *removing the laptop from the field*. In June 2018 SpeedyBee shipped the F4 AIO FC with Bluetooth, described by Liang as "the first of its kind to have a built-in bluetooth module," allowing pilots to configure a quad from a phone. One month later (18 July 2018) came the USB-to-Bluetooth Adapter, solving the same problem for flight controllers that had no spare UART. Sources: [Oscar Liang, page 3](#) (<https://oscarliang.com/page/3/?s=speedybee>).

This is the company's actual strategic DNA and it explains everything since: SpeedyBee is a *configuration-experience* company that sells hardware, which is why its corporate registration says "software and IT services" and why the SpeedyBee App — not any single board — is the real moat.

Phase 2 — 2019, video transmission. The TX500 VTX (24 January 2019), a 20x20mm-stack-fitting VTX with built-in mic and VTX control, and the DVR VTX (25 January 2019), a ~600mW transmitter with onboard DVR recording to survive signal breakup. Source: [Oscar Liang, page 3](#) (<https://oscarliang.com/page/3/?s=speedybee>).

Phase 3 — 2020, the Adapter becomes a category. The WiFi Adapter 2 (7 August 2020) moved from Bluetooth to WiFi and let pilots run full Betaflight setup from a phone alone. Liang later called the Adapter 2 "the MUST HAVE tool" ([Adapter 3 review](#) (<https://oscarliang.com/speedybee-adapter-3/>)). This product made the brand.

Phase 4 — 2021, incorporation and the flight-controller pivot. The Zhuhai entity is incorporated 6 January 2021. In March 2021 the F7 V2 FC stack launched with integrated Bluetooth — "you don't need a computer" ([Oscar Liang, page 2](#) (<https://oscarliang.com/page/2/?s=speedybee>)). The V2 used a dual-MCU design: STM32F722 for the FC plus a dedicated STM32F051 for ESC duties ([SoarSky lineup comparison](#) (<https://soarsky.com>)).

[rc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-specs-amp-speedybee-full-lineup-comparison](https://oscarliang.com/page/2/?s=speedybee))).

Phase 5 — 2022, the value-stack breakout. October 2022 was the company's decisive month: F405 V3 (26 Oct, "the best value stack"), F7 V3 (27 Oct, US\$119.99 full stack), Adapter 3 (21 Oct), and TX800 VTX (20 Oct) all landed within eight days, plus the Flex25 whoop frame in July ("possibly the smallest 2.5" whoop ever existed"). Source: [Oscar Liang, page 2 \(https://oscarliang.com/page/2/?s=speedybee\)](https://oscarliang.com/page/2/?s=speedybee). The F405 V3 became, by common consensus in subsequent coverage, the default budget 5-inch stack of its generation.

Phase 6 — 2023, verticalization into complete aircraft. SpeedyBee stopped selling only components and began selling flyable drones and frames: Master 5 V2 BNF (19 July 2023), Goggles BEC (31 May 2023), F405 Mini stack (15 Aug 2023), F405 WING APP (25 Dec 2023 — its first fixed-wing FC, "currently the cheapest wing FC you can get"), Bee35 Cinewhoop (27 Dec 2023), Mario Fold 8 long-range folding drone (30 Dec 2023). Sources: [page 1 \(https://oscarliang.com/?s=speedybee\)](https://oscarliang.com/?s=speedybee), [page 2 \(https://oscarliang.com/page/2/?s=speedybee\)](https://oscarliang.com/page/2/?s=speedybee).

Phase 7 — 2024, platform breadth. TX1600 Ultra VTX (19 Jan), F405 Wing Mini (18 Feb), Mario 5 frame (28 Feb, pitched against the TBS Source One), F405 V4 (13 Apr), Bee25 Cinewhoop (21 Oct, built around DJI O3), Mario 5 complete drone (11 Nov), F405 AIO (18 Nov).

Phase 8 — 2025–2026, modularity and O4. F405 V5 OX32 (30 Aug 2025) and the current modular Mario MiniX / Master 3X / Bee25 / Bee35 families built around DJI O3 and O4 air units ([speedybee.com \(https://www.speedybee.com/\)](https://www.speedybee.com/)).

1.3 The RunCam relationship (important and under-reported)

The US trademark "SPEEDY BEE" (Serial No. 88478719, Registration No. 6121981, filed 18 June 2019) is registered not to the Zhuhai entity but to RunCam Technology (Shenzhen) Co., Ltd. — covering downloadable software for controlling and analyzing flight-controller parameter settings, vehicle radios, and cases. Source: [Justia Trademarks \(https://trademarks.justia.com/884/78/speedy-88478719.html\)](https://trademarks.justia.com/884/78/speedy-88478719.html).

Corroborating evidence of the tie:

- SpeedyBee's own storefront hosts RunCam product pages, e.g. [RunCam 5 Orange \(https://www.speedybee.com/runcam-5-orange/\)](https://www.speedybee.com/runcam-5-orange/) and [RunCam Thumb \(https://www.speedybee.com/runcam-thumb/\)](https://www.speedybee.com/runcam-thumb/), under "PARTNER PRODUCTS" sections for [action cameras \(https://www.speedybee.com/action-camera/\)](https://www.speedybee.com/action-camera/) and [analog FPV systems \(https://www.speedybee.com/analog-camera/\)](https://www.speedybee.com/analog-camera/).
- SpeedyBee's own BNF drones ship with RunCam optics as standard: the Master 5 V2 uses a RunCam Phoenix 2 SE ([review \(https://oscarliang.com/speedybee-master-5-v2-drone/\)](https://oscarliang.com/speedybee-master-5-v2-drone/)), and the Bee35 analog variant is sold at Pyrodrone explicitly as "(RunCam Phonix SE)" ([Pyrodrone \(https://pyrodrone.com/collections/speedybee-products/\)](https://pyrodrone.com/collections/speedybee-products/)).
- The SpeedyBee App's camera firmware flasher lists RunCam hardware as its example targets — "RunCam Hybrid, Racer2 etc." ([SpeedyBee App page \(https://www.speedybee.com/speedy-bee-app/\)](https://www.speedybee.com/speedy-bee-app/)).
- Product launches were historically synchronized — the SpeedyBee Adapter 2 launched alongside the RunCam Split 4 and RunCam Webcam.

Interpretation, stated as inference not fact: SpeedyBee originated inside or immediately adjacent to RunCam's orbit as a software/accessory brand around 2018–2019 (hence RunCam holding the US mark for the *software* class), and was subsequently spun into its own Zhuhai operating company in January 2021 as the hardware business scaled. The two now operate as sister/partner brands with deliberately non-overlapping catalogs: RunCam owns cameras, SpeedyBee owns flight controllers, ESCs, VTX, frames, adapters and the app. Separate storefronts (shop.runcam.com vs [speedybee.com](https://www.speedybee.com)) obscure this from most buyers.

Caveat: the US mark's ownership has not been confirmed as reassigned to the Zhuhai entity. Meanwhile the Zhuhai entity is filing its own marks in China (see §6.2). A reader should not assume a current parent-subsidary structure — only a documented shared-IP history.

1.4 Mission and positioning

The company's public mission statements are unusually terse:

- Site tagline: "Speedy Bee, Simplifying FPV." ([about-us \(https://www.speedybee.com/about-us/\)](https://www.speedybee.com/about-us/))
- Homepage tagline: "Simplify FPV drone racing!" ([speedybee.com \(https://www.speedybee.com/\)](https://www.speedybee.com/))
- Full About Us text, in its entirety: *"We are the new choice for you, but always reliable. What you want is what we want to do."*

That is the whole page. There is no published founding narrative, no leadership page, no investor relations, no press room. The English about-us page contains no founding year, no address, no team size, no founders, no contact details ([verified \(https://www.speedybee.com/about-us/\)](https://www.speedybee.com/about-us/)). The Chinese-site equivalent at [speedybee.cn/about-us.html](https://www.speedybee.cn/about-us.html) is indexed by search engines but currently returns HTTP 404 on live fetch; a Wayback snapshot exists (timestamp 20260316085427) but web.archive.org was not fetchable from this environment.

Positioning in one line: SpeedyBee is the Xiaomi of FPV electronics — aggressive price-performance, high release cadence, ecosystem lock-in via a first-party mobile app, and a deliberate strategy of making the **setup experience** the differentiator rather than raw specs.

2. FINANCIALS

Bottom line: there are no financials. SpeedyBee is private, unlisted, unfunded, and files nothing public.

What can be established:

- No public listing. No stock exchange, no ticker, no prospectus, no IPO filing appears in any source consulted. SpeedyBee does not appear in any securities filing database.
- No revenue disclosure. [Tracxn \(https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIDm-lkDTNdM\)](https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIDm-lkDTNdM) records revenue as not disclosed. [Crunchbase \(https://www.crunchbase.com/organization/speedybee\)](https://www.crunchbase.com/organization/speedybee) (403 on direct fetch; summarized via search index) likewise carries no revenue estimate.
- No valuation. Never externally priced, because never externally funded.
- Registered capital: RMB 5,000,000, fully paid in ([水滴信用 \(https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html\)](https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html)). Note that in the PRC, registered capital is a legal ceiling on shareholder liability, not a measure of assets, revenue or enterprise value. A RMB 5M registered capital is small and typical of a bootstrapped founder-owned firm.
- No annual-report financial data published. The registry aggregators expose shareholder and contribution data from the 2025 annual report (年报) but not P&L; figures — Chinese LLCs of this size are not required to publish financial statements publicly.
- No bond issuance, no LEI, no filings in the Legal Entity Identifier system under this name.

2.1 Indirect financial indicators (the only quantitative proxies available)

Since no statements exist, here are the defensible proxies:

- 132 social-insurance enrollees ([爱查ICP \(https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html\)](https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html)). For a Guangdong hardware SME with a design-and-outsource model, this is a mid-size operation, not a garage shop.
- 81 concurrent open roles (ibid.) — inconsistent with a contracting business; suggests active expansion as of mid-2026.
- Catalog breadth as revenue proxy: at least 43 distinct SKUs at a single US retailer ([Pyrodrone \(https://pyrodrone.com/collections/speedybee-products\)](https://pyrodrone.com/collections/speedybee-products)), 15 at [NewBeeDrone \(https://newbeedrone.com/collections/speedybee\)](https://newbeedrone.com/collections/speedybee), 11 flight controllers alone on the first-party site ([speedybee.com/flight-controller \(https://www.speedybee.com/flight-controller/\)](https://www.speedybee.com/flight-controller)), spanning US\$0.99 (a beeper) to US\$611.99 (a Bee35 PRO HD complete drone).
- Price band: components cluster US\$30–140; complete drones US\$249–612. The company plays the high-volume mid-market, not the premium niche.

- Distribution footprint: first-party e-commerce (speedybee.com), AliExpress, Amazon, plus Western specialist retail — GetFPV, RaceDayQuads (RDQ), Pyrodrone, NewBeeDrone, KiwiQuads (NZ), SoarSky. Multi-channel global distribution across at least three continents.
- Paid-in capital equals registered capital — both RMB 5M, contributed 19 January 2024 in cash (货币) per the 2025 annual report (爱查ICP (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>)). A later-dated full cash paydown is a mild positive liquidity signal.

2.2 Material financial risk disclosed on the company's own storefront

SpeedyBee is currently collecting a 54% prepaid tax on orders shipped to US addresses, refundable against an official duty receipt if customs assesses less. This notice appears on multiple product pages — e.g. [Mario MiniX](#)

(<https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/>) and [Bee25](#) (<https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/>) — and is corroborated by third-party coverage. This is the single largest identifiable threat to SpeedyBee's direct-to-consumer economics: the US is a core FPV market, and a 54% prepaid surcharge on a price-led brand is close to an existential pressure on the direct channel. It also creates a structural incentive to push volume through US-domiciled distributors instead.

A second logistics constraint: batteries ship separately and only to nine countries (US, Germany, France, Switzerland, UK, Spain, Sweden, Italy, Australia), and standard postal air mail is unavailable on drone SKUs — express delivery is mandatory ([Bee25 product page](#) (<https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/>)).

3. FUNDING HISTORY

There is none. This section is complete as written.

- Rounds raised: zero. [Tracxn](#) (https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIdm-lkDTNdM): "Speedybee has not raised any funding yet." Status classified as Unfunded.
- Investors: none. No angels, no seed, no venture, no strategic, no government fund, no state-backed vehicle. Note that many Guangdong drone firms take municipal or provincial industrial-fund money; SpeedyBee has not.
- Amounts: n/a. Dates: n/a. Post-money valuation: n/a.
- Acquisitions by SpeedyBee: none. Tracxn: "Speedybee has made no investments or acquisitions yet."
- SpeedyBee as an acquisition target: no recorded transaction. No exit, no merger, no trade sale in any source consulted.
- Ownership is 100% natural-person held, split between two individuals (see §4), per the 2025 annual report filing (爱查ICP (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>), 水滴信用 (<https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html>)). There is no holding company, no VIE structure, no offshore Cayman/BVI vehicle visible in the registry — which is itself evidence that no foreign VC round was ever contemplated, since a China-based startup planning USD venture funding would typically have erected an offshore structure.

Competitive funding context (for benchmarking, from [Tracxn](#)

(https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIdm-lkDTNdM)):

Tracxn tracks 77 active competitors, of which only 5 are funded and 5 have exited. SpeedyBee ranks 6th of 77 by Tracxn score (36/100) and 7th by total funding — i.e. it outranks most funded peers on the composite score despite raising nothing. Funded comparables cited: uAvionix (acquired, \$10M raised), Sagetech (Series B, \$12M), Zerodrag (Seed, \$1.23M), Defendron Technologies (Seed, \$422K), with ARK Electronics unfunded. Other listed competitors: Starlogic Intelligent Technology, Team BlackSheep, Skynetics, Falcon Skyworks. Note these comparables skew toward avionics/ADS-B rather than hobby FPV, so the competitor set is imperfect — SpeedyBee's real rivals are iFlight, HGLRC, JHEMCU, Foxeer, BetaFPV, Diatone, T-Motor and Holybro.

4. LEADERSHIP AND PEOPLE

SpeedyBee publishes no leadership page, no executive bios, no founder interviews, and no named spokespeople. Everything below comes from the PRC corporate registry, which is the only authoritative source.

4.1 Shareholders and officers

Name	Role	Stake	Subscribed / paid capital	Date
杨宗元 (Yang Zongyuan)	Legal Representative; Executive Director (执行董事) and Manager (经理)	92%	RMB 4,600,000, cash	2024-01-19
颜文军 (Yan Wenjun)	Shareholder	8%	RMB 400,000, cash	2024-01-19
袁新星 (Yuan Xinxing)	Finance Officer (财务负责人)	—	—	—

Sources: 水滴信用 (<https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html>), 爱查ICP (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>). A third listed key person exists in the registry record but was not detailed in the retrieved source.

4.2 What this governance structure tells you

- Yang Zongyuan is effectively the company. In a PRC LLC, the 法定代表人 (legal representative) holds binding signature authority; combining that with 执行董事 (sole executive director, i.e. a one-person board in lieu of a full board) *and* 经理 (general manager) concentrates all governance, board and operational authority in one individual. Functionally he is founder, chairman, CEO and controlling shareholder simultaneously. At 92% ownership he also cannot be outvoted on any matter.
- No board of directors exists. The 执行董事 structure is the PRC alternative to a board, permitted for smaller LLCs. This confirms there is no investor board seat, no independent director, and no outside governance — consistent with the zero-funding finding.
- Yan Wenjun's 8% is most plausibly a co-founder or key-employee stake, though the registry does not state a role or title. There is no public information tying him to any operational function.
- Yuan Xinxing (Finance Officer) is the only named finance contact; no CFO title is used.
- No published bios exist for any of the three. I found no interviews, conference appearances, LinkedIn profiles, academic backgrounds, or prior-employment records for Yang Zongyuan, Yan Wenjun or Yuan Xinxing in any English- or Chinese-language source consulted. This is normal for Chinese component-industry founders who market a brand rather than a personality — a deliberate contrast with DJI's Frank Wang (汪滔) or Fat Shark's Greg French, both of whom have substantial public profiles.
- No named CTO, COO, CMO or VP-level executives are public. The engineering organization is anonymous.
- Community-facing presence is institutional, not personal. The brand runs a [SpeedyBee FPV YouTube channel](https://www.youtube.com/@speedybeefpv1088) (<https://www.youtube.com/@speedybeefpv1088>) and communicates product changes through official statements (e.g. the image-only F405 V4 discontinuation notice reported by [Oscar Liang](https://oscarliang.com/speedybee-f405-v5/) (<https://oscarliang.com/speedybee-f405-v5/>)), never through a named individual.

4.3 Workforce

- 132 employees on the Zhuhai social-insurance roll ([爱查ICP](https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html) (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>)).
- 81 active job postings, recruiting via BOSS直聘. Sample: AI Application Engineer, posted 2026-08-13, RMB 10–15k/month (ibid.).
- Registered under a software/IT industry code with a licensed scope that includes UAV manufacturing, implying a mixed software-and-hardware staff profile.

5. PRODUCTS — FULL CATALOG

SpeedyBee's catalog spans six lines: (A) flight controllers and stacks, (B) ESCs, (C) VTX, (D) configuration hardware and software, (E) complete drones and frames, (F) motors, power and accessories. There is one

notable absence: SpeedyBee does not make FPV goggles — it makes goggle *accessories*. See §5.7.

5.1 (A) Flight controllers and FC/ESC stacks — the core business

The definitive comparison, compiled from SpeedyBee's own documentation by retailer SoarSky ([full lineup comparison, updated August 2026](https://soarskyrc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-specs-amp-speedybee-full-lineup-comparison) (<https://soarskyrc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-specs-amp-speedybee-full-lineup-comparison>)), covers 11 models:

Model	Type	MCU	Gyro/IMU	Size (mm)	ESC	UARTs	Blackbox	9V BEC	Price
F405 AIO V2	AIO	STM32F405	ICM42688P	25.5	40A Bluejay	4	8MB	External	\$80.99
F405 V3	Stack	STM32F405	BMI270	30.5	50A BLHeli_S	4	MicroSD	No	\$65.99
F405 V4	Stack	STM32F405	ICM42688P	30.5	55A BLHeli_S	6	MicroSD	3A	\$98.99
F405 V5	Stack	STM32F405	ICM42688P	30.5	55A OX32	6	16MB	2.5A	\$115.99
F745 AIO	AIO	STM32F745	MPU6000	25.5	35A BLHeli_S	4	8MB	No	\$139.99
F7 AIO	FC only	STM32F722	ICM20689	25.5	—	5	32MB	Yes	\$72.99
F7 Mini	Stack	STM32F722	BMI270	20	35A Bluejay	3	8MB	2A	\$115.99
F7 V2	Stack	STM32F722	MPU6000	30.5	45A BLHeli32	5	16MB	2.5A	\$129.99
F7 V3	FC only	STM32F722	BMI270	30.5	—	5	500MB	4A	\$69.99
F405 WING APP	Wing FC	STM32F405	ICM42688P	30.5	90A PDB	6	MicroSD	1.8A	\$74.99
F405 WING MINI	Wing FC	STM32F405	ICM42688P	25.5	—	6	MicroSD	Yes	\$79.99

Per-model notes (same source unless stated):

- F405 V4 — the volume champion. F405 MCU, built-in Bluetooth for wireless Betaflight config via the SpeedyBee app, WiFi, 6 full UARTs, ICM-42688-P gyro, plug-and-play connectors that eliminate most soldering, 5V 3A BEC, 55A continuous / 70A burst (10s), 6S capable. Reviewed [13 April 2024](https://oscarliang.com/speedybee-f405-v4/) (<https://oscarliang.com/speedybee-f405-v4/>). Now discontinued — see §7.
- F405 V5 OX32 — the current flagship budget stack, US\$74.99–\$115.99 depending on configuration and retailer. Square board, relocated LED pads for soft-mount gummies, dedicated GPS and receiver connectors, wireless firmware flashing (V4 offered wireless config only), onboard 16MB flash replacing the microSD slot, DSHOT1200, SB-LED/ST-LED dual mode, PINIO remote on/off for the 9V rail, and Ardupilot support added alongside Betaflight and iNav. FC weight 8.1g (down from V4's 10.5g); ESC 19.1g (down from 23.5g). Trade-offs: motor outputs cut from 8 to 4 (kills hexa/octo builds), servo outputs up from 1 to 2, and a closed-source OX32 ESC firmware. Source: [Oscar Liang, 30 Aug 2025](https://oscarliang.com/speedybee-f405-v5/) (<https://oscarliang.com/speedybee-f405-v5/>).
- F7 V3 — premium/long-range/X8 board. Marketing claims 33% faster wireless firmware flashing via an upgraded Bluetooth/WiFi 2-in-1 chip, ~1-minute smartphone firmware updates, and 500MB onboard blackbox flash — roughly 16× a typical FC, about 100 complete flights. Highest 9V BEC in the Betaflight lineup at 4A. Launched 27 Oct 2022 at US\$119.99 for the full stack; now discounted to \$45.99–\$69.99. Sources: [Oscar Liang](https://oscarliang.com/speedybee-f7-v3/) (<https://oscarliang.com/speedybee-f7-v3/>), [SpeedyBee](https://www.speedybee.com/speedybee-f7-v3-flight-controller/) (<https://www.speedybee.com/speedybee-f7-v3-flight-controller/>).
- F7 V2 — dual-MCU: STM32F722 for FC plus a separate STM32F051 dedicated to ESC; WiFi + Bluetooth onboard. Launched March 2021.
- F745 AIO — the only board in the range with combined WiFi *and* Bluetooth; 10.1g; wireless firmware flashing and blackbox download over WiFi.
- F7 Mini — 20×20mm stack, dual M2/M3 mounting compatibility, 5.7g board.
- F405 Mini BLS 35A 20×20 — budget micro stack, launched 15 Aug 2023, \$31.99–\$64.99.
- F405 AIO 40A Bluejay 25.5×25.5 3-6S — STM32F405 + ICM-42688P + 40A Bluejay on a 25.5mm board for 2–3.5" cinewhoops and light builds. \$55.99–\$80.99 depending on channel.
- F405 WING APP — first fixed-wing FC (25 Dec 2023), "the cheapest wing FC you can get," runs iNAV or ArduPilot (not Betaflight), ships with a separate PDB rated 90A continuous, 8.9g.
- F405 WING MINI — compact wing FC (18 Feb 2024), 5.6g, ArduPilot/iNAV, targeted at sub-250g fixed-wing. Sold in China as a development board through distributors such as [Spotpear](https://spotpear.cn/shop/SpeedyBee-F405-WING-MINI-Flight-Controller-Driver-Board-FC-ArduPilot-iNAV-DJI-RC-FPV-Fixed-) (<https://spotpear.cn/shop/SpeedyBee-F405-WING-MINI-Flight-Controller-Driver-Board-FC-ArduPilot-iNAV-DJI-RC-FPV-Fixed->

[Wing-Airplane-Drone-Quadcopter.html](#)).

- F4 AIO with Bluetooth (2018) — the historic first, "first of its kind to have a built-in bluetooth module." Discontinued.
- F435-OX32-25A-AIO — AT32/Artery-based AIO used inside the Mario MiniX modular platform ([product page](#) (<https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/>)); note the shift to domestic AT32 MCUs, also seen in Chinese community build guides ([ChinaFlier forum build](#) (<https://uav.chinaflier.com/thread-34960-1-1.html>)).

Current first-party store pricing ([speedybee.com/flight-controller](#) (<https://www.speedybee.com/flight-controller/>)) runs \$31.99–\$115.99, with F7 V3 discounted from \$62.99 to \$45.99 and F405 V3 from \$40.99 to \$34.99 — clearance pricing consistent with end-of-life on the older generation.

5.2 (B) ESCs (sold standalone and in stacks)

- F7 V3 BL32 50A 30×30 4-in-1 ESC (also sold as a full stack) — [speedybee.com](#) (<https://www.speedybee.com/>)
- 55A OX32 32-bit 4-in-1 (F405 V5) — closed-source OX32 firmware, ESC telemetry, 19.1g
- 55A BLHeli_S 4-in-1 (F405 V4) — 55A continuous / 70A burst
- 50A BLHeli_S 4-in-1 (F405 V3)
- 45A BLHeli_32 4-in-1 (F7 V2)
- 40A / 35A Bluejay 4-in-1 (AIO boards; Bluejay = open-source BLHeli_S derivative adding bidirectional DShot)
- 35A BLHeli_S (F7 Mini, F745, Bee25 AIO)
- 25A (Mario MiniX F435-OX32-25A AIO)

Firmware positioning matters commercially here: SpeedyBee's historical advantage was shipping open-source Bluejay/BLHeli_S. The V5's move to closed-source OX32 reverses that and is the brand's live reputational issue (§7).

5.3 (C) VTX — video transmitters

- TX1600 Ultra / "TX Ultra" (Jan 2024) — 5.8GHz, 48 channels across 6 bands; power PIT / 25mW / 200mW / 800mW / 1600mW MAX; input 7–28V (up to 6S); output 5V @ 300mA max; 12V @ 90–560mA draw; MMCX connector; 75Ω video input impedance; 29×28×6mm, 20×20mm M2 mounting; 3.2g bare, 5.7g with heatsink, 7.0g with heatsink and antenna; removable heatsink; three status LEDs (blue/red/orange for channel, band, power); button control; Betaflight VTX control via IRC Tramp only, not SmartAudio; pit-mode switch; ships locked, requiring a 10-second button hold to unlock full channels and power. Liang estimated ~36% range gain over the TX800 and found the labelled power steps did not always match measured output. Note: TX Ultra and TX1600 are the same product under two marketing names. Source: [Oscar Liang review](#) (<https://oscarliang.com/speedybee-tx1600-ultra-vtx/>).
- TX800 (Oct 2022) — 800mW analog VTX, "a budget VTX with decent performance," 5V input only. US\$44.99 at [NewBeeDrone](#) (<https://newbeedrone.com/collections/speedybee>). Ships as the stock VTX in the Master 5 V2 analog BNF. Source: [Oscar Liang](#) (<https://oscarliang.com/speedybee-tx800/>).
- TX500 (Jan 2019) — 20×20mm stack-mountable VTX with built-in microphone and VTX control.
- DVR VTX (Jan 2019) — ~600mW transmitter with integrated DVR recording.

On the digital side, SpeedyBee does not manufacture its own HD link; it integrates DJI O3 and O4 air units into its airframes (O4 Lite, O4 WA, O4 Pro variants on the Mario MiniX; O3 Air Unit on Bee25/Bee35/Master 3X). This is a deliberate dependency on DJI's digital ecosystem.

5.4 (D) Configuration hardware and the SpeedyBee App — the strategic core

Adapter line:

- USB-to-Bluetooth Adapter (Jul 2018) — the original.

- WiFi Adapter 2 (Aug 2020) — WiFi-based FC configuration from a phone; required manual WiFi network switching in phone settings.
- Adapter 3 (Oct 2022), US\$35.99 — the mature product. Connects to Betaflight / EmuFlight / iNav over WiFi *and* Bluetooth; handles firmware flashing, ExpressLRS configuration, and BLHeli_S/BLHeli_32 ESC configuration (none of which Adapter 2 could do); doubles as a blackbox log viewer/downloader, a battery voltage checker with LED display, and a 30W USB fast charger. Input 1–6S LiPo via XT60 or PH2.0; supports both USB-C and Micro-USB FC interfaces; Android 7.0+ / iOS 12.0+; max FC output 700mA @ 5V; fast charge to 30W/3.5A with QC3.0, AFC, FCP, SCP protocol support; 24.8 × 21.8 × 53.6mm; 35g; ABS+PC housing. Source: [Oscar Liang, 21 Oct 2022 \(https://oscarliang.com/speedybee-adapter-3/\)](https://oscarliang.com/speedybee-adapter-3/).

SpeedyBee App (iOS and Android; [App Store listing](#)

<https://apps.apple.com/cn/app/speedybee/id1150315028>); [product page](#)

<https://www.speedybee.com/speedy-bee-app/>). Tagline: "The FPV APP you must have" — configure quads "in the field, no need to bring the laptop." Connects via Bluetooth or OTG cable (Android only).

Supports Betaflight, INAV and EMU firmware ecosystems. Features:

- FC Firmware Flasher — flashes and downloads multiple FC firmware types. Boards without a built-in SpeedyBee chip need Adapter 2/3; boards with integrated Bluetooth/WiFi (F7 V2, F745, F405 V4/V5 etc.) flash directly.
- Blackbox Analyzer — reviews blackbox logs on-phone for tuning; supports Betaflight/EMU formats.
- BLHeli_S ESC Firmware Flasher.
- Camera Firmware Flasher — for analog FPV cameras, e.g. RunCam Hybrid, Racer2.
- Full configuration tabs: PID tuning, motor direction, OSD, VTX settings, setup, ports, configuration, power/battery, failsafe, receiver, flight modes, adjustments, GPS, motor test, blackbox, LED, and a CLI tab.

Versions circulating publicly include 2.4.0 (Android, [安粉丝 https://www.anfensi.com/download/429782.html](https://www.anfensi.com/download/429782.html))

and 3.0.3 (iOS, [soft112 listing https://speedybee-app-ios.soft112.com/](https://speedybee-app-ios.soft112.com/)).

5.5 (E) Complete drones and frames

Current flagship drones ([speedybee.com https://www.speedybee.com/](https://www.speedybee.com/)):

- Mario MiniX Modular O4 Lite / O4 Pro HD Digital VTX FPV Drone — \$2.90–\$249.00 configurable; base listing US\$119.00, SKU SB-MARIO-MINI25-PW-W. A full modular build-your-own system: PowerKits (Mario Mini25, Mini25 Pro, Mini3, Mini3 Pro, Mini35 Pro, Mini4 Pro in black/white/red/purple), VTX kits (O4 Lite, O4 WA, O4 Pro), motor protection parts, props (HQProp T4X2X3, HQProp T3X2.9X3, Gemfan 3630-3, HQ RT2.5X2.1X3), batteries (4S 880mAh, 4S 680mAh 150C), T700 arm kits, 7075 CNC frame colorways, F435-OX32-25A-AIO board, BeeStorm 1505 4000kv motors, illuminated shells, battery straps. AIO board is 36.5 × 36.5 × 7.0mm; recommended for 3.5"–5" cinewhoop/long-range or lightweight quads. [Product page https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/](https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/).
- Bee25 2.5" Drone HD O3 Air Unit FPV — \$11.99–\$399.99; "just 249g, barely the size of your palm"; 122g±2g PNP / 124g±2g ELRS-TBS bare weight; 120mm wheelbase; STM32F405; 35A 8-bit Bluejay ESC; 1404-V2-4600KV motors; ICM-42688-P gyro; AT7456E OSD; 8MB blackbox; HQProp DT63mm tri-blade; XT30; 4S 16.8V (650–1100mAh recommended); dual air-intake cooling; optional aluminium-alloy head for crash resistance; "Meteor LED 2.0" status system signalling errors, RTH, low voltage and lost-drone alerts; wireless tuning via app; exclusive Bluetooth antenna; open-source 3D-printable parts. Variants: TBS / ELRS / PNP, plus upgraded O4P ELRS and O4P PNP, plus alloy head upgrade kits. [Product page https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/](https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/); [review, 21 Oct 2024 https://oscarliang.com/speedybee-bee25-cinewhoop/](https://oscarliang.com/speedybee-bee25-cinewhoop/).
- Master 3X Modular O3/O4 Pro HD Digital VTX FPV Drone — \$139.99–\$429.99; modular 3–3.6" platform with individually replaceable top/middle/bottom plates and arms, a Type-C extension module, and a bundled 3-in-1 screwdriver ([Pyrodrone https://pyrodrone.com/collections/speedybee-products](https://pyrodrone.com/collections/speedybee-products)).
- Bee35 3.5" Drone HD O3 Air Unit FPV — \$169.99–\$429.99; also sold as Bee35 Analog 6S CineWhoop with RunCam Phoenix SE camera (\$377.99–\$390.99) and Bee35 PRO HD O3 Air Unit 6S (\$586.99–\$611.99). [Review, 27 Dec 2023 https://oscarliang.com/speedybee-bee35-cinewhoop/](https://oscarliang.com/speedybee-bee35-cinewhoop/).

- Master 5 V2 BNF — 226mm wheelbase; 5mm arms, 2.5mm top/bottom plates; SpeedyBee F7 V3 FC (STM32F722, BMI270); 50A 4-in-1 BLHeli_32 (fw 32.9); Emax ECO2 2306 1900KV 6S motors; Gemfan MCK Hurricane 51466 tri-blade props; RunCam Phoenix 2 SE camera (analog version); SpeedyBee TX800 VTX; receiver options TBS Crossfire Nano / ExpressLRS / PNP 4-pin; 426g dry, 625g with a 6S 1050mAh pack; onboard WiFi/Bluetooth; 500MB blackbox; buzzer; LEDs; GPS holder (BZ-181 M10 GPS option added post-launch); ~4–5 min flight time; Betaflight 4.4.0, target SPEEDYBEEF7V3. DJI O3 variant costs \$280 more than analog — which Liang judged poor value. [Review, 19 Jul 2023](#) (<https://oscarliang.com/speedybee-master-5-v2-drone/>).
- Mario 5 — budget 5" BNF and frame; the frame was positioned against the TBS Source One. [Frame review, 28 Feb 2024](#) (<https://oscarliang.com/speedybee-mario-5-frame/>); [drone review, 11 Nov 2024](#) (<https://oscarliang.com/speedybee-mario-5/>).
- Mario Fold 8 — foldable 8" long-range platform, frame or BNF, US\$127.99 for the frame kit. [Review, 30 Dec 2023](#) (<https://oscarliang.com/speedybee-mario-fold-8/>).
- Flex25 — micro whoop frame, "possibly the smallest 2.5" whoop ever existed." [Review, 24 Jul 2022](#) (<https://oscarliang.com/speedybee-flex25/>).

Frames and frame parts ([Pyrodrone catalog](https://pyrodrone.com/collections/speedybee-products/) (<https://pyrodrone.com/collections/speedybee-products/>)): Master 5 HD 5" Frame Kit w/ 200mm coaxial cable (\$98.99); Master 5 V2 Frame (\$79.99); Master3X Modular Frame 3–3.6" (\$41.99–\$51.99) with replacement bottom (\$3.99), middle (\$5.99), top (\$4.99) plates, arms (\$5.99) and Type-C extension module (\$2.99); Mario 5 DC/XH frames in O4 Advanced (\$50.99) and O4 Pro (\$71.99) versions with individual arms (\$7.99), plates (\$5.99–\$7.99) and Upgrade Kits A (\$12.99) and B (\$22.99); Bee25 Wireless Tuning Frame — Pro (\$46.99) and Standard (\$35.99); Bee35 3.5" Frame Kit (\$48.99) and Bee35 PRO Kit (\$50.99); Mario Fold 8 DC Long Range Frame Kit (\$127.99).

5.6 (F) Motors, power and accessories

Motors ([Pyrodrone](https://pyrodrone.com/collections/speedybee-products/) (<https://pyrodrone.com/collections/speedybee-products/>), [speedybee.com](https://www.speedybee.com/) (<https://www.speedybee.com/>)):

- 1404 4600KV (Bee25 2.5"), 4-pack \$63.99
- 2006 1950KV (Bee35 3.5"), 4-pack \$86.99
- 2306.5 1800KV (Master 5 V2 5"), 4-pack \$89.99, single \$17.99
- 2807 1050KV 6S (Mario Fold 8)
- BeeStorm 1505 4000KV (Mario MiniX)

Power and field tools:

- Goggles BEC (May 2023), from \$22.99 — external power solution addressing DJI Goggles 2 battery-life limits ([review](#) (<https://oscarliang.com/speedybee-goggles-bec/>))
- Discharger for LiPo battery (Feb 2024), \$25.99 — a 60W USB-C unit that turns LiPo/Li-ion packs into portable power banks; Liang characterized it as "a power adapter" more than a discharger ([review](#) (<https://oscarliang.com/speedybee-discharger-60w/>))
- Short-Circuit Protector Smart Power Tester ([speedybee.com](https://www.speedybee.com/) (<https://www.speedybee.com/>))
- JST-XH 2-6S LiPo balance / charging extension cable, 22AWG
- Tie Down Strap for LiPo battery, \$6.99

RF and navigation:

- 5.8GHz Goggles Receiver Module, \$114.99–\$140.99 — a receiver module for goggles, *not* a goggle
- 5.8GHz Antenna V2 SMA, \$18.99
- BZGNSS BZ-181 GPS — resold/bundled GPS module

Optics and lighting:

- ND/UV/CPL Filter Set for O4 Air Unit Pro, \$25.99
- Bee35 Meteor LED V2, \$15.99–\$20.99
- Master 5 V2 LED protection tube set (\$2.99), motor wire protection tubes (\$2.99), screw kit (\$3.99), beeper + cable (\$0.99), CNC head part L+R (\$20.99), DJI O3 Air Unit antenna base (\$2.99)
- Bee25/Master 3X O4 Pro aluminium alloy head (new version)

- Bee25 multi-color injection-molded parts
- Master3X black 3-in-1 screwdriver, \$3.99
- PDB board for F405 Wing APP FC, \$19.99

5.7 Products SpeedyBee does NOT make (worth stating explicitly)

Contrary to the brief's framing, the public catalog shows no SpeedyBee-branded FPV goggles and no SpeedyBee-branded radio transmitters. What exists is:

- a Goggles BEC (powers someone else's goggles),
- a 5.8GHz Goggles Receiver Module (a module that slots into someone else's goggles),
- and RunCam cameras resold under "Partner Products."

The homepage's own category navigation is limited to FPV Drones, Frame, Stack, Motors, Accessories ([speedybee.com \(https://www.speedybee.com/\)](https://www.speedybee.com/)). The "Partner Products" category pages for [action cameras \(https://www.speedybee.com/action-camera/\)](https://www.speedybee.com/action-camera/) and [analog FPV systems \(https://www.speedybee.com/analog-camera/\)](https://www.speedybee.com/analog-camera/) exist, though the [partner-pro page \(https://www.speedybee.com/partner-pro/\)](https://www.speedybee.com/partner-pro/) currently returns "There are no products listed under this category" — suggesting the partner program is being wound down or restructured.

6. PATENTS AND INTELLECTUAL PROPERTY

6.1 Patents — activity summary (individual documents not publicly retrievable)

Count: The Chinese business-information aggregator 爱查ICP records 6 patents (专利) under 珠海市快蜂科技有限公司 in its intellectual-property section ([source \(https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html\)](https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html)). The 水滴信用 record for the same entity does not display a patent section at all.

Individual patent numbers, titles and filing dates could not be retrieved. Google Patents returns JavaScript-rendered results that do not resolve to text via fetch; its XHR query endpoint returned HTTP 503; Espacenet and PatentGuru both blocked automated retrieval (403 / 468). Retrieving the six documents requires a direct session on CNIPA's 专利检索及分析系统 (pss-system.cnipa.gov.cn) or a paid database (企查查 / 天眼查 / 智慧芽 PatSnap / Lens.org) searching applicant 珠海市快蜂科技有限公司. I am flagging this as an unresolved gap rather than fabricating numbers.

What can be said about the patent posture:

- Six patents is a low count for a 132-person hardware company with 60+ SKUs. For calibration, DJI holds patents in the thousands. This is characteristic of the FPV component sector generally, where product cycles are 6–18 months, designs are frequently open-hardware-adjacent, and enforcement against Shenzhen/Zhuhai clone manufacturers is impractical — so firms compete on cadence, price and ecosystem rather than IP exclusivity.
- Chinese filings of this profile are typically 实用新型 (utility models) and 外观设计 (design patents) rather than 发明专利 (invention patents). Utility models in the PRC are granted without substantive examination in roughly 6–12 months, are cheap, and are the standard defensive instrument for mechanical hardware. SpeedyBee's likely filings are mechanical and layout claims, not algorithmic ones.
- Probable subject matter, inferred from the product features SpeedyBee markets as proprietary:
 1. Wireless-configuration flight-controller architecture — an integrated Bluetooth/WiFi 2-in-1 chip on the FC enabling app-based configuration *and* firmware flashing without a PC. This is the company's signature claim and the most likely invention-patent candidate. ([App page \(https://www.speedybee.com/speedy-bee-app/\)](https://www.speedybee.com/speedy-bee-app/), [F7 V3 \(https://www.speedybee.com/speedybee-f7-v3-flight-controller/\)](https://www.speedybee.com/speedybee-f7-v3-flight-controller/))
 2. Anti-shock fixed stack mounting structure — the Master 5 series' stack mounting designed to isolate motor-induced vibration from the FC gyro.
 3. Modular frame architecture — the Master3X / Mario MiniX modular systems with independently replaceable plates, arms, PowerKits, VTX kits and Type-C extension modules. Design-patent territory.

4. Aluminium-alloy head / crash-protection assemblies for O3/O4 air units on Bee25 and Master 3X.
 5. Dual air-intake cooling on the Bee25.
 6. Meteor LED status-signalling system — LED-encoded error, RTH, low-voltage and lost-drone states.
 7. Multi-function adapter combining FC configuration, ESC flashing, blackbox retrieval, voltage checking and 30W USB fast charging in one 35g device (Adapter 3).
- Licensing/litigation: no patent litigation, licensing agreement, infringement action or opposition proceeding involving SpeedyBee surfaced in any source consulted.

6.2 Trademarks — documented in detail

China (Zhuhai entity): 26 trademark records under 珠海市快蜂科技有限公司 (爱查ICP (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>), 水滴信用 (<https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html>)). Two are itemized in the retrieved records:

Mark	Application No.	Filing date	Nice class	Status
SPEEDY BEE	93927839	2026-09-09	Class 28 — 健身器材 (sporting/fitness goods)	商标其他情形 (other circumstances)
快蜂	92741296	2026-07-02	Class 12 — 运输工具 (vehicles/transport)	申请中 (pending)

Both filings are from 2026 — i.e. the company is actively expanding trademark coverage right now, into Class 12 (vehicles — the correct class for drones as aircraft) and Class 28 (sporting goods — the correct class for model aircraft/toys). This is a defensive brand-consolidation push, and notably it is the Zhuhai entity filing in its own name, distinct from the RunCam-held US mark.

United States: SPEEDY BEE, Serial No. 88478719, Registration No. 6121981, filed 18 June 2019, owned by RunCam Technology (Shenzhen) Co., Ltd. Goods/services: downloadable computer software for controlling and analyzing flight controller parameter settings; vehicle radios; cases. Source: [Justia Trademarks](https://trademarks.justia.com/884/78/speedy-88478719.html) (<https://trademarks.justia.com/884/78/speedy-88478719.html>). The Justia owner-portfolio page for RunCam returned HTTP 403 and could not be enumerated.

The US/China ownership split is the clearest documentary artifact of the RunCam relationship described in §1.3, and is a genuine IP-structure risk worth noting: the Zhuhai operating company does not appear to own its own brand in its largest export market.

6.3 Other IP and domains

- ICP filing: 粤ICP备2023067832号 for speedybee.cn / www.speedybee.cn, approved 2023-06-19 (爱查ICP (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>)). Only one ICP record — so speedybee.com is hosted outside the PRC filing regime, consistent with an export-first business.
- Software copyrights (软件著作权): not displayed in either registry record retrieved. Given the App's centrality, filings likely exist but are unconfirmed.
- Open-source posture: SpeedyBee publishes open-source 3D-printable TPU parts for the Bee25 ([product page](https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/) (<https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/>)) and builds on open-source firmware (Betaflight, INAV, ArduPilot, Bluejay, BLHeli_S). Its 2025 move to closed-source OX32 ESC firmware is a notable departure — see §7.

7. LATEST NEWS AND DEVELOPMENTS (2024–2026)

7.1 The F405 V4 discontinuation and the OX32 closed-source controversy (August 2025 → ongoing)

The biggest brand-level story of the period. On 30 August 2025, SpeedyBee launched the F405 V5 OX32 stack and issued an official statement (delivered as an image, not text) confirming the discontinuation of the F405 V4 — until then arguably the single most widely deployed budget stack in 5" FPV. Price moved to \$74.99 from the V4's \$65.99. Source: [Oscar Liang](https://oscarliang.com/speedybee-f405-v5/) (<https://oscarliang.com/speedybee-f405-v5/>).

The controversy is the ESC firmware. The V5's 32-bit ESC runs OX32, a closed-source firmware. Liang, the sector's most-read reviewer, wrote that he was "not thrilled about this move," noting OX32 is "very new" and "still largely untested by the community," and drew an explicit parallel to the earlier BLHeli_32 loss of open-source status. He argued SpeedyBee could have stayed on 8-bit hardware with open-source Bluejay, since 32-bit ESCs deliver little practical benefit to most FPV pilots, and flagged as "unclear" whether OX32 hardware can be reflashed to AM32, the open-source alternative.

Secondary complaints from the same coverage and its comment thread:

- Mandatory activation: first connection via the SpeedyBee App requires activating the FC and supplying an email address — unprecedented for an FPV flight controller. (Not required if the user only ever uses Betaflight Configurator.) This is the closest thing to a data-collection story in SpeedyBee's history.
- Motor outputs cut from 8 to 4, breaking hexa/octocopter builds. One commenter (Patrick) called the stack "unusable for my octocopter" and criticized the absence of an upgrade path.
- INAV support reported missing in practice despite being advertised (commenter Chris Nugent).
- Conformal coating present but not waterproof; hand-coating of solder joints still advised.
- Competitive defection recommendations surfaced in comments (HGLRC F460, JHEMCU H7 at similar prices).

This matters strategically: SpeedyBee's entire brand equity was built on *openness plus convenience*. The V5 keeps the convenience and trades away the openness, in a community that is unusually ideological about open-source firmware. Watch whether AM32 flashing becomes possible — that single technical outcome will likely determine whether this becomes a durable reputational problem or a passing complaint.

7.2 The 54% US prepaid tariff surcharge (2025–2026, ongoing)

SpeedyBee is collecting a 54% prepaid tax on all orders shipped to US addresses, in response to new US tariff policy, refundable on presentation of an official duty receipt showing a lower actual assessment. This notice is live across product pages including the [Mario MiniX](https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/)

(<https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/>) and [Bee25](https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/) (<https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/>), and is corroborated by retailer commentary (SoarSky (<https://soarskyrc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-specs-amp-speedybee-full-lineup-comparison>)). For a price-led brand this is the most consequential commercial development of the period and materially disadvantages SpeedyBee's direct channel versus US-stocked distributors.

7.3 Move to DJI O4 and modular architecture (2025–2026)

The current product generation is organized around DJI O4 air units (O4 Lite, O4 WA, O4 Pro) and around modularity as a design philosophy. The four "new"-tagged products on the homepage as of this research are all from this wave ([speedybee.com](https://www.speedybee.com) (<https://www.speedybee.com/>)):

- Mario MiniX Modular O4 Lite/O4 Pro (\$2.90–\$249.00 configurable)
- Bee25 2.5" HD O3 Air Unit (\$11.99–\$399.99)
- Master 3X Modular O3/O4 Pro (\$139.99–\$429.99)
- Bee35 3.5" HD O3 Air Unit (\$169.99–\$429.99)

The modular approach — individually purchasable plates, arms, PowerKits, VTX kits, motor guards, shells, and colorways down to \$0.99 spare beepers — is a recurring-revenue play: sell the platform once, sell crash replacements forever. It also neatly dodges the tariff problem on complete-aircraft imports by moving volume into low-value spare parts.

7.4 Domestic silicon migration

The Mario MiniX ships an F435-OX32-25A-AIO board, and Chinese community build guides explicitly highlight "国产AT32飞控" (domestic AT32 flight controller) SpeedyBee builds ([ChinaFlier forum](https://uav.chinaflir.com/thread-34960-1-1.html) (<https://uav.chinaflir.com/thread-34960-1-1.html>)). The shift from STMicroelectronics STM32 to Artery AT32 parts is a supply-chain and cost story with geopolitical undertones, and it is happening quietly across the Chinese FPV sector.

7.5 Fixed-wing expansion matured

Having entered fixed-wing in December 2023 with the [F405 WING APP](https://oscarliang.com/speedybee-f405-wing-app/) (<https://oscarliang.com/speedybee-f405-wing-app/>) ("currently the cheapest wing FC you can get"), SpeedyBee now fields two wing FCs (WING APP and WING MINI) running INAV/ArduPilot, and the WING MINI is distributed in China as a development board ([Spotpear](https://spotpear.cn/shop/SpeedyBee-F405-WING-MINI-Flight-Controller-Driver-Board-FC-ArduPilot-INAV-DJI-RC-FPV-Fixed-Wing-Airplane-Drone-Quadcopter.html) (<https://spotpear.cn/shop/SpeedyBee-F405-WING-MINI-Flight-Controller-Driver-Board-FC-ArduPilot-INAV-DJI-RC-FPV-Fixed-Wing-Airplane-Drone-Quadcopter.html>)). This diversifies the company beyond multirotor and into the ArduPilot community — a different, more technical, less price-sensitive audience.

7.6 Hiring into AI

The AI Application Engineer posting (2026-08-13, RMB 10–15k/month, [爱查ICP](https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html) (<https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>)) among 81 open roles is the clearest forward-looking signal in the dataset. For a company whose differentiator has always been software-mediated setup, the plausible read is AI-assisted tuning, automated blackbox diagnosis, or conversational configuration in the app — extending the original 2018 "leave the laptop at home" thesis.

7.7 Component-level trend context

The ICM-42688-P gyro (32kHz sampling, lower noise floor than MPU6000) has become the 2025–2026 standard, and SpeedyBee has adopted it across F405 AIO V2, F405 V4, F405 V5, WING APP and WING MINI ([SoarSky](https://soarskyrc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-specs-amp-speedybee-full-lineup-comparison) (<https://soarskyrc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-specs-amp-speedybee-full-lineup-comparison>), [UAVMODEL 2026 guide](https://blog.uavmodel.com/top-5-fpv-flight-controllers-in-2026-betaflight-vs-inav-vs-ardupilot/) (<https://blog.uavmodel.com/top-5-fpv-flight-controllers-in-2026-betaflight-vs-inav-vs-ardupilot/>)). SpeedyBee remains a fast follower on silicon rather than a leader.

7.8 Quality perception — the persistent counter-narrative

Not all coverage is favorable, and it would be misleading to omit this. Reviewer comment threads contain repeated durability complaints: one commenter on the Master 5 V2 review alleged over \$1,500 in losses from SpeedyBee electronics failures and advised buying other brands ([Oscar Liang](https://oscarliang.com/speedybee-master-5-v2-drone/) (<https://oscarliang.com/speedybee-master-5-v2-drone/>)); another on the V5 piece cited receiver compatibility problems and frame quality ([Oscar Liang](https://oscarliang.com/speedybee-f405-v5/) (<https://oscarliang.com/speedybee-f405-v5/>)). These are third-party comments, not reviewer assessments — the reviewers themselves have been broadly positive on value. The pattern is nonetheless a real brand risk for a company competing on price. Counterbalancing it: SpeedyBee demonstrably responds to feedback, e.g. adding a BZ-181 M10 GPS option to the Master 5 V2 after launch in response to a reader request (*ibid.*).

8. RESEARCH GAPS AND CONFIDENCE NOTES

State these plainly rather than papering over them:

1. The six patent documents were not retrievable. Count confirmed (6, via [爱查ICP](https://www.aichaicp.com)); numbers, titles and dates require a CNIPA session or a paid database. Everything in §6.1 about subject matter is labelled inference.
2. The Chinese About Us page (speedybee.cn/about-us.html) is indexed but returns 404 live. A Wayback snapshot exists (20260316085427) but web.archive.org was not reachable from this environment. The company's own self-told history remains unread.
3. Crunchbase and GetFPV returned HTTP 403; their data is included only as summarized by the search index, not verified directly.
4. Tracxn's "founded 2005" is demonstrably wrong and should not be propagated. Likewise its unnamed founders field.
5. The full 26-trademark Chinese portfolio was not enumerated — only two examples were exposed by the aggregator.
6. The RunCam corporate relationship is documented but not fully characterized. The shared trademark, shared components and shared launch cadence are facts; a formal

parent/subsidiary/common-ownership structure is an inference and has not been confirmed in any registry.

7. Revenue, margin, unit volumes and channel mix are entirely unknown and cannot be derived from available sources.
8. Web search quota was exhausted mid-research (200/200 calls), so later gap-filling relied on direct fetches only. Remaining avenues if this needs extending: CNIPA patent search, 企查查/天眼查 paid records, Wayback via a reachable mirror, the SpeedyBee YouTube channel's announcement videos, and USPTO TSDR for the full RunCam mark file wrapper.

9. SOURCE INDEX

Company-controlled:

- <https://www.speedybee.com/> — homepage, catalog, taglines
- <https://www.speedybee.com/about-us/> — mission statement (full text)
- <https://www.speedybee.com/flight-controller/> — FC lineup and pricing
- <https://www.speedybee.com/speedy-bee-app/> — app features
- <https://www.speedybee.com/speedybee-f7-v3-flight-controller/>
- <https://www.speedybee.com/speedybee-bee25-2-5-inch-drone-hd-o3-air-unit-fpv/>
- <https://www.speedybee.com/speedybee-mario-minix-modular-o4-lite-o4-pro-hd-digital-vtx-fpv-drone/>
- <https://www.speedybee.com/partner-pro/> , /action-camera/ , /analog-camera/
- <https://www.speedybee.com/runcam-5-orange/> , /runcam-thumb/
- <https://www.speedybee.cn/> (ICP 粤ICP备2023067832号)
- <https://www.youtube.com/@speedybeefpv1088>
- <https://apps.apple.com/cn/app/speedybee/id1150315028>

Registry and IP:

- <https://m.shuidi.cn/company-91c66a46b7a78c6a34bf7fb14f7c1901.html>
- <https://www.aichaicp.com/cd-91c66a46b7a78c6a34bf7fb14f7c1901.html>
- <https://trademarks.justia.com/884/78/speedy-88478719.html>

Databases:

- https://tracxn.com/d/companies/speedybee/_tadsCzWnNqxyzY6kybm9CSBAr75hDILWtIDm-lkDTNdM
- <https://www.crunchbase.com/organization/speedybee>

Independent reviews and retail:

- <https://oscarliang.com/?s=speedybee> (and /page/2/, /page/3/) — plus individual reviews: speedybee-f405-v5, speedybee-f405-v4, speedybee-f405-v3, speedybee-f7-v3, speedybee-f7-v2, speedybee-f405-aio-fc, speedybee-f405-mini-fc-stack, speedybee-f405-wing-app, speedybee-f405-wing-mini-fc, speedybee-adapter-2, speedybee-adapter-3, speedybee-tx800, speedybee-tx1600-ultra-vtx, speedybee-goggles-bec, speedybee-discharger-60w, speedybee-master-5-v2-drone, speedybee-mario-5, speedybee-mario-5-frame, speedybee-mario-fold-8, speedybee-bee25-cinewhoop, speedybee-bee35-cinewhoop, speedybee-flex25, iflight-blitz-atf435-speedybee-f405-v4
- <https://soarskyrc.com/blogs/technology-sharing/fpv-flight-controller-complete-guide-how-it-works-key-pecs-amp-speedybee-full-lineup-comparison>
- <https://pyrodrone.com/collections/speedybee-products>
- <https://newbeedrone.com/collections/speedybee>
- <https://www.kiwiquads.co.nz/brand/speedybee/>
- <https://blog.uavmodel.com/top-5-fpv-flight-controllers-in-2026-betaflight-vs-inav-vs-ardupilot/>
- <https://uav.chinaflier.com/thread-34960-1-1.html>
- <https://spotpear.cn/shop/SpeedyBee-F405-WING-MINI-Flight-Controller-Driver-Board-FC-ArduPilot-INAV-DJI-RC-FPV-Fixed-Wing-Airplane-Drone-Quadcopter.html>

Three corrections carried through the whole report: HQ is Zhuhai, not Shenzhen; the company is unfunded with zero rounds, so §2 and §3 are short by necessity, not by omission; and SpeedyBee does not make goggles — only goggle power and receiver accessories.

RESEARCH REPORT: Starlogic Intelligent Technology (星逻智能 / SKYSYS)

0. METHODOLOGY AND A CRITICAL NAMING NOTE — READ FIRST

"Starlogic Intelligent Technology" is the English name used by Western startup databases (principally Tracxn) for the Chinese company 星逻智能科技（苏州）有限公司 — *Xingluo Zhineng Keji (Suzhou) Co., Ltd.* The name is a direct calque: 星 (xing, "star") + 逻 (luo, "logic").

The company does not market itself as "Starlogic" in English. Its own English-language brand is SKYSYS (skysys.cn). If you search only for "Starlogic Intelligent Technology" you will find essentially one thin database entry; the substantive record lives under 星逻智能 and SKYSYS.

Explicit caveat on English-language availability: very little reliable English-language material exists on this company. The single English aggregator profile (Tracxn) contains several claims that are flatly contradicted by primary Chinese sources and by the company's own website — see Section 8. The bulk of this report is sourced from Chinese-language tech press, the company website, and Chinese venture databases. Where figures conflict, I state both and flag the conflict rather than reconciling them silently.

Sources I attempted and could not access (so you know what is missing rather than assuming it does not exist): Google Patents (HTTP 503), Espacenet (HTTP 403), FreePatentsOnline (connection reset), USPTO legacy search (DNS failure), Qichacha company registry (blocked). As a result, Section 6 gives a patent *activity summary* with aggregate counts from secondary sources but no individual patent numbers or titles — I could not verify any, and I am not going to invent them.

1. COMPANY OVERVIEW

1.1 Legal identity

- Chinese legal name: 星逻智能科技（苏州）有限公司
- English brand: SKYSYS (registered marks appear as UltraHive®, SkyCab®, SkyScout®)
- English name in Western databases: Starlogic Intelligent Technology; also transliterated "Xingluo Intelligent Technology"
- Entity type: 有限责任公司（自然人投资或控股）— limited liability company, natural-person invested/controlled
- Date of incorporation: 19 July 2017 ([news.qq.com](https://news.qq.com/rain/a/20240606A07RGP00) (<https://news.qq.com/rain/a/20240606A07RGP00>), [innohere](https://www.innohere.com/ir/111839/finance.html) (<https://www.innohere.com/ir/111839/finance.html>))
- Legal representative / 实控人 (ultimate controller): 王海滨 Wang Haibin
- Registered capital: RMB 4,166.880 (416.688万元) ([news.qq.com](https://news.qq.com/rain/a/20240606A07RGP00) (<https://news.qq.com/rain/a/20240606A07RGP00>))
- ICP licence: 苏ICP备17054692号-1
- Website: <https://www.skysys.cn> (Chinese, English, Spanish, French, Arabic versions)

1.2 Founding story

The company was founded in 2017 by three people who met in industry, two of them at General Electric. The origin story, told in detail by The Paper ([thepaper.cn](https://www.thepaper.cn/newsDetail_forward_21336616) (https://www.thepaper.cn/newsDetail_forward_21336616)), is that co-founders 钱茂冬 (Qian Maodong) and 王海滨 (Wang Haibin) were on an overseas business trip and observed a contradiction: mining haul trucks had already been automated, yet drone operations still depended entirely on human pilots, making them expensive and non-scalable. They identified 2016 as the inflection point at which drones began migrating from consumer/entertainment products into industrial production settings — chemicals, power, security, environmental inspection — and judged the enabling layer to be a market worth "hundreds of billions" (千亿级) of RMB.

They left GE and began operating as a project team within three months. Dr. 肖素枝 (Xiao Suzhi) joined roughly three months after that, supplying the software and algorithms capability the two hardware engineers lacked.

1.3 Strategic positioning — the defining decision

The single most important strategic fact about this company is what it refuses to build. Co-founder Qian Maodong has stated the position bluntly ([thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616)):

"如果我们做无人机，相当于站在了所有无人机的对立面" — *"If we made drones, we would effectively be standing in opposition to every drone maker."*

SKYSYS therefore builds everything around the aircraft but never the aircraft: docking stations/hangars, automatic charging and battery-swap systems, onboard AI compute modules, autonomous-flight middleware, cloud dispatch, and vertical AI inspection software. Its hardware is deliberately compatible with mainstream commercial airframes — DJI M30, M350 RTK, M3D/M3TD and others.

The stated ambition is to become infrastructure, not a product vendor: Qian has described the goal as building "全国最大的无人机低空调度网络" — the country's largest low-altitude drone dispatch network — and has analogised the hangar network to petrol stations for ground vehicles.

A key technical differentiator claimed is 纯网联技术 ("pure network-connected technology"): the drones connect over cellular networks rather than depending on a ground radio link to a local controller. This allows point-to-point missions — take off from Hangar A, land at Hangar B — rather than the out-and-back-only pattern that radio-tethered competitors are limited to. For transmission-line inspection, this enables "leapfrog" operation between hangars spaced along a power corridor ([skysys.cn \(https://www.skysys.cn/DockingStation/\)](https://www.skysys.cn/DockingStation/), [thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616)).

1.4 Mission and current framing

The company's self-description has evolved. Early framing (2018-2021) was generic drone automation and smart city. Current framing, post-2023, is explicitly new-energy focused:

"以AI驱动无人机与机器人完成能源检测与运维闭环，提升发电效率" — *driving drones and robots with AI to close the loop from energy inspection through to O&M; raising generation efficiency.* — Wang Haibin ([stcn.com \(https://www.stcn.com/article/detail/1224287.html\)](https://www.stcn.com/article/detail/1224287.html))

Wang has articulated the underlying market thesis: compared with building new power stations, raising the output efficiency of the existing installed base is the cheaper and faster path — which is the entire commercial premise of the inspection-plus-cleaning-robot product line.

The company describes itself as "国内头部的低空智能化系统供应商" (a leading domestic supplier of low-altitude intelligent systems) and claims to be "国内首家出口海外的无人机自动化供应商" — the first Chinese drone-automation supplier to export overseas ([36kr pitchhub \(https://pitchhub.36kr.com/project/1678486488167426\)](https://pitchhub.36kr.com/project/1678486488167426)).

1.5 Headquarters and footprint

Headquarters: Suzhou Industrial Park, Jiangsu Province — AI Industrial Park (人工智能产业园) E1, Jinji Lake Avenue. An earlier registered address was 苏州工业园区星湖街328号创意产业园4-B604 ([kchuhai.com \(https://www.kchuhai.com/company/detail.aspx?id=5457\)](https://www.kchuhai.com/company/detail.aspx?id=5457)). Front desk: 0512-87816318; hotline 400-860-8819.

Offices / centres listed by the company ([skysys.cn/AboutSkysys \(https://www.skysys.cn/AboutSkysys/\)](https://www.skysys.cn/AboutSkysys/)):

Location	Function
Suzhou, Jiangsu	Headquarters; business and hardware R&D; centre
Shanghai Lingang (临港)	R&D; centre
Shanghai Jinqiao / Pudong	Office
Nanjing, Jiangsu (Pukou)	R&D; centre

Location	Function
Hangzhou, Zhejiang	R&D; centre
Beijing (Chaoyang)	Office
Hohhot, Inner Mongolia	Marketing centre
Qingyuan, Guangdong	R&D; centre
Fuzhou, Fujian (Minhou)	Office
Xuzhou, Jiangsu (Quanshan)	Office

Note the geographic logic: Hohhot sits in China's largest wind and solar build-out region; Qingyuan and Xuzhou are distributed-PV heavy; Lingang is the origin smart-city deployment site.

1.6 Headcount

Reported figures, in chronological order — the trend is more informative than any single number:

- ~100+ people, 70% R&D; — The Paper, 2023 ([thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616))
- ~160 people, 70% R&D; — multiple June 2024 reports ([news.qq.com \(https://news.qq.com/rain/a/20240606A07RGP00\)](https://news.qq.com/rain/a/20240606A07RGP00), [36kr pitchhub \(https://pitchhub.36kr.com/project/1678486488167426\)](https://pitchhub.36kr.com/project/1678486488167426))
- 50–150 — kchuhai directory (stale range)
- "fewer than 50" — innohere (clearly stale)
- "3 employees as of 30 June 2026" — Tracxn. This figure is not credible and should be disregarded: it is contradicted by the company's own office footprint, a nine-figure B+ round in 2025, and continuing 2026 product launches and conference activity.

Team academic provenance cited by the company: Zhejiang University, Shanghai Jiao Tong University, Durham University (UK), University of California, University of Illinois, Georgia Tech, Tongji University, and the Chinese Academy of Sciences ([leaderobot.com \(https://www.leaderobot.com/news/4321\)](https://www.leaderobot.com/news/4321)).

1.7 Corporate timeline

Year	Event
2017	Founded (19 July); enters drone-AI industry
2018	Builds what it claims was the largest drone hangar cluster then known, in Shanghai Lingang — 7 hangars (later 9) covering the 95 km ² core urban area; first to apply hangars to large-scale PV plant inspection; wins MIIT "绽放杯" national 5G prize
2019	Named one of "China's Top 10 AI Innovation Enterprises"; top-3 in the transportation track of WAIC's AIWIN competition
2020	Series A; wins Huawei Cloud Cup AI competition enterprise track; enters "Global Top 50 Drone Companies" list
2021	A+ and A++ rounds; co-drafts IEEE 1937.1 with Huawei and DJI — described as the world's first international standard for drone payload systems; begins Lantern cleaning-robot R&D; refocuses on three verticals (smart city, grid, PV)
2022	Launches UltraHive Mk4 Pro charge/swap hangar; releases 星曜取光 PV platform; rolls out four vertical software products (wind, solar, grid, emergency)
2023	Launches Skysys Lantern PV-panel cleaning robot; begins building out a robotics product line
2024	B round (>RMB 100M, June); Lantern wins two international awards at the 8th World Drone Congress; extends from low-altitude *inspection* into low-altitude *O&M*;
2025	B+ round (>RMB 100M, June); named national-level 专精特新 "little giant"; wins China Innovation & Entrepreneurship Competition low-altitude final; WISE2025 and 金翼奖 recognitions
2026	Named to Suzhou unicorn-cultivation list; MoU with France's JPA Group; Lantern 小蓝 wins German Red Dot and French FDA design awards; chief scientist named a national "最美职工"

2. FINANCIALS

Bottom line: SKYSYS is a private, venture-backed company. No revenue figures, gross margins, profitability, or valuations have been publicly disclosed in any source I could access, Chinese or English. There are no financial filings because it is not listed on any exchange and no IPO process has been announced.

What *is* on the public record:

Registered capital: RMB 4,166,880 ([news.qq.com \(https://news.qq.com/rain/a/20240606A07RGP00\)](https://news.qq.com/rain/a/20240606A07RGP00)). Note that in Chinese corporate practice registered capital is a legal formality and bears no relationship to capital actually raised — this company has raised orders of magnitude more than its registered capital.

Ownership: Equity-penetration analysis reported by Tencent News shows Wang Haibin as the ultimate controller with an ultimate beneficial stake of 22.3501% ([news.qq.com \(https://news.qq.com/rain/a/20240606A07RGP00\)](https://news.qq.com/rain/a/20240606A07RGP00)). No other shareholder percentages have been disclosed. A ~22% founder stake after eight rounds is consistent with a heavily institutionally-financed cap table.

Cumulative capital raised (estimate): Aggregating the disclosed and banded round sizes in Section 3 produces roughly RMB 300 million or more (very approximately USD 40–45 million at prevailing rates). This is an estimate, not a disclosed figure, because five of the eight rounds were announced only in bands ("数千万" = tens of millions; "超亿元" = over one hundred million).

Valuation: Not disclosed for any round. Tracxn's valuation fields were masked/redacted. However, two circumstantial data points bracket it upward:

- Inclusion in the 《2025年度苏州市"独角兽"培育企业名单》 (Suzhou 2025 unicorn-*cultivation* enterprise list), announced January 2026 ([skysys.cn/News \(https://www.skysys.cn/News/\)](https://www.skysys.cn/News/)). Note carefully: "培育" means *cultivation* — this is a pipeline list of companies deemed capable of *becoming* unicorns, not a certification that the company is valued above USD 1 billion. Do not report this as unicorn status.
- Two consecutive nine-figure RMB rounds (B and B+) within twelve months, with strategic and state-linked investors participating.

Revenue signals (indirect only): The Paper reported in 2023 that the 2021 refocus onto three verticals produced "台阶式增长" — step-change growth — in both revenue and headcount, but gave no figures ([thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616)). The business-model shift disclosed at the B round is material for revenue quality: the wind product 御风 is moving to a "产品即服务与服务分润" model — product-as-a-service with revenue sharing — via franchise partnerships with wind O&M providers, which implies a transition from one-off hardware sales toward recurring service revenue ([stcn.com \(https://www.stcn.com/article/detail/1224287.html\)](https://www.stcn.com/article/detail/1224287.html)).

Tracxn's "\$1.49M total raised" figure is wrong and should not be used. It omits the 2024 B round and 2025 B+ round entirely, each of which alone exceeded RMB 100 million (~USD 14M). Tracxn's own news section even cites the June 2024 "over 100 million yuan" report while the aggregate field contradicts it.

3. FUNDING HISTORY

Eight disclosed transactions between 2018 and 2025. The consolidated table below is drawn primarily from [innohere \(https://www.innohere.com/ir/111839/finance.html\)](https://www.innohere.com/ir/111839/finance.html) and cross-checked against [36kr pitchhub \(https://pitchhub.36kr.com/project/1678486488167426\)](https://pitchhub.36kr.com/project/1678486488167426), [leaderobot \(https://www.leaderobot.com/news/4321\)](https://www.leaderobot.com/news/4321), [stcn \(https://www.stcn.com/article/detail/1224287.html\)](https://www.stcn.com/article/detail/1224287.html), [chinaerospace \(https://www.chinaerospace.com/article/24066\)](https://www.chinaerospace.com/article/24066), and [aliyun startup \(https://startup.aliyun.com/info/102033.html\)](https://startup.aliyun.com/info/102033.html).

#	Date	Round	Amount (RMB)	Investors
1	2018-07-06	Angel (天使轮)	10,000,000	邀问创投 (CM Venture Capital)
2	2019-01-25	Equity financing (股权融资)	Undisclosed	中关村发展启航产业投资基金 (Zhongguancun Development Qihang Industrial Investment Fund)
3	2019-03-11	Pre-A	千万级 (tens of millions)	Lead: 远瞻资本 (Lighthouse Capital); 邀问创投
4	2020-04-29	Series A	数千万 (several tens of millions)	常春藤资本 (Ivy Capital), 邀问创投, 远瞻资本
5	2021-03-30	Series A+	数千万	Lead: 启航投资 (Zhongguancun Qihang); 邀问创投, 常春藤资本, 远瞻资本
6	2021-11-12	Series A++	数千万	元禾控股 (Oriza/Yuanhe Holdings), 中关村发展启航基金, 远瞻资本

#	Date	Round	Amount (RMB)	Investors
7	2024-06-03	Series B	超亿元 (>100M)	Co-leads: 广东粤科母基金 (Guangdong Yueke Fund-of-Funds), 临港前沿阿特斯扬州基金 (Lingang Qianyan / Arctech Yangzhou Fund); follow-on: 宁波诺登创投 (Ningbo Nuodeng), 苏州工业园区科创基金 / 苏州领军创投; angel investors adding on: 遛问创投, 远瞻资本, 常春藤投资; 粤科金融集团
8	2025-06-23	Series B+	超亿元 (>100M)	苏州港航集团 (Suzhou Port & Shipping Group), 临港前沿投资

3.1 Investor name mapping (Chinese ↔ English)

This mapping matters because English databases list the English names and Chinese press lists the Chinese names, making the same investor appear twice:

- 遛问创投 = CM Venture Capital
- 远瞻资本 = Lighthouse Capital Management (sometimes "Yuanjian Capital")
- 常春藤资本 = Ivy Capital (Tracxn: "Shanghai Ivy Capital")
- 元禾控股 = Oriza Holdings (Tracxn: "Yuanhe Holdings") — the Suzhou Industrial Park state investment platform
- 中关村发展启航产业投资基金 = Zhongguancun Development Qihang Fund (Beijing state-linked)
- 广东粤科母基金 = Guangdong Yueke Fund-of-Funds (Guangdong provincial state capital)
- 苏州港航集团 = Suzhou Port & Shipping Group (municipal SOE)

3.2 Reading the cap table

Three observations worth flagging:

(a) Extraordinary insider persistence. 远瞻资本 (Lighthouse) participated in the Pre-A, A, A+, A++ and B rounds — five consecutive rounds. 遛问创投 (CM Venture) led the angel and then followed in Pre-A, A, A+ and B. 常春藤 (Ivy) joined at A and was still adding at B. Angel-round investors doubling down at a nine-figure B round six years later is a meaningful positive signal.

(b) Progressive shift to state and strategic capital. The early rounds are pure financial VC. From the A+ onward, state-linked vehicles appear (Zhongguancun Qihang, Oriza/Suzhou Industrial Park), and by B and B+ the leads are provincial state fund-of-funds (Guangdong Yueke), a solar-industry strategic (the Arctech / 阿特斯 Yangzhou fund — Arctech is a major PV tracker manufacturer, so this is a customer-adjacent strategic), and a municipal SOE (Suzhou Port & Shipping). This pattern is typical of Chinese hard-tech companies approaching pre-IPO positioning.

(c) The Arctech tie is strategically significant. 临港前沿阿特斯扬州基金 links SKYSYS to the PV tracker/module supply chain — precisely the installed base its inspection and cleaning robots service.

3.3 Investor commentary on the B round

- 曾漳 (Zeng Wei), GM, Guangdong Yueke Fund-of-Funds: as new-energy installed capacity rises rapidly, intelligent O&M; is becoming an emerging blue-ocean track; called the team "务实、高效" (pragmatic and efficient) and "引领这个新兴赛道的新锐力量."
- 单秀红 (Shan Xiuhong), Chairman, Lingang Qianyan Investment: praised the company's accumulation of "大量飞行数据和应用经验" — a large body of flight data and application experience — and described it as a representative low-altitude-economy enterprise.
- 章钊 (Zhang Zhao), Partner & GM, Ningbo Nuodeng: described SKYSYS as the company that first achieved drone inspection requiring no manual operation, solving the sector's high-cost/low-efficiency pain point, with the PV cleaning robot adding a value-added service layer — the stated reason for following on after the A+ round.
- 陈希麟 (Chen Xilin), Suzhou Lingjun Venture Capital: expects the company to become "行业里的创新者、开拓者及引领者."

(All quotes from [stcn.com](https://www.stcn.com/article/detail/1224287.html) (<https://www.stcn.com/article/detail/1224287.html>) and [chinaerospace.com](https://www.chinaerospace.com/article/24066) (<https://www.chinaerospace.com/article/24066>).)

3.4 Use of proceeds

- B round (2024): primarily mass production and technical iteration of the Lantern PV cleaning robot, specifically the goal of a "flying" cleaning robot capable of working on 45° -tilt arrays, completing the loop "从低空智能检测向低空智能运维" (from low-altitude intelligent inspection to low-altitude intelligent O&M);. Secondly, moving the 御风 wind-inspection product into accelerated service-model rollout.
- B+ round (2025): use of proceeds not disclosed in the sources I could reach.

4. LEADERSHIP AND PEOPLE

4.1 王海滨 — Wang Haibin — Founder, Chairman, CEO, legal representative

- Education: B.Eng. in Automation, Zhejiang University; M.Eng. in Vehicle Engineering, Shanghai Jiao Tong University
- Prior role: R&D; Director at GE 科技学社 (GE's technology institute/academy) ([news.qq.com](https://news.qq.com/rain/a/20240606A07RGP00) (<https://news.qq.com/rain/a/20240606A07RGP00>))
- Domain expertise: industrial drone applications, and specifically battery management and charging technology — this is the technical seed of the whole company, since SKYSYS entered the market through the charging/battery-swap module rather than through aircraft or software
- Prior engineering work: led battery management system design for Segway's two-wheeled vehicles, and sodium-salt (sodium-nickel-chloride) battery systems ([36kr pitchhub](https://pitchhub.36kr.com/project/1678486488167426) (<https://pitchhub.36kr.com/project/1678486488167426>))
- Experience: variously reported as 16 years ([36kr pitchhub](https://pitchhub.36kr.com/project/1678486488167426) (<https://pitchhub.36kr.com/project/1678486488167426>)) and "nearly 20 years" ([leaderobot.com](https://www.leaderobot.com/news/4321) (<https://www.leaderobot.com/news/4321>)) in drone-related industries
- Ownership: ultimate beneficial stake of 22.3501%; ultimate controller of the company
- Public positioning: frames the company around China's "3060" dual-carbon goals and rising electricity demand from computing/AI growth; argues that efficiency gains on the existing generation fleet beat new-build economics

4.2 钱茂冬 — Qian Maodong — Co-founder

- Born: 1991
- Education: B.Eng. Mechatronic Engineering, Zhejiang University (2013); M.Sc. Communications Engineering, Durham University, UK (2016)
- Prior roles: Teradyne and GE, working on analogue/digital circuit hardware development
- Expertise: mechanical design, small aircraft development, multiple programming languages; holds several Chinese national invention patents in his own name
- Role: co-leads hardware technology direction with Wang Haibin; has historically been the company's principal external spokesperson — the "we will never build the aircraft" strategic doctrine is articulated by him
- Source: [thepaper.cn](https://www.thepaper.cn/newsDetail_forward_21336616) (https://www.thepaper.cn/newsDetail_forward_21336616)

4.3 肖素枝 博士 — Dr. Xiao Suzhi — Co-founder and Chief Scientist

- Joined approximately three months after the other two founders, filling the software and algorithms gap on an otherwise hardware-heavy founding team
- Built and leads the team that developed the company's AI algorithm engine
- Personally led delivery of the Shanghai Lingang smart-city project, for which she received the 2021 Shanghai May Day Labour Medal (上海市五一劳动奖章)
- April 2026: named one of the national top ten "最美职工" ("Most Beautiful Workers") — a high-profile national honour conferred by the All-China Federation of Trade Unions ([skysys.cn/News](https://www.skysys.cn/News) (<https://www.skysys.cn/News/>))

- Current title: Chief Scientist (首席科学家)
- Active as the company's technical public voice in 2026: keynote at the PSGEC power conference (24 Aug 2026); interview published 14 Sept 2026 headlined **"无人机只是'手脚'，新能源运维真正拼的是AI'大脑'"** — **"Drones are only the hands and feet; what new-energy O&M; really competes on is the AI 'brain'"**
- Sources: [thepaper.cn](https://www.thepaper.cn/newsDetail_forward_21336616) (https://www.thepaper.cn/newsDetail_forward_21336616), [skysys.cn/News/](https://www.skysys.cn/News/) (<https://www.skysys.cn/News/>)

Important reconciliation: Tracxn lists a single co-founder named "Victoria Xiao." Given that Dr. Xiao Suzhi is the only founder surnamed Xiao and the only one whose profile matches a database-registered founder record, "Victoria Xiao" is almost certainly her English name. Tracxn's omission of Wang Haibin and Qian Maodong makes its founder data incomplete rather than wrong.

4.4 Other named personnel

No CFO, COO, CTO or board composition has been publicly disclosed in any source I could access. Investor-side individuals named in connection with the company (not employees) are listed in Section 3.3. The company reports approximately 160 staff, 70% in R&D.;

5. PRODUCTS

SKYSYS organises its portfolio into four layers: hangars/docking stations (hardware), onboard compute (edge AI), flight and dispatch software (OS layer), and vertical AI inspection applications — plus a newer fifth line, embodied robots. All descriptions and specifications below are as published by the company at [skysys.cn](http://www.skysys.cn/) (<http://www.skysys.cn/>).

5.1 Drone hangars / docking stations — the UltraHive family

The company claims to have built China's first productised drone hangar and the first Chinese drone hangar exported overseas ([chinaerospace.com](https://www.chinaerospace.com/article/24066) (<https://www.chinaerospace.com/article/24066>)). A core enabling technology is proprietary 气冷闪充 (air-cooled flash-charging).

(1) UltraHive® Mk4 Pro — charge/battery-swap combined hangar ([specs](https://www.skysys.cn/Mk4Pro/) (<https://www.skysys.cn/Mk4Pro/>))

The flagship fixed hangar. Provides safe charging, storage and data exchange for drones in field environments.

Parameter	Value
Weight	790 kg
Structure	High-strength stainless steel + magnesium-aluminium alloy frame; lightweight, non-magnetic
Ingress protection	IP54 (industrial outdoor rain/dust standard)
Wind resistance	Level 6
Drone compatibility	Multirotors with wheelbase ≤ 1200 mm; automatic charging *and* automatic battery swap for DJI M350 RTK
Charging interface	Quick-charge connector, multi-model compatible, forward-compatible with future airframes
Single sortie duration	30 minutes
Daily sortie frequency	10–30
Power supply	220 V mains
Power draw	2,300 W peak + 350 W standby
Network	100 Mb+ Ethernet; 4G/5G or local RF ground-station video link; public or private network
Weather sensing	Wind speed, wind direction, rainfall, temperature, humidity, barometric pressure
Night operations	Supported (night landing)
Security	NFC identity authentication for drone recognition/authorisation; dynamic authentication for remote commands; HD internal and external monitoring; image-recognition intrusion alerts; landing-attitude confirmation
Readiness	24-hour standby, supporting a "5-minute response radius" concept

Exterior dimensions are not published.

(2) UltraHive® MkY / MkX — lightweight drone charging hangar

Positioned as light, flexible to deploy and quick to launch/recover, aimed at lowering end-user cost. Both MkX and MkY designations appear on the site; MkY is the one currently shown on the homepage product grid, suggesting MkX is the earlier variant.

(3) UltraHive® MkZ 千巢 — fixed-wing / eVTOL drone airport ([specs \(https://www.skysys.cn/MkZ/\)](https://www.skysys.cn/MkZ/))

Claimed as the industry's first hangar able to accommodate multiple eVTOL models and large multirotors. Uses an AGV shuttle cart to physically transport the aircraft between a sealed storage bay and the takeoff pad.

Parameter	Value
Weight	≤ 900 kg (including AGV)
Ingress protection	IP54
Daily sortie frequency	10–30
Power supply	220 V mains
Power draw	3,500 W peak + 750 W standby
Single sortie duration	80–120 minutes
Network	4G/5G or RF box
Night operations	Supported
Weather sensing	Wind speed, direction, rainfall, temperature, humidity, pressure
Bay opening	≤ 60 s to prepare
AGV lift	≤ 5 min to raise the aircraft
Transport to pad	≤ 60 s to deliver and release at the takeoff point
Sealing	Wing-profile conforming seal devices accommodate longer-wingspan aircraft
Capacity	Can house multiple aircraft, of the same or different types
Charging	Robotic arm performs contact charging on returning aircraft; air-cooled flash-charge system

(4) UltraPad Mk1 岚 — micro drone hangar

Compact, stable, low operational risk, high safety. Targeted at construction sites, industrial parks and similar confined or urban settings.

(5) MobileHive Mk3 — mobile drone carrier (无人机移动航母)

A vehicle-mounted/portable hangar operable by a single person, supporting automatic takeoff, monitoring and return, enabling routine cloud-edge coordinated inspection away from fixed infrastructure.

(6) DJI DOCK 2 integration

SKYSYS resells/integrates DJI's own Dock 2 as a lightweight, easily deployed unattended platform paired with the dedicated Matrice 3D / 3TD aircraft; 34 kg excluding the aircraft. Notable as evidence of the non-competitive posture toward DJI.

Deployment concept: hangars are intended for gridded (网格化) deployment to create a "无人机五分钟出勤圈" — a five-minute drone response circle — with a stated target of two minutes to launch and three minutes to arrive on scene for emergency response. For transmission and substation inspection, hangars are spaced along the line so drones leapfrog between them.

5.2 Onboard compute — SkyCab® 天斗 ([specs \(https://www.skysys.cn/SkyCab/\)](https://www.skysys.cn/SkyCab/))

A lightweight airborne AI computing terminal — effectively the edge-inference brain that turns a stock commercial drone into an autonomous inspection platform.

Parameter	Value
Compute module	NVIDIA Xavier NX; up to 21 TOPS; upgrade path to Orin NX stated
Deep learning accelerators	2× NVIDIA engines
Vision accelerator	7-channel VLIW vision processor

Parameter	Value
Memory	8 GB / 16 GB, 128-bit LPDDR4x, 59.7 GB/s bandwidth
Storage	128 GB eMMC, TF card expandable; industrial-grade SSD options in 8G/16G specs
Weight	~145 g including antenna and basic bracket
Dimensions	84 × 56.5 × 30 mm
Ingress protection	IP55 — all-weather outdoor operation
Drone interface	DJI OSDK / PSDK standard hardware interfaces; direct coaxial-cable connection to M30, M350 and similar
Payload interface	Proprietary XSDK integrating USB, UART and Ethernet for third-party payloads
External ports	Micro HDMI 2.0 ×1, USB 3.0 ×1, USB 2.0 ×1, OTG ×1, TTL UART ×3
Wired network	10/100/1000Base-T adaptive Ethernet (via OSDK connector expansion)
Wireless	5G / 1.4 GHz private-network modules; 700 MHz "golden band" communication — low propagation loss, strong penetration, wide coverage
Video	GDR hard-decode terminal to compress data volume for weak-network operation

Supported third-party payloads cited: high-altitude loudspeakers, dual-light (visible + thermal) cameras, parachutes.

5.3 Flight and dispatch software — SkyScout® 祺云 ([page \(https://www.skysys.cn/SkyScout/\)](https://www.skysys.cn/SkyScout/))

The remote drone operating system / dispatch layer. Web client plus PAD (tablet) client. Supports defining and dispatching inspection tasks; central operators remotely control both drones and hangars and monitor operations; on-site personnel can manually take over an aircraft via the PAD client for real-time response. Handles route planning, task scheduling, remote control, and real-time cloud upload of data.

5.4 Vertical AI inspection software — the four-energy suite

The company markets four vertical applications covering what it calls 光电风气 — solar, grid, wind, and gas/emergency ([page \(https://www.skysys.cn/AISoftware/\)](https://www.skysys.cn/AISoftware/)):

- (1) 星逻驭光 — Skysys PV Inspector (flagship). Flies predefined waypoint routes covering an entire solar site, delivering a closed O&M; loop: fully automated inspection → data upload → diagnostic report → defect rectification → re-inspection verification. Combines drone thermal/visual data with inverter diagnostics to produce whole-lifecycle diagnosis of a PV plant.
- (2) 星逻巡电 — Skysys Electricity Inspector. Grid and transmission-line inspection: automated route import, tower route editing and generation, autonomous flight execution, AI defect analysis, one-click defect report generation. Marketed for conductor and ground-wire (导/地线) inspection.
- (3) 星逻御风 — Skysys Wind Turbine Inspector. Wind turbine blade/tower inspection. Productised for two years as of 2024. Supports three operating modes: unattended (无人值守), turbine-stopped service (停机服务), and turbine-running service (不停机服务). Delivered at multiple power-generation group sites. From 2024 being pushed through a franchise/partnership model with wind O&M; service providers on a product-as-a-service and revenue-share basis.
- (4) 星逻万象 — Skysys Universal Inspector. Emergency response and security platform. Frees industrial drones from single-platform lock-in and combines aerial and ground perspectives into a dual-view picture, for multi-scenario low-altitude urban coverage.

5.5 Embodied robots — the Skysys Lantern line

Categorised on the site as 低空具身机器人 ("low-altitude embodied robots") — the 2023-onward expansion from *detection* into *maintenance*, and the primary use of B-round proceeds.

Skysys Lantern PV panel cleaning robot. Currently marketed in two models: Lantern 小蓝 (Small Blue) and Lantern 大蓝 (Big Blue). Earlier reporting described a blue "dry-clean" variant for dust and mud in arid regions

and a red variant using cleaning fluid for stubborn residue such as bird droppings ([kjc.shlingang.com \(https://kjc.shlingang.com/kjc/xwzxs/yqxw/202405/t20240531_31574.shtml\)](https://kjc.shlingang.com/kjc/xwzxs/yqxw/202405/t20240531_31574.shtml)).

Distinguishing characteristics:

- Drone-deployed and drone-retrieved. The robot is airlifted onto and off panels by drone, enabling automated placement on complex terrain — rooftops, hillsides, tidal flats — that ground crews cannot easily reach. The company calls this the world's first deep integration of a drone with a cleaning robot.
- Negative-pressure adhesion conforming to panel tilt angle.
- Traverses inclines up to 45° — the explicit engineering target of the B-round funding.
- Autonomous operation: depth algorithms drive self-positioning, path planning and cleaning, with real-time status telemetry.
- Verification loop: a follow-up drone inspection pass confirms cleaning quality — closing the loop with 驭光.
- Claimed outcome: 8%–30% increase in power generation ([stcn.com \(https://www.stcn.com/article/detail/1224287.html\)](https://www.stcn.com/article/detail/1224287.html)). Note this is a company claim, not independently verified, and the range is wide enough to suggest heavy site-dependence (desert sites with heavy soiling would sit at the top of the range).
- Lightweight design for one-person transport.

R&D; began in 2021; the product went through "两年多次迭代" (two years of repeated iteration) before launch in 2023.

5.6 Application domains

As listed by the company across Chinese and English sites: PV inspection; wind power inspection; grid / conductor and ground-wire inspection; petrochemical pipeline inspection (appears only on the English site); emergency and security inspection; PV cleaning; plus historical/other applications documented in directory listings — smart city management, algae bloom monitoring, illegal fishing detection, and during COVID, crowd density, mask-wearing and traffic monitoring ([kchuhai.com \(https://www.kchuhai.com/company/detail.aspx?id=5457\)](https://www.kchuhai.com/company/detail.aspx?id=5457), [thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616)).

5.7 Deployment metrics over time

These are the closest available proxies for commercial traction. Note the growth trajectory and the internal inconsistency between two 2024 sources:

Reported	PV stations	Installed capacity covered	Strings inspected
June 2024 (stcn (https://www.stcn.com/article/detail/1224287.html) , chinaerospace (https://www.chinaerospace.com/article/24066))	427	8.3 GW	—
2024, alternate report (leaderobot (https://www.leaderobot.com/news/4321))	~400 ("近400")	>6,000 MW	>850,000
Later / current (36kr pitchhub (https://pitchhub.36kr.com/project/1678486488167426) , leaderobot (https://www.leaderobot.com/news/4321) updated text)	540+	>10 GW	—

Other traction indicators:

- ~200 drone automation systems deployed nationwide, product/service through six iterations (2023, [thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616))
- 100+ projects deployed across dozens of cities in China and abroad ([skysys.cn/AboutSkysys \(https://www.skysys.cn/AboutSkysys/\)](https://www.skysys.cn/AboutSkysys/))
- Named deployment sites: Shanghai Chongming; Tongliao, Inner Mongolia; Ōsaki, Japan; Shanghai Lingang (9 hangars / 95 km²); a drone hangar security project in London, UK ([startup.aliyun.com \(https://startup.aliyun.com/info/102033.html\)](https://startup.aliyun.com/info/102033.html)); Shanghai, Jiangsu, Yunnan, and Europe
- Named customers: Alibaba, Huawei, and other Fortune 500 companies

6. PATENTS AND INTELLECTUAL PROPERTY

Direct statement of limitation: I was unable to retrieve a single verified patent number or patent title. Google Patents, Espacenet, FreePatentsOnline and the USPTO search endpoint were all inaccessible during this research (503 / 403 / connection reset / DNS failure respectively), and the Qichacha registry page was blocked. What follows is an aggregate activity summary from secondary sources. Do not treat any of these counts as independently verified, and do not let any downstream system infer specific patent identifiers from this section — none are known.

6.1 Aggregate counts (conflicting)

Source	Claim	Date
36kr pitchhub (https://pitchhub.36kr.com/project/1678486488167426)	150+ national patents, including applications pending; 120+ software copyrights (软著); 30+ national-level awards	current profile
leaderobot (https://www.leaderobot.com/news/4321)	国家专利 (含申请中) 150余项; 软著超120项	2024 (updated)
innohere (https://www.innohere.com/ir/111839/finance.html)	63 patents	database field, undated

The gap between 63 and 150+ is best explained by the qualifier "含申请中" — the 150+ figure includes pending applications, while 63 likely reflects granted patents captured by that database at some earlier snapshot date. Both can be true simultaneously.

Individual holdings: co-founder 钱茂冬 (Qian Maodong) is described as holding multiple Chinese national invention patents in his own name ([thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616)) — though again, no numbers are given.

6.2 Technology areas covered

Chinese reporting describes the patent portfolio as concentrated in ([leaderobot \(https://www.leaderobot.com/news/4321\)](https://www.leaderobot.com/news/4321)):

1. 无人机自动充电 — automated drone charging (the company's founding technical domain; Wang Haibin's battery-management background)
2. 机器视觉 — machine vision
3. 调度算法 — dispatch/scheduling algorithms
4. 数据处理 — data processing

Additional proprietary technologies named in product literature that are plausible patent subjects:

- 气冷闪充技术 — air-cooled flash-charging (Mk4 Pro, MkZ)
- 机翼仿形密封装置 — wing-profile conforming sealing device for the MkZ bay
- AGV-based aircraft handling for fixed-wing/eVTOL hangars
- 纯网联 cross-hangar cellular flight architecture (take off from A, land at B)
- Negative-pressure adhesion and 45° -incline traversal for the Lantern robot
- The company states its charging pile (充电桩) has independent/proprietary IP (自主知识产权) ([thepaper.cn \(https://www.thepaper.cn/newsDetail_forward_21336616\)](https://www.thepaper.cn/newsDetail_forward_21336616))

6.3 Standards participation — arguably more significant than the patent count

In 2021, SKYSYS co-drafted IEEE 1937.1 together with Huawei and DJI — reported as the world's first international standard for drone payload systems (无人机载荷系统国际标准) ([startup.aliyun.com \(https://startup.aliyun.com/info/102033.html\)](https://startup.aliyun.com/info/102033.html)). For a company of this size, sitting at the standards table alongside Huawei and DJI is a stronger signal of technical standing than a raw patent count.

6.4 Design rights

The Lantern 小蓝 robot won the German Red Dot Award and a French FDA design award in June 2026 ([skysys.cn/News \(https://www.skysys.cn/News/\)](https://www.skysys.cn/News/), [skysys.cn/en \(https://www.skysys.cn/en/\)](https://www.skysys.cn/en/)) — implying registered industrial-design protection, though no design registration numbers are published.

6.5 Qualifications

- 国家级专精特新"小巨人"企业 — national-level "Specialised, Refined, Differentiated and Innovative Little Giant" enterprise. This designation is granted by MIIT and typically requires demonstrated IP holdings, R&D intensity and niche market leadership, so it independently corroborates a substantive patent portfolio ([skysys.cn](http://www.skysys.cn/) (<http://www.skysys.cn/>)).

7. LATEST NEWS AND DEVELOPMENTS (2024 – SEPTEMBER 2026)

Compiled from the company's own news index ([skysys.cn/News](https://www.skysys.cn/News/) (<https://www.skysys.cn/News/>)) plus the press sources cited throughout. Dates as published.

2024

- 24 May 2024 — Two international awards at the 8th World Drone Congress, Shenzhen. The newly launched Skysys Lantern PV robot won both the International Unmanned System Design Award (国际无人系统设计奖) and the International Unmanned System Innovation Product Award (国际无人系统创新产品奖), judged on innovation, aesthetics, practicality and functionality. The event drew participants from 100+ countries and 500+ exhibitors. (kjc.shlingang.com (https://kjc.shlingang.com/kjc/xwzs/yqxw/202405/t20240531_31574.shtml))
- 3–6 June 2024 — Series B closed, over RMB 100 million. Co-led by Guangdong Yueke Fund-of-Funds and the Lingang Qianyan / Arctech Yangzhou Fund, with A-round and angel investors following on. Proceeds directed to Lantern mass production and the 御风 wind service rollout. Widely covered across Chinese business media. (stcn.com (<https://www.stcn.com/article/detail/1224287.html>), chinaerospace.com (<https://www.chinaerospace.com/article/24066>), news.qq.com (<https://news.qq.com/rain/a/20240606A07RGP00>), dikongjie.com (<https://dikongjie.com/273.html>), tidesight.com (<https://www.tidesight.com/news/1369891.html>))
- 2024 — Product line extension into robotics. Launch of PV and wind-power robot series, formally extending the company from low-altitude intelligent *detection* into low-altitude intelligent *O&M*.

2025

- 23 June 2025 — Series B+ closed, over RMB 100 million. Investors: Suzhou Port & Shipping Group (苏州港航集团) and Lingang Qianyan Investment (临港前沿投资). ([innohere](http://innohere.com) (<https://www.innohere.com/ir/111839/finance.html>)) *Note: this round is recorded in the innohere financing database; I was unable to locate a corresponding press release in the sources I could reach, so treat the details as single-sourced.*
- 17 Nov 2025 — Named an "AI 100 应用标杆" (AI 100 Application Benchmark) at the 2025 AI+ Conference.
- 28 Nov 2025 — First place in the low-altitude sector final of the 2025 China Innovation & Entrepreneurship Competition (中国创新创业大赛低空领域决赛第一名).
- 4 Dec 2025 — Named "WISE2025 商业之王 — 年度低空经济创新企业" (36Kr's WISE awards, Low-Altitude Economy Innovation Enterprise of the Year) — a repeat win.
- 14 Dec 2025 — Won the 金翼奖 (Golden Wing Award) again at the 2025 Global Unmanned Systems Conference.
- Undated 2025 — Awarded national-level 专精特新"小巨人" (Little Giant) enterprise status.

2026

- 13 Jan 2026 — Named to the 《2025年度苏州市"独角兽"培育企业名单》, Suzhou's 2025 unicorn-cultivation enterprise list. (Again: *cultivation* list, not confirmed unicorn valuation.)
- 16 Mar 2026 — Strategic cooperation MoU signed with France's JPA Group (法国JPA集团). The most significant international development in the period — consistent with the company's longstanding overseas-export positioning and its French-language website. Terms not disclosed in the sources available.
- 24 Apr 2026 — Exhibited at the 2026 Jiangsu PV Innovation & Development Conference, winning the "卓越智能运维企业" (Outstanding Intelligent O&M; Enterprise) award.

- 30 Apr 2026 — Chief Scientist Dr. Xiao Suzhi named one of the national top ten "最美职工" ("Most Beautiful Workers").
- 24 Jun 2026 — Lantern 小蓝 robot wins the German Red Dot Award and the French FDA Design Award. Referenced on the English site as the "Skysys Lantern B1 Robot."
- 10 Aug 2026 — Won the "赋能电站运维奖" (Empowering Plant O&M; Award) at 2026 工商业光伏之星 (Commercial & Industrial PV Star awards).
- 24 Aug 2026 — Dr. Xiao Suzhi delivered a keynote at the PSGEC power conference.
- 25 Aug 2026 — Delegation from Fudan University School of Economics visited the company.
- 28 Aug 2026 — Exhibited at the Qingdao Power Conference, under the theme "掘金光伏下半场" (mining the second half of the PV cycle) — thematically aligned with the thesis that value is shifting from installation to operations.
- 14 Sept 2026 — Interview published with Dr. Xiao Suzhi: "无人机只是'手脚'，新能源运维真正拼的是AI'大脑'" — "Drones are only the hands and feet; what new-energy O&M; really competes on is the AI 'brain'." This is the clearest recent statement of strategic direction: the company is positioning its differentiation in the AI/diagnostics layer rather than in hardware.

Note on news coverage depth: the company's news index shows 19 pages of listings. I retrieved only the first page (nine items, skewed to 2026 because the page appears to sort with pinned/featured items first). Substantially more 2024–2025 item-level news exists on that index and was not enumerated here.

8. DATA QUALITY ASSESSMENT AND CONTRADICTIONS

For a RAG stress-test corpus, the contradictions are as informative as the facts. Documented conflicts:

(1) Headquarters: Shanghai vs Suzhou. Tracxn says Shanghai. The company's own site and its legal registration say Suzhou Industrial Park, with Shanghai (Lingang and Pudong) as R&D;/office sites. The likely explanation is that early operations centred on the Shanghai Lingang project (2018–2020) before the entity was established/relocated in Suzhou; the legal entity name itself contains " (苏州) ". Suzhou is correct for HQ.

(2) Employee count: 3 vs ~160. Tracxn's "3 employees as of 30 June 2026" is not credible against ten offices, a nine-figure 2025 round, a national Little Giant designation, and continuous 2026 activity. ~160 with 70% R&D; is the credible figure.

(3) Total funding: \$1.49M vs RMB 300M+. Tracxn's aggregate omits both nine-figure rounds. Tracxn is wrong.

(4) Founders: one vs three. Tracxn lists only "Victoria Xiao." Chinese sources consistently name three co-founders: Wang Haibin (CEO/chairman), Qian Maodong, and Dr. Xiao Suzhi. "Victoria Xiao" ≈ Xiao Suzhi.

(5) Wang Haibin's experience: "16 years" vs "nearly 20 years." Different sources, likely different vintages of the same boilerplate.

(6) PV deployment scale: 427 stations / 8.3 GW vs ~400 / 6 GW vs 540+ / 10 GW. Partly genuine growth over time, partly inconsistent source vintages. The 540+ station / 10 GW figure is the most recent.

(7) Tracxn's competitor set (uAvionix, Sagetech, ARK Electronics, Zerodrag, Defendron) is a category mismatch — those are avionics/transponder/ADS-B companies, not drone-hangar or energy-inspection companies. SKYSYS's real competitive set would be drone-in-a-box vendors (DJI Dock itself, 复亚智能 Fuya, 中科云图 ZhongkeYuntu, 道通 Autel), PV O&M; robotics firms, and energy inspection software vendors. Tracxn's "6th of 68 competitors" ranking should be disregarded.

(8) Products described by Tracxn — "a proprietary drone with payload up to 1 kg and 25 minutes endurance," "a docking system in the form factor of a drawer," "GPS and VPS dual positioning" — describe an early (circa 2018–2019) product generation and directly contradict the company's explicit, repeatedly stated policy of never building drones. This is stale data from the company's earliest seed-stage description.

Recommended source hierarchy for this company: (1) skysys.cn product pages for specifications; (2) Chinese business press (stcn.com, 36Kr, Tencent News) for funding and metrics; (3) thepaper.cn for founder and history detail; (4) innohere / 36Kr pitchhub for round-by-round financing tables; (5) Tracxn — low

reliability, use only for the English name mapping.

9. SUMMARY

Starlogic Intelligent Technology / 星逻智能 / SKYSYS is an eight-year-old, Suzhou-headquartered Chinese company of roughly 160 people that sells the infrastructure layer for autonomous industrial drone operations — hangars, chargers, onboard AI compute, dispatch software — plus vertical AI inspection applications, and since 2023 a line of drone-deployed PV cleaning robots. It deliberately does not manufacture aircraft, positioning itself as a neutral enabler to DJI and other airframe makers rather than a competitor.

It has raised eight rounds since 2018, with the last two (B in June 2024, B+ in June 2025) each exceeding RMB 100 million, backed increasingly by provincial state funds, a PV-industry strategic (Arctech), and a municipal SOE. Founder-CEO Wang Haibin retains ~22.4%. No revenue, valuation or filings are public.

Its commercial centre of gravity has shifted decisively from smart-city/security (2018–2021) to new-energy asset O&M; (2022–present), anchored by the 驭光 PV inspection platform (540+ stations, 10+ GW covered) and the award-winning Lantern cleaning robot (claimed 8–30% generation uplift). It holds 150+ patents including pending applications and 120+ software copyrights — though no individual patent numbers could be verified for this report — and co-drafted IEEE 1937.1 with Huawei and DJI. It is on Suzhou's unicorn-cultivation list, holds national Little Giant status, and in 2026 signed an MoU with France's JPA Group while its Lantern robot won a Red Dot award.

English-language coverage of this company is genuinely sparse and, where it exists, substantially inaccurate. Anyone building an information product on this company must work from Chinese sources.

SOURCES

- Tracxn — Starlogic Intelligent Technology company profile (https://tracxn.com/d/companies/starlogic-intelligent-technology/_dB9FxFVRJeS9FwhbVNJW9l48phFLvocP0KV1Y2QoUHCE) *(low reliability — see Section 8)*
- SKYSYS official site — homepage (<http://www.skysys.cn/>)
- SKYSYS — About Skysys (<https://www.skysys.cn/AboutSkysys/>)
- SKYSYS — English site (<https://www.skysys.cn/en/>)
- SKYSYS — Drone docking stations (<https://www.skysys.cn/DockingStation/>)
- SKYSYS — UltraHive Mk4 Pro (<https://www.skysys.cn/Mk4Pro/>)
- SKYSYS — UltraHive MkZ (<https://www.skysys.cn/MkZ/>)
- SKYSYS — SkyCab 天斗 (<https://www.skysys.cn/SkyCab/>)
- SKYSYS — SkyScout 祺云 (<https://www.skysys.cn/SkyScout/>)
- SKYSYS — AI inspection software (<https://www.skysys.cn/AISoftware/>)
- SKYSYS — DJI Dock 2 (<https://www.skysys.cn/DJIDock2/>)
- SKYSYS — News index (<https://www.skysys.cn/News/>)
- SKYSYS — Resources/manuals (<https://www.skysys.cn/Information/>)
- The Paper — 如何让无人机实现真“无人”？三学霸的无人机智能化巡检系统 (https://www.thepaper.cn/newsDetail_forward_21336616) *(founder bios, founding story, strategy)*
- 36Kr PitchHub — 星逻智能项目信息 (<https://pitchhub.36kr.com/project/1678486488167426>) *(funding table, patents, headcount)*
- 36Kr — 无人机机库服务商「星逻智能」完成千万级Pre-A轮融资，远瞻资本领投 (<https://36kr.com/p/1723333918721>)
- InnoHere — 星逻智能融资历史 (<https://www.innohere.com/ir/111839/finance.html>) *(eight-round financing table incl. B+)*
- 证券时报 STCN — 低空领航企业星逻智能完成超亿元B轮融资 (<https://www.stcn.com/article/detail/1224287.html>)
- ChinaErospace — 低空领航企业星逻智能完成超亿元B轮融资 (<https://www.chinaerospace.com/article/24066>)

- Tencent News — 星逻智能完成超亿元B轮融资，实控人王海滨曾任GE科技学社研发总监 (<https://news.qq.com/rain/a/20240606A07RGP00>) *(registered capital, equity stake)*
- 机器人大讲堂 Leaderobot — 国内低空智能机器人头部企业「星逻智能」完成超亿元B轮融资 (<https://www.leaderobot.com/news/4321>) *(full product list, patent counts)*
- 阿里云创新中心 — 星逻智能获数千万元A+轮融资 (<https://startup.aliyun.com/info/102033.html>) *(A+ round, IEEE 1937.1, London project)*
- 低空界 Dikongjie — 星逻智能完成超亿元B轮融资 (<https://dikongjie.com/273.html>)
- 观潮新消费 Tidesight — 星逻智能完成超亿元B轮融资 (<https://www.tidesight.com/news/1369891.html>)
- 上海临港科技创新城 — 星逻Lantern斩获双项国际奖 (https://kjc.shlingang.com/kjc/xwzs/yqxw/202405/t20240531_31574.shtml) *(World Drone Congress awards, Lantern specs)*
- 快出海 kchuhai — 星逻智能科技（苏州）有限公司 (<https://www.kchuhai.com/company/detail.aspx?id=5457>) *(awards history 2018-2020, overseas markets)*

Inaccessible during research (gaps noted in report): Google Patents (503), Espacenet (403), FreePatentsOnline (ECONNRESET), USPTO legacy search (DNS), Qichacha registry (blocked), SKYSYS Honor/Lantern product sub-pages (404), news index pages 2–19 (pagination URL not resolved).

TEAM BLACKSHEEP (TBS) — RESEARCH REPORT

Compiled 2026-09-24.

0. CORRECTION TO BRIEF

"Founder Michael 'Fishpepper'/Raphael Pirker" conflates two unrelated people. Raphael Pirker ("Trappy") is the actual founder/CEO. "fishpepper" = Simon Schulz, an independent German open-source hardware developer (tinyFISH FC, tinyPEPPER ESC, OpenSky firmware) with no relationship to TBS. Also: Tracxn's "Blacksheep" profile is an unrelated Chennai media production company (180 employees) — a false-positive trap for retrieval.

1. COMPANY OVERVIEW

Origin: early-2000s Swiss FPV scene. Raphael Pirker, a University of Zurich student, began FPV flying ~2003 in the Swiss mountains with a foam-wing aircraft and hacked baby-monitor electronics, eventually setting an 89km long-range FPV record. Name/team origin dated 2007 by the company itself. Early collaborators: Christoph Dunkel and Remo Masina.

Founding date is contested across sources: 2007 (brand origin), 2009 (fpv24.com), 2010 (Crunchbase), 2013 (BlackSheep Avionics HK incorporation). Best reading: informal team from ~2007, commercial entity from ~2010, current HK manufacturing entity from 2013.

Current corporate structure — three entities, three jurisdictions:

- BlackSheep IP Holdings Pte Ltd (Singapore) — HQ, R&D; support; Pirker is CEO here.
- BlackSheep Avionics Co. Ltd (Hong Kong/Kowloon) — factory/logistics, also filed as TBS Avionics Limited on FCC grants.
- Conceptos Pirker GmbH (Austria, St. Anton am Arlberg) — EU office/returns.

Size: 14 employees at the HK entity per Bounce Watch (likely HK-only, not group-wide). Self-reported: "10+ Years of Experience," "2M+ Systems installed," 1M+ flight hours across the Tracer ecosystem. Engineers described as spanning USA, Europe, Asia.

Mission/positioning: storefront tagline "Serious Toys." About page: "redefine the FPV hobby one flight at a time," in an industry "dominated by imitation." Singapore holdco frames itself more corporately: "a defining R&D; company in advanced radio control, telemetry, and IoT hardware" — signaling ambitions beyond hobby FPV. 10+ years of free, continuous firmware updates as a durable commercial promise.

The FAA case — defining public episode

Pirker became famous flying a fixed-wing FPV aircraft over Lower Manhattan/Statue of Liberty (posted to YouTube), and TBS filmed similar stunts in the Taj Mahal, Dubai, Miami, Rio, London, Hong Kong. Labeled "aerial anarchists" by Australian media. Legal chain:

- 2011: FAA proposes \$10,000 penalty against Pirker for a commercial shoot over University of Virginia, calling the flight "careless or reckless."
- Oct 2013: motion to dismiss filed, arguing FAA had no enforceable rule covering model aircraft.
- 6 Mar 2014: NTSB Administrative Law Judge dismisses the penalty — widely treated as the foundational US drone-regulation precedent (Huerta v. Pirker). Gave FPV pilots ~8 months of relatively unrestricted flying before rules tightened again.
- TBS publicly stated it reinvests considerable revenue into R&D; for safer, more reliable civilian drones.

2. FINANCIALS

No public financials exist anywhere — privately held, founder-controlled, apparently bootstrapped. Crunchbase, Craft.co, ZoomInfo all show a profile exists but no revenue/valuation disclosed (and blocked automated retrieval). Bounce Watch: "Revenue: not disclosed." No filings, no equity listing, no reported valuation, never reported acquired or as an acquirer.

Unverified inference only: "2M+ systems installed" self-claim + catalog price points (\$3.95–\$208.95, most volume SKUs under \$150) sold direct plus through a dealer network (NewBeeDrone, Pyrodrone, BuildYourOwnDrone, FPV24, etc.) — a 14-person manufacturing entity at that footprint plausibly sits in the low-to-mid eight figures USD annually at peak. Treat as pure estimate.

3. FUNDING HISTORY

No funding rounds, no investors, no VC — ever, in any accessible database. Bounce Watch states flatly "No funding data available... yet." Crunchbase carries no funding rounds. Consistent with the company's own narrative: grew from a hobby team funded off product margin, later restructured into Singapore/HK/Austria entities for tax/logistics reasons rather than investor-driven structuring.

4. LEADERSHIP AND PEOPLE

Raphael "Trappy" Pirker — Founder & CEO. Swiss-Austrian, studied at University of Zurich, ties to St. Anton am Arlberg, Austria. Titles: "Founder and CEO" (Crunchbase), "Technology Pioneer, CEO" of BlackSheep IP Holdings (Singapore site). Notably says outright in a Bard Center interview: "I don't have any technical background... It developed as a hobby and I learned quickly." Began making drone videos in 2008. Respondent in *Huerta v. Pirker*, the first FAA enforcement action of its kind in the US. Also an officer of Conceptos Pirker GmbH (Austria).

Co-founders: Christoph Dunkel and Remo Masina — early FPV experimenters who joined at formation; current involvement unconfirmed, don't appear in any current corporate listing.

Organizational notes: historically had a dedicated video editor distinct from pilots. Engineering distributed across USA/Europe/Asia. No CTO, CFO, COO, or board member named anywhere publicly — unusually flat, founder-centric public face. Currently hiring (live signal): Embedded Software Engineer, Electronics Engineer, RF Engineer, Production Manager, Customer Support Manager — the RF+embedded+production mix indicates continued in-house hardware development.

Adjacent ecosystem names (not TBS employees): Simon Schulz "fishpepper" (independent); BrainFPV and Achilles (partners on TBS Fusion goggle module — image processing and OSD/UI respectively); "Mr Steele" (Steele Davis, signature motor line); ETHIX (co-produced sub-brand).

5. PRODUCTS — FULL CATALOG

Store organized into ~18 categories: Ready to Fly, Frames, Electronics, FPV Equipment, RC Link, TBS Crossfire, TBS Tracer, Props, Battery, Motors, Wings, Hardware and Tools, Software, Swag, Clearance, ETHIX, HNY Tech, Lessons. FPV24 counts 102 TBS catalog items.

TBS CROSSFIRE (flagship long-range RC link, 868/915MHz): -130dB receiver sensitivity, full RF diversity, self-healing DSSS/FHSS frequency hopping, proprietary CRSF protocol (now a de facto industry standard beyond TBS hardware), range 60mi+/100km+, framerate 50-150Hz, latency 6.5ms, 4-8 simultaneous pilots, RF output 10mW-2W, MAVLink tunneling, receiver beacon mode for crash location, OTA firmware updates, CRSF/SBUS/PPM/PWM outputs, compatible with Betaflight/KISS/Raceflight/ArduPilot/OpenTX. Transmitters: TX full-size \$208.95 (out of stock), TX LITE \$149.95 (out of stock), Micro TX V2 \$69.95 (out of stock), Nano TX \$69.95 (in stock, current focus). Receivers: Nano RX \$24.95 (volume seller), Nano RX SE \$29.95, Nano RX Pro \$34.95, Micro RX V2 \$34.95, Diversity Nano RX \$49.95, 8Ch Diversity RX \$123.95, 6ch PWM Nano RX \$32.95, Sixty9 (integrated RX+VTX) \$49.95. Plus antenna/accessory line (\$2.95-\$24.95 range).

TBS TRACER (2.4GHz low-latency link, "Crossfire's little brother"): 250Hz update rate, 3ms end-to-end latency, 12-channel digital stream, range 25km+/15mi+, up to 20 simultaneous pilots, antenna diversity. Lineup: Micro TX/Nano TX \$69.95 each, Starter Sets \$129.95, Nano RX \$29.95 (0.5g), 6CH PWM Nano RX \$32.95, Sixty9 \$49.95, Fusion Packs \$229.95 each.

Shared ecosystem across both links: OTA wireless firmware updates, auto-bind/multi-bind, LUA scripts for FC PID config from the radio, TBS Cloud for device sync/telemetry sharing — all claimed as industry firsts. Configuration software: TBS Agent M (browser), Agent X (desktop), Agent Lite (radio-side).

Radios: TBS Tango 2 \$159.95 (smallest RC remote available, built-in Crossfire, FreedomTX open-source OpenTX fork, 565g); Tango 2 Pro \$199.95 (foldable gimbal sticks); TBS Mambo \$139.95 (first radio with Tracer built in + JR bay for Crossfire modules); Ethix Mambo \$189.95 (co-branded).

FPV video: TBS does NOT make complete goggles — only modules that go inside other goggles. TBS Fusion \$134.95: analog video receiver module using active video fusion (merging two signals) to cut breakup/extend range, -96dBm sensitivity, built-in OSD, integrates with Crossfire, fits Fatshark Dominator/HD1-3/HDO/HDO2. Built with BrainFPV + Achilles. Audio omitted due to global filter-component shortage. Video transmitters: Unify Pro 5G8 HV line (\$24.95-\$50.95) plus legacy Unify/Unify Pro/Race/Nano/Evo. Video receivers: Groundstation 2G4, Fusion Diversity Receiver System. Antennas: Triumph, Triumph Pro, Triumph Stab, AircREW RHCP, patch/micro antennas. Cameras: Kamera 59, Kamera 69, Tiny Kamera, ZeroZero, ChipChip.

Flight controllers/ESCs/power — TBS Lucid line (current): Lucid H7 FC \$59.95, Lucid 8S 4in1 ESC \$59.95, Lucid 16S 200A ESC \$149.95 (high-voltage/heavy-lift, notable for a hobby brand), Lucid PDB 500A \$14.95. Legacy: TBS PowerCube family, Bulletproof 25A ESC, Caipirinha 2 ESC+PDB. GPS: TBS M10 GPS Glonass \$15.95 (top FPV24 seller by volume), M10N Mini GPS.

Airframes/wings: TBS Discovery (the quad that made TBS a hardware name), Discovery Pro, Oblivion, Vendetta, Gemini, Source One (open-source frame), Source Two, Chupito (gated behind "Secret Project" login), X-Class frames. RTF categories: Race, Freestyle, Long Range, Cinematic, Whoops. Wing line: Chupito, Mojito (current generation), Caipirinha/Caipirinha 2 (long-running flying wing, full parts ecosystem still sold).

Motors/props/batteries/tools: TBS Master Pilot, Micro Brushless, Mr Steele signature line, Endurance line; third-party MAD heavy-lift motors (Taurus, Swift, Polar, up to 6-12S) point at a push into larger/industrial airframes; props 2"-17"; battery accessories; soldering/hardware tools.

ETHIX sub-brand: heated LiPo bags, Ethix Mambo radio, power straps, goggle straps, equipment bags.

Software/services/lifestyle: Velocidrone license resale; FreedomTX (OpenTX fork); paid Lessons category; Swag including FPV Chair, Love Seat, FPV Table.

B2B services ("Hire us" page): video capture services (real estate to full events) and aerial-video-system consulting for companies building UAV products/fleets. Media exposure cited: ~30M viewers/year. No OEM/white-label or IP-licensing offering advertised despite the "IP Holdings" corporate name.

FLARM for drones (July 2017): collision-avoidance/electronic-conspicuity system — TBS's clearest excursion into safety/regulatory-facing avionics, aligning with its post-FAA-case positioning.

6. PATENTS

No patents attributable to Team BlackSheep, TBS Avionics, BlackSheep Avionics, or BlackSheep IP Holdings were found; none naming Raphael Pirker as inventor. Patent-title searches surfaced only unrelated documents (Parrot, other RF/antenna patents). Google Patents and Justia both blocked automated retrieval (503/403) — the negative result is partially search-limited, not conclusive.

TBS's actual regulatory paper trail is equipment authorization, not patents: FCC ID QOS-TX2W (TBS Crossfire, applicant TBS Avionics Limited, registered 2016-11-08), FCC ID QOS-TANGO2, EU Declarations of Conformity for TBS Avionics Co Ltd products.

Assessment: TBS's moat is protocol lock-in (CRSF), firmware/cloud ecosystem, brand/community, and RF type-approval engineering — not patents. Company has historically operated in an open/community-adjacent way (open-source Source One frame, FreedomTX as an OpenTX fork, free perpetual firmware). "IP Holdings" in the corporate name most likely refers to trademarks/copyrighted firmware, not patents.

7. LATEST NEWS

TBS runs no press operation — no dated blog, no press-release page; announces via Facebook/Discord/store. No new Crossfire or Tracer hardware platform launch found for 2025-2026.

Live catalog signals (Sept 2026):

- Full-size Crossfire TX, TX LITE, and entire Micro TX V2 line out of stock; Nano TX/RX generation in stock — consolidation onto Nano form factor.
- Tango 2 and Tango 2 Pro out of stock (new orders ship in 2-3 weeks); Mambo/Ethix Mambo in stock — radio business shifting to Mambo/Tracer platform.
- Tracer Sixty9, Nano RX, Fusion Packs, and the TBS Fusion module itself all out of stock, with Fusion's audio feature permanently removed due to a global filter-component shortage — a visible supply-chain scar.
- Lucid electronics family (H7 FC, 8S/16S ESCs, PDB) is the current active development line; the 16S 200A ESC plus MAD heavy-lift motors indicate a deliberate move toward larger, higher-voltage, industrial/cinelifter-class airframes, away from pure 5-inch hobby racing.
- Chupito/Mojito are the current wing push; Chupito gated behind a "Secret Project" login, indicating unreleased/partner-restricted development.
- Store increasingly carries third-party inventory (Axis Flying, MAD, FpvCycle, HQ Props, Velocidrone) — evolving into a full-line retailer, not just a house brand.
- Active hiring for an RF Engineer is the strongest signal a next-gen link product is in development.

Competitive context — ExpressLRS: the single most consequential development in TBS's market is external. ExpressLRS (open-source, GPLv3, LoRa/FSK on Semtech transceivers) began 2018 at 900MHz/200Hz, added 2.4GHz plus cheap commercial hardware in 2020 driving mass adoption, now supports up to 1000Hz packet rates and 100km+ range. FPV pilots have gravitated to ExpressLRS over Crossfire/FrSky due to lower cost and better latency; it's near de facto standard status in the FPV community by the mid-2020s. Since 2022, ExpressLRS has also become dominant in Ukraine-conflict FPV strike drones (EW resistance, open-source adaptability). TBS pioneered long-range-link/telemetry/OTA-update/cloud-sync features and has since watched an open-source project commoditize the category — the Tracer's speed/latency positioning, CRSF's ubiquity (ELRS itself speaks CRSF), and the Lucid high-voltage push all read as responses to that squeeze.

Historical timeline: 2011 FAA penalty proposed; Oct 2013 motion to dismiss; 6 Mar 2014 NTSB dismisses penalty; Jul 2017 FLARM collision-avoidance released; Oct 2017 Oscar Liang reviews PowerCube Elite; 21 Jun 2018 TBS-associated pilot races Marc Márquez's F1 car with an FPV drone; Mar 2020 "Quarantine Masters" Velocidrone racing series (pandemic-era); Feb 2022 TBS Agent Lite released; 14 Feb 2023 Oscar Liang's consolidated Crossfire/Tracer setup guide.

8. ASSESSMENT

TBS is a founder-controlled, bootstrapped, three-jurisdiction (Singapore IP holdco / Hong Kong manufacturing / Austria EU office) FPV hardware company built on a YouTube-famous origin and a landmark FAA legal case (Huerta v. Pirker). No public financials, no funding history, no patents (moat is protocol lock-in via CRSF, firmware/cloud ecosystem, and brand). Currently consolidating hardware onto Nano-form-factor links and the Mambo/Tracer radio platform, pushing into higher-voltage/industrial airframe support (Lucid 16S ESC, MAD motors), and facing real competitive pressure from the open-source ExpressLRS project that has commoditized much of what TBS pioneered.

Sources: team-blacksheep.com, blacksheep.com.sg, dronecenter.bard.edu, suasnews.com, fccid.io, oscarliang.com, en.wikipedia.org/wiki/ExpressLRS, fpv24.com, buildyourowndrone.co.uk, bouncewatch.com.

I now have comprehensive material across all 7 sections. Compiling the final report.

uAvionix Corporation — Research Report

Disambiguation (Read This First)

There are at least two distinct companies with similar names that are easy to confuse. This report is about uAvionix Corporation (styled "uAvionix" in some branding), the aviation/UAS avionics company headquartered in Bigfork, Montana. It should not be confused with:

- UVIONIX / Uvionix Aerospace Corporation — a completely different company, founded in 2014 in Rancho Cordova, California by Boris Iskrev and Yordan Iskrev (with Stanislav Darmonski later involved). UVIONIX builds autonomous *indoor warehouse* drones (its flagship aircraft is called "U-VeE," a compact coaxial drone) for inventory management, paired with "Uvionix.Cloud" software. It has raised roughly \$3.54M across 2 rounds from 4 investors — an entirely separate market (warehouse robotics/logistics) from uAvionix's aviation-safety/avionics focus. (Sources: [Crunchbase](https://www.crunchbase.com/organization/uvionix) (<https://www.crunchbase.com/organization/uvionix>), [Tracxn](https://tracxn.com/d/companies/uvionix-aerospace-corporation/_weRRSuTLZ7_73pwPQRW7JeTXwxqoLaU2tb7r7-tApbQ) (https://tracxn.com/d/companies/uvionix-aerospace-corporation/_weRRSuTLZ7_73pwPQRW7JeTXwxqoLaU2tb7r7-tApbQ), [uvionix.com](https://www.uvionix.com/about) (<https://www.uvionix.com/about>))
- Search-index noise: CB Insights lists a separate profile under "Univonix," which appears to simply be a data-entry variant/typo of "Uvionix" rather than a third distinct company — no independent entity by that exact name was found.

Everything below concerns uAvionix Corporation, the Bigfork, Montana-based ADS-B/transponder/avionics company.

1. Company Overview

Legal name: uAvionix Corporation Founded: 2015 (a minority of secondary sources, including some LinkedIn/CB Insights listings, cite 2014, but the company's own press materials and the majority of press coverage — AOPA, AVweb, PitchBook, DC Capital Partners' own release — consistently use 2015 as the founding year) Founder: Paul Beard, who served as founding CEO and later transitioned to Chief Technology Officer (CTO) while remaining a company co-founder/principal. Headquarters: Dual-headquartered at 300 Pine Needle Ln, Bigfork, Montana 59911, and 880 Harrison St SE, Leesburg, Virginia 20175. Montana is generally cited as the primary/founding HQ; Leesburg appears to have become more prominent after the 2022 DC Capital Partners acquisition, reflecting closer proximity to Washington, D.C.-area government/defense customers. Mission: uAvionix describes its mission as "connecting everything that flies" — building small-form-factor communications, navigation, and surveillance (CNS) avionics that enable safe integration of unmanned aircraft systems (UAS/drones) into the National Airspace System (NAS) alongside crewed aircraft, while also serving the general aviation (GA) and defense/government markets. Industry: Aerospace & Defense — avionics, ADS-B (Automatic Dependent Surveillance–Broadcast), transponders, detect-and-avoid (DAA), command-and-control (C2) infrastructure for BVLOS (Beyond Visual Line of Sight) drone operations, and airport surface surveillance. Size: Estimated at roughly 40–66 employees depending on the data source and date (Tracxn: 41 employees as of July 2024; LinkedIn: 11–50; LeadIQ: ~66 as of March 2026 across North America and Europe; PitchBook: 57; RocketReach: ~60). It is a private company and does not disclose headcount officially. uAvionix has a secondary/European office presence (uavionix.store / European distribution) supporting UK CAA Electronic Conspicuity requirements. Ownership status: Formerly venture-backed and founder-led; since March 2022, majority-owned/controlled by private equity firm DC Capital Partners, LLC. Notable positioning: uAvionix bills itself as providing an "unparalleled engineering and management team" spanning avionics, surveillance, airport services, UAS aircraft development, RF engineering, and semiconductor design backgrounds. It has grown via both organic product development and acquisition (Aero Vonics in 2019, Iris Automation in 2023) into a broader CNS/detect-and-avoid/airport-surface-safety company rather than a single-product ADS-B vendor.

Sources: [uAvionix Press Room](https://uavionix.com/press/) (<https://uavionix.com/press/>), [PitchBook profile](https://pitchbook.com/profiles/company/157321-54) (<https://pitchbook.com/profiles/company/157321-54>), [AOPA — DC Capital acquisition](https://www.aopa.org) (<https://www.aopa.org>).

[org/news-and-media/all-news/2022/march/14/uavionix-acquired-by-dc-capital-partners](https://www.dccp.com/2022/03/09/dc-capital-partners-makes-strategic-investment-in-uavionix/)), DC Capital Partners release (<https://www.dccp.com/2022/03/09/dc-capital-partners-makes-strategic-investment-in-uavionix/>)

2. Financials

uAvionix is a privately held company and does not publish audited financial statements, so all figures below are third-party estimates and should be treated as directional, not authoritative:

- Revenue estimates: Highly inconsistent across data aggregators — Owler: \$5.0M–\$25M annually; Growjo: ~\$20.4M annually; RocketReach: ~\$7.7M (cited for 2025). Given the spread, a reasonable working estimate is single-digit to low-tens-of-millions of dollars in annual revenue, consistent with a specialty avionics hardware manufacturer serving GA, UAS, and now government/FAA infrastructure contracts.
- Valuation: PitchBook reported a \$28 million valuation tied to uAvionix's \$5 million Series A financing round (2016) — the only concrete, sourced valuation figure available. No later private valuation has been publicly disclosed; the 2022 DC Capital Partners buyout and the 2023 Iris Automation acquisition were both for undisclosed sums.
- M&A-driven; revenue growth: The Iris Automation acquisition (October 2023) and prior Aero Vonics acquisition (October 2019) were structured as adding complementary product lines (optical detect-and-avoid; aviation displays) rather than disclosed as revenue-accretive dollar figures.
- Government contract revenue (concrete, disclosed figures): In a 2026 procurement disclosed by the FAA, uAvionix's VTU-20 ADS-B Vehicle Movement Area Transmitter (VMAT) contract was valued at \$16.5 million, covering deployment across roughly 2,000 FAA Technical Operations vehicles nationwide (installation plus ongoing service). This is one of the few hard dollar figures publicly tied to the company. In addition, a November 2025 FAA Surface Awareness Initiative (SAI) award added 55 more airport tower deployments to an already-operational base of 14 towers, though the dollar value of that specific award was not disclosed in available coverage.
- Total disclosed funding raised (equity, pre-buyout): \$10 million across 2 rounds (see Funding History below); some aggregators (e.g., PitchBook) cite \$12M when including smaller/undisclosed tranches or grant-like Innovate UK funding.

Sources: PitchBook — uAvionix valued at \$28M with \$5M financing (<https://pitchbook.com/newsletter/uavionix-valued-at-28m-with-5m-financing>), PitchBook company profile (<https://pitchbook.com/profiles/company/157321-54>), FAA VMAT/VTU-20 contract — uAvionix press release (<https://uavionix.com/press/faa-to-equip-vehicles-with-uavionix-vehicle-ads-b-systems-to-improve-airport-safety/>), Owler (<https://www.owler.com/company/uavionix>), Growjo (<https://growjo.com/company/uAvionix>), RocketReach (https://rocketreach.co/uavionix-corporation-profile_b5580c82f6acb7fc)

3. Funding History

Round	Date	Amount	Lead / Investors	Notes
Series A	April 2016	\$5,000,000	Led by Playground Global (Silicon Valley VC firm founded by Andy Rubin, Bruce Leak, Matt Hershenson, Peter Barrett)	Reported valuation of \$28M post-money per PitchBook.
Series B	November 2017	\$5,000,000	Led by new investor Airbus Ventures, with continued participation from Playground Global and Redpoint Ventures	Brought cumulative funding to \$10M. Airbus Ventures CEO Thomas d'Halluin joined uAvionix's Board of Directors as part of the deal.

Round	Date	Amount	Lead / Investors	Notes
Strategic/grant funding	Various (undated in available sources)	Undisclosed	Innovate UK	Cited by PitchBook as an investor/funding source, likely tied to UK-market Electronic Conspicuity (SkyEcho) development and UK CAA compliance work; pushes some aggregate totals to ~\$12M.
Buyout / Majority Acquisition	March 9, 2022	Undisclosed	DC Capital Partners, LLC	Not a VC round — a private-equity acquisition/strategic investment giving DC Capital Partners control. Houlihan Lokey advised on the transaction. This was DC Capital's second acquisition for its (still-raising) third fund, following its August 2021 acquisition of Sionyx Corp.
Bolt-on acquisition (by uAvionix, as acquirer)	October 8, 2019	Undisclosed	uAvionix acquired Aero Vonics (Albuquerque, NM-based aviation display maker, founded 2018)	Product-line acquisition, not a funding round.
Bolt-on acquisition (by uAvionix, as acquirer)	October 24, 2023	Undisclosed	uAvionix acquired Iris Automation (Reno/San Francisco-based optical detect-and-avoid company)	Structured as a buyout/LBO under DC Capital Partners ownership; resulted in Iris CEO Jon Damush becoming uAvionix CEO.

Total disclosed pre-buyout equity funding: \$10 million (2 priced rounds, 3–4 named investors: Playground Global, Redpoint Ventures, Airbus Ventures, plus Innovate UK).

Sources: uAvionix Series B press release (<https://uavionix.com/press/uavionix-leader-in-drone-communications-navigation-and-surveillance-systems-secures-series-b-funding-from-airbus-ventures-playground-global-and-redpoint-ventures/>), PRNewswire — Series B (<https://www.prnewswire.com/news-releases/uavionix-leader-in-drone-communications-navigation-and-surveillance-systems-secures-series-b-funding-from-airbus-ventures-playground-global-and-redpoint-ventures-300562173.html>), VC News Daily — Series A (<https://www.vcnewsdaily.com/access/getarticle.php?aid=pfcfrldrcl>), DC Capital Partners release (<https://www.dccp.com/2022/03/09/dc-capital-partners-makes-strategic-investment-in-uavionix/>), AVweb — DC Capital purchase (<https://avweb.com/aviation-news/dc-capital-partners-purchases-uavionix/>), Crunchbase (<https://www.crunchbase.com/organization/uavionix>), CB Insights financials (<https://www.cbinsights.com/company/uavionix/financials>)

4. Leadership and People

- Paul Beard — Founder; original CEO; currently Chief Technology Officer (CTO) and co-founder. Beard has been publicly credited as the technical architect behind uAvionix's core ADS-B/transcoder patents and has been the company's most consistent public spokesperson through its history (including during the Garmin litigation and the DC Capital acquisition, where he was quoted affirming alignment with the new owners).
- Jon Damush — President and CEO of uAvionix Corporation, appointed in October 2023 concurrent with uAvionix's acquisition of Iris Automation, where Damush had been CEO. Damush brings 30+ years of aviation technology experience, including background at Insitu (Boeing subsidiary, maker of ScanEagle UAS) and Boeing itself. His appointment was described by DC Capital Partners' David Page as continuing the company's leadership strength while adding optical-DAA expertise.
- Christian Ramsey — Chief Commercial Officer, leading business development and growth across uAvionix's crewed aviation, uncrewed aviation, aviation-networks, and defense market segments.
- Ryan Braun — Chief Operating Officer. Braun was also the company spokesperson quoted in 2018 explaining the technical operation of the "Power Transcoder" patent (detecting transponder-induced noise on aircraft power wires rather than over-the-air signals).

- John Howard — Chief Financial Officer (per aggregated executive-team listings; not independently confirmed via a primary uAvionix press source in this research pass).
- David Page — Partner at DC Capital Partners, publicly engaged with uAvionix as an investor-side representative on strategic announcements (e.g., commenting on Damush's CEO appointment).
- Thomas J. Campbell — Founder and President of DC Capital Partners, quoted at the time of the March 2022 acquisition praising uAvionix's "highly skilled and talented workforce."
- Thomas d'Halluin — CEO of Airbus Ventures, joined uAvionix's Board of Directors in 2017 as part of the Series B investment.
- A full, current, independently verified board-of-directors roster could not be located in public sources; DC Capital Partners' control since 2022 implies its representatives (potentially including Page and/or Campbell) hold board seats, but this is inferred rather than directly confirmed by a primary source.

Sources: uAvionix — Iris Automation acquisition press release (<https://uavionix.com/press/a-new-era-for-aviation-safety-and-efficiency-uavionix-acquires-iris-automation/>), Commercial UAV News — interview with Jon Damush (<https://www.commercialuavnews.com/regulations/uavionix-acquires-iris-automation-an-interview-with-jon-damush>), DroneLife — uAvionix acquires Iris Aviation (<https://dronelife.com/2023/10/24/uavionix-acquires-iris-aviation/>), DC Capital Partners release (<https://www.dccp.com/2022/03/09/dc-capital-partners-makes-strategic-investment-in-uavionix/>), FlyingMag — Power Transcoder patent (<https://www.flyingmag.com/uavionix-granted-power-transcoder-patent/>), uAvionix Series B release (<https://uavionix.com/press/uavionix-leader-in-drone-communications-navigation-and-surveillance-systems-secures-series-b-funding-from-airbus-ventures-playground-global-and-redpoint-ventures/>)

5. Products

uAvionix's catalog spans three broad market segments: (A) General Aviation / crewed-aircraft avionics, (B) Uncrewed Aircraft Systems (UAS/drone) avionics and safety systems, and (C) Airport/ATC ground infrastructure. Below is the fullest catalog assembled from company materials and industry coverage.

A. General Aviation (Crewed Aircraft) Products

- skyBeacon — uAvionix's original certified wing-tip-mounted ADS-B Out solution; integrates ADS-B radio, GPS, and antennas into a replacement position/strobe light housing, cutting installation time to minutes versus traditional panel-mount ADS-B installs. This product line was the subject of the Garmin patent lawsuit (see Patents section).
- tailBeacon / tailBeaconX — Tail-mounted counterpart to skyBeacon; tailBeaconX is a certified ADS-B transponder solution leveraging tail-mounted antenna placement for optimal signal performance to both ground-based and space-based (satellite) surveillance infrastructure. Covered by U.S. Patent 11,011,830 ("Wireless Communication System within an External Enclosure for Attachment to a Vehicle," issued June 8, 2021).
- echoUAT — A 978 MHz Universal Access Transceiver (UAT)-based ADS-B solution, also implicated in the Garmin litigation alongside skyBeacon.
- ping978EC — A newly launched (March 2026) 978 MHz UAT ADS-B transceiver, part of a broader Portable Electronic Conspicuity (PEC) push, launched alongside a new "Pilot Awareness Website" educating pilots on PEC devices and aligning with the FAA's proposed Part 108 BVLOS rule.
- SkyEcho / SkyEcho 2 — A portable ADS-B In/Out transceiver conforming to the UK CAA's Electronic Conspicuity standard; broadcasts an aircraft's position, altitude, course, and speed to nearby aircraft. Popular in the UK/European general aviation and gliding markets (distributed via uavionix.store, the company's European storefront).
- AV-30-C (certified) / AV-30-E (experimental) — A panel-mounted, 3½-inch multifunction electronic flight instrument replacing legacy vacuum-driven gauges. Functions as an Attitude Indicator, Directional Gyro, Slip/Skid indicator, Angle-of-Attack (AoA) indicator, or multifunction display (MFD), with 4 base modes and 14+ functions. Mounts from behind the panel to preserve the classic "round dial" look without panel cutting.

- Technical specs: attitude rate limit $\pm 250^\circ$ /sec; attitude operational range 360° roll / 180° pitch; AV-30-E airspeed range 40–300 knots; altitude range –1,000 to +25,000 ft; AoA reliable up to $\pm 8G$, operational range -10° to $+30^\circ$, valid at 25–300 knots.
- AV-HSI — Expansion module adding IFR-approved lateral/vertical (glideslope) deviation display for enroute and approach navigation; integrates with Trio and TruTrak/Aerocruze autopilots; connects up to four AV-30-C units and provides an ARINC 429 interface to digital autopilots. Received FAA certification (PMA) around June 2025.
- AV-APA — Autopilot-integration module, certified alongside AV-HSI, described by the company as a "major step forward in IFR flight capability for general aviation."
- AV-Link — Wi-Fi bridge module (~\$349) enabling live ADS-B traffic display, wireless firmware updates, and (as of February 2026, via software v3.2.1) ForeFlight route-building/transfer and autopilot coupling on certified AV-30-C installs.
- SkyPlay — A jointly announced uAvionix/ForeFlight product unveiled around AirVenture Oshkosh 2026; further technical details were not available in this research pass.
- NanoHead — Transponder control head/interface product (per company patent/product listings).
- truFYX / truFYX EXT — GPS position-source products supporting ADS-B position integrity requirements.

B. Uncrewed Aircraft Systems (UAS/Drone) Products

- ping200X — FAA TSO-certified Mode S ADS-B Out transponder for UAS, enabling detection by Secondary Surveillance Radar (SSR), Traffic Collision Avoidance Systems (TCAS), and ADS-B In receivers — effectively making drones visible to crewed-aircraft traffic systems.
- ping200XR — Combines the TSO-certified ping200X transponder with an internal TSO-certified aviation-grade GPS receiver in one unit; achieved FAA TSO certification around August 2024.
- pingRX / pingRX Pro — A Detect-and-Avoid (DAA) ADS-B receiver for professional UAS, detecting cooperative aircraft on both 978 MHz (UAT) and 1090 MHz (Mode S/ES) frequencies. Marketed as providing "nearly 100x greater range" than radar, camera, or FMCW-based detect-and-avoid alternatives, giving operators more reaction time. Compatible with ArduPilot, PixHawk, and Auterion Skynode autopilot/flight-controller platforms for rapid integration. Described by the company as the evolution of the skyBeacon/tailBeacon lineage applied to the UAS market.
- pingStation — Ground-based ADS-B receiver station product (per company catalog listings).
- pingUSB — USB-based ADS-B receiver, likely aimed at developers/integrators and ground control station use.
- pingEFB — ADS-B receiver product oriented toward Electronic Flight Bag (EFB) integration.
- Legacy ping2020 / ping2020i / ping1090 / ping1090i — Earlier-generation ADS-B transceiver/receiver products in the "ping" family (predecessors to the current ping200X/pingRX lines).
- Casia X — Airborne, onboard optical Detect-and-Avoid (DAA) system (originating from the Iris Automation acquisition/technology lineage) that senses non-cooperative aircraft (those not broadcasting ADS-B) using patented computer-vision and AI, fused with integrated ADS-B for cooperative-aircraft coverage. Automatically executes collision-avoidance maneuvers via turnkey autopilot integration; reports detected intruders to ground station/pilot-in-command in real time; low Size-Weight-and-Power (SWaP) footprint for small UAS; detects/classifies small single-engine aircraft, single-rotor helicopters, birds, and multirotors; designed/assembled in the USA; FCC Part 15 and Industry Canada compliant.
- Casia G — Ground-based (ground-station-mounted) optical/AI surveillance system providing 360° full-optical-field-of-view detection and classification of GA aircraft and rotorcraft to enable automated drone-in-a-box (DFR — Drone as First Responder) surveillance and BVLOS operations. Fully passive (does not transmit), so it requires no FCC Station License or Special Temporary Authorization; expandable via mesh-networked nodes covering large areas without added FCC approval; integrates with uAvionix's FlightLine Portal and third-party UTM/DFR visualization platforms; mounts on walls/poles, requires 110V power plus Ethernet data and clear sky view. As of March 2025, Casia G gained nighttime aircraft detection capability, enabling 24/7 BVLOS operations. (Note: per uAvionix's support site, "Casia G" as a standalone SKU is now listed as discontinued, likely superseded/folded into newer FlightLine-integrated

ground-surveillance offerings — exact successor SKU naming was not fully confirmed in this pass.)

- SkyLink — Command-and-Control (C2)/CNPC (Control and Non-Payload Communications) radios for BVLOS drone links.
- SkyStation — Ground-based C2 radio infrastructure counterpart to SkyLink.
- Scout and Sentry — Additional named products in uAvionix's catalog/patent listings (specific specs not independently confirmed beyond naming in aggregator/patent-page summaries).
- skyLight and skySensor — Additional named UAS avionics products per company patent/product-page listings.

C. Airport / Air Traffic / Ground Infrastructure Products

- FlightLine (Surface Awareness Initiative / SAI system) — uAvionix's surface-surveillance system for airport runway/taxiway situational awareness, sold to and deployed by the FAA. As of the most recent reporting, FlightLine SAI is fully operational at 14 FAA-towered airports, with a November 2025 FAA award expanding deployment to 55 additional towers (12-month rollout target), supporting the FAA's broader goal of equipping all its towers with surface-awareness technology within two years, aligned with the "Brand New Air Traffic Control System" (BNATCS) initiative.
- VTU-20 — Vehicle Movement Area Transmitter (VMAT): an ADS-B transponder for airport ground vehicles (e.g., FAA Tech Ops trucks), making them visible on the same controller/pilot displays as aircraft. In May 2026, the FAA procured VTU-20 units for roughly 2,000 Technical Operations vehicles nationwide in a deal valued at \$16.5 million including installation and ongoing service.
- FlightLine Airborne Position Reference Tool — First installed at Glacier Park International Airport (Kalispell, Montana) around October 2025, per AVweb coverage of the SAI expansion.
- FlightLine Portal — Software/data platform into which Casia G and other ground sensors feed; also the integration point cited in the 2026 Airwise Solutions/Skyway Range partnership, where uAvionix FlightLine surveillance data feeds into Airwise's "Nexus" common operating picture, eliminating the need for BVLOS operators to deploy their own local sensors and instead consuming "FAA-grade" airspace traffic data as a service.
- Command-and-Control (C2) network services — uAvionix has also functioned as an infrastructure/service provider (not just a hardware vendor), e.g., providing the terrestrial C2 infrastructure and cloud-based C2 service for Vantis, North Dakota's statewide BVLOS UAS network operated by the Northern Plains UAS Test Site — the network on which uAvionix itself received the FAA's first BVLOS approval in January 2023.

Sources: uAvionix pingRX Pro announcement (<https://uavionix.com/press/introducing-pingrx-pro/>), uAvionix products page (<https://uavionix.com/products/>), AV-30-C product page (<https://uavionix.com/product/av-30-c-certified/>), AV-30-E product page (<https://uavionix.com/product/av-30-exp/>), Casia G support page (<https://support.uavionix.com/hc/en-us/sections/47107221806099-Casia-G-Discontinued>), Casia X datasheet PDF (https://uavionix.com/wp-content/uploads/dlm_uploads/2024/12/DATASHEET-Casia-X-8.9M-P-GigE-3.4-XACM-0100_HEOC-0009-V1.0.pdf), AV-HSI/AV-APA FAA certification press release (<https://uavionix.com/press/uavionix-announces-faa-certification-for-av-hsi-and-av-apa-major-step-forward-in-ifr-flight-capability-for-general-aviation/>), AVweb — AV-30-C ForeFlight integration (<https://avweb.com/avionics/uavionix-adds-foreflight-certified-av-30-c/>), AOPA — AV-30-C ForeFlight/autopilot (<https://www.aopa.org/news-and-media/all-news/2026/february/26/av-30-c-gains-for-eflight-integration-advanced-autopilot>), AVweb — FAA adds 55 towers to SAI (<https://avweb.com/aviation-news/faa-adds-55-towers-to-uavionix-sai-program/>), Airports International — SAI contract (<https://www.airportsinternational.com/article/uavionix-wins-large-faa-sai-contract>), FAA VMAT contract release (<https://uavionix.com/press/faa-to-equip-vehicles-with-uavionix-vehicle-ads-b-systems-to-improve-airport-safety/>), Airwise Solutions partnership (https://airwisesolutions.com/company/airwise_solutions_uavionix_partner/), uAvionix/Vantis C2 press release (<https://uavionix.com/press/uavionix-to-provide-bvlos-c2-infrastructure-and-service-to-vantis-north-dakotas-uas-network/>), DroneLife — BVLOS Vantis approval (<https://dronelife.com/2023/01/09/uavionix-win-s-faa-approval-for-bvlos-flights-in-north-dakota-drone-network-a-first-for-vantis/>)

6. Patents

uAvionix has actively built and defended a patent portfolio, and patent strength has been directly relevant to a major legal dispute in its history.

- U.S. Patent 10,156,627 — "Aircraft Navigation Light ADS-B Radio" (the "Power Transcoder" patent). Application accepted by USPTO announced December 4, 2018; issued December 18, 2018. Covers technology synchronizing squawk code and Mode C altitude data between an aircraft's secondary-surveillance-radar (SSR) transponder replies and its ADS-B transmissions — notably by detecting electrical noise induced on the aircraft's power wiring during transponder transmission, rather than capturing the data over the air. This design underlies the skyBeacon/tailBeacon/echoUAT product family and was central to defending against Garmin's infringement claims (see below).
- U.S. Patent 11,011,830 — "Wireless Communication System within an External Enclosure for Attachment to a Vehicle." Issued June 8, 2021. Covers leveraging wing/tail mounting locations on an aircraft to optimize transponder performance for both ground-based and satellite-based surveillance infrastructure; underlies the tailBeaconX product.
- U.S. Patent 10,991,260 — "Intelligent Non-Disruptive ADS-B Integration for Unmanned Aircraft Systems (UAS)." Granted April 27, 2021. Covers a method allowing UAS to gain ADS-B safety benefits ("inert and alert" capability) while minimizing radio-spectrum utilization — relevant to the UAS/drone ADS-B receiver product line (pingRX, ping200X family).
- U.S. Patent 10,890,657 — "ADS-B transponder system and method." Relates to vehicle-control systems incorporating 978 MHz UAT transponders for aircraft that must also carry 1090 MHz Mode A/C transponders, addressing FAA ADS-B dual-frequency compliance.
- Patented computer-vision/AI detection technology underlying the Casia X/Casia G optical detect-and-avoid systems (inherited via the Iris Automation acquisition) — used to sense and classify non-cooperative aircraft (those without ADS-B) via camera and machine-learning models; specific patent numbers for this technology were not identified in available public sources during this research pass and would require direct USPTO assignee-name searches under "Iris Automation" for full enumeration.

The Garmin Litigation (IP Moat Context): In June 2018, Garmin International sued uAvionix in Montana federal court, alleging that uAvionix's skyBeacon and echoUAT products infringed Garmin's U.S. Patent 8,102,301, specifically regarding methods of obtaining transponder Mode 3/A code and altitude data for ADS-B broadcast. uAvionix denied infringement, asserting it used a different, patent-pending method (the Power Transcoder approach later patented as US 10,156,627). The case was closely watched industry-wide as a proxy for how market leader Garmin (dominant in GA panel/remote-mount avionics) would respond to lower-cost new entrants. The dispute settled on confidential terms on October 17, 2019 — no injunction was issued, no damages were publicly disclosed, and uAvionix retained the right to continue manufacturing and selling skyBeacon, tailBeacon, and echoUAT. uAvionix's own December 2018 patent grant (during the litigation) was publicly framed by the company as evidence the USPTO recognized "distinct and innovative differences" between its solution and Garmin's.

IP strategy / moat assessment: uAvionix's patent strategy centers on protecting the specific *engineering methods* that let it build low-cost, minutes-to-install ADS-B hardware (the transcoder/power-line-noise-detection approach, and wing/tail-mount RF engineering) rather than broad platform patents. This is a defensible but narrow moat — it protects specific implementations against direct copying (as demonstrated by successfully surviving the Garmin suit) but does not obviously block competitors from building ADS-B hardware via different engineering approaches. The 2023 Iris Automation acquisition materially broadened the IP moat by adding computer-vision/AI-based non-cooperative-aircraft detection — a different technical approach (optical vs. RF/transponder-based) that diversifies uAvionix's defensible technology base and reduces reliance on any single patent family. Given the company's now-diversified position across (a) certified GA ADS-B/EFIS hardware, (b) UAS-specific ADS-B/DAA hardware, (c) optical DAA (via Iris), and (d) government airport-surface-surveillance contracts (FlightLine/VTU-20), its moat is increasingly a combination of patented hardware techniques, FAA certification/TSO approvals (a significant regulatory barrier to entry in itself), and incumbency in government procurement (e.g., the sole-source-feeling FAA SAI and VMAT contracts), rather than patents alone.

Sources: AOPA — transcoder patent approved (<https://www.aopa.org/news-and-media/all-news/2018/december/06/uavionix-transcoder-patent-approved>), FlyingMag — Power Transcoder patent granted (<https://www.flyingmag.com/uavionix-granted-power-transcoder-patent/>), uAvionix — tailBeaconX patent (<https://uavionix.com/uavionix-receives-patent-for-tailbeaconx-transponder/>), uAvionix — ADS-B Inert & Alert patent for UAS (<https://uavionix.com/uavionix-receives-patent-for-ads-b-inert-alert-capability-for-uas/>), USPTO — Patent 10,890,657 PDF (<https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10890657>), AOPA — uAvionix sued for patent infringement (<https://www.aopa.org/news-and-media/all-news/2018/august/21/uavionix-sued-for-patent-infringement>), FlyingMag — Garmin/uAvionix lawsuit background (<https://www.flyingmag.com/garmin-uavionix-patent-lawsuit/>), AOPA — Garmin, uAvionix settle lawsuit (<https://www.aopa.org/news-and-media/all-news/2019/october/17/uavionix-garmin-settle-suit>), AVweb — Garmin, uAvionix resolve patent lawsuit (<https://avweb.com/avionics/adsb/garmin-uavionix-resolve-patent-lawsuit/>), uAvionix — statement on patent resolution (<https://uavionix.com/uavionix-statement-patent/>)

7. Latest News (2024–2026)

- August 2024 (approx.): uAvionix achieved FAA TSO certification for the ping200XR, its combined Mode S transponder and internal TSO-certified GPS receiver for UAS.
- October 2024: uAvionix's FlightLine Airborne Position Reference Tool was first installed at Glacier Park International Airport (Kalispell, Montana), an early proof point for its airport-surface-safety expansion.
- June 2024: uAvionix received its first approvals under the FAA Surface Awareness Initiative (SAI) program, aimed at improving controller situational awareness and reducing runway-incursion risk.
- March 2025: uAvionix announced that its Casia G ground-based optical detect-and-avoid system gained nighttime aircraft-detection capability, enabling round-the-clock (24/7) BVLOS drone operations — a significant operational unlock for commercial drone operators previously restricted to daylight BVLOS flights.
- June 2025: uAvionix announced FAA certification of its AV-HSI and AV-APA modules for the AV-30-C, described as "a major step forward in IFR flight capability for general aviation," enabling full instrument-approach navigation guidance and digital-autopilot integration on legacy GA aircraft equipped with the AV-30-C.
- November 2025: The FAA awarded uAvionix a contract to expand FlightLine SAI deployment to 55 additional air-traffic-control towers (on top of 14 already operational), with a 12-month installation/full-operational-status target, plus deployment of VTU-20 VMAT units on FAA Technical Operations vehicles so ground vehicles appear on the same displays as aircraft. This rollout is explicitly tied to the FAA's push to equip all its towers with surface-awareness technology within two years and aligns with the agency's "Brand New Air Traffic Control System" (BNATCS) modernization vision.
- January 2026: *Avionics News* (Aircraft Electronics Association publication) ran a feature headlined "FAA Approves uAvionix," consistent with continued regulatory momentum (specific certification not fully detailed in available excerpts).
- February 2026: uAvionix released AV-30 software v3.2.1, bringing ForeFlight integration and autopilot coupling to the certified AV-30-C (previously available only on the experimental AV-30-E), letting pilots build/modify flight routes in ForeFlight and wirelessly transfer them via the optional AV-Link Wi-Fi bridge module (\$349).
- March 2026: uAvionix launched the ping978EC, a new 978 MHz UAT ADS-B transceiver, alongside a new "Pilot Awareness Website" educating pilots on Portable Electronic Conspicuity (PEC) devices — positioned explicitly around the FAA's proposed Part 108 rulemaking for BVLOS drone operations.
- May 2026: The FAA procured uAvionix VTU-20 VMAT units for a \$16.5 million contract (including installation and ongoing service) to equip roughly 2,000 FAA Technical Operations vehicles nationwide with ADS-B, closing a long-standing surface-visibility gap between ground vehicles and aircraft/controllers.

- August 2026: uAvionix and Airwise Solutions announced a strategic partnership to support Skyway Range, integrating uAvionix's FlightLine surveillance data into Airwise's "Nexus" common operating picture — pitched as letting BVLOS operators avoid deploying their own local sensors in favor of consuming "FAA-grade" airspace-traffic data as a service. Around the same time (AirVenture Oshkosh 2026), uAvionix and ForeFlight unveiled a jointly branded product called "SkyPlay," though granular product details were not available in this research pass.

Overall 2024–2026 narrative: uAvionix has continued to diversify beyond its original consumer/GA and drone-hobbyist ADS-B hardware roots into being a significant FAA infrastructure contractor (airport surface surveillance, vehicle ADS-B) while simultaneously deepening its general aviation avionics ecosystem (AV-30 family plus ForeFlight integrations) and its BVLOS-enabling detect-and-avoid technology (Casia nighttime capability, Part 108-aligned PEC push). This reflects the strategic logic of the 2022 DC Capital Partners buyout and 2023 Iris Automation acquisition: leveraging private-equity backing and broadened optical/AI DAA capability to win larger, higher-value government and infrastructure contracts rather than relying solely on smaller-ticket avionics hardware sales.

Sources: uAvionix Press Room (<https://uavionix.com/press/>), uAvionix — AV-HSI/AV-APA certification (<https://uavionix.com/press/uavionix-announces-faa-certification-for-av-hsi-and-av-apa-major-step-forward-in-ifr-flight-capability-for-general-aviation/>), AVweb — FAA adds 55 towers to SAI program (<https://avweb.com/aviation-news/faa-adds-55-towers-to-uavionix-sai-program/>), Airports International — SAI contract win (<https://www.airportsinternational.com/article/uavionix-wins-large-faa-sai-contract/>), Avionics News — FAA Approves uAvionix (Jan 2026) (<https://digital.avionicsnews.net/articles/faa-approves-uavionix/>), AVweb — AV-30-C ForeFlight/autopilot (<https://avweb.com/avionics/uavionix-adds-foreflight-certified-av-30-c/>), AOPA — AV-30-C ForeFlight coverage (<https://www.aopa.org/news-and-media/all-news/2026/february/26/av-30-c-gains-foreflight-integration-advanced-autopilot>), uAvionix — FAA VMAT/VTU-20 contract (<https://uavionix.com/press/faa-to-equip-vehicles-with-uavionix-vehicle-ads-b-systems-to-improve-airport-safety/>), Airwise Solutions partnership announcement (https://airwisesolutions.com/company/airwise_solutions_uavionix_partner/)

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- uAvionix — FAA Grants Initial BVLOS Approval press release (<https://uavionix.com/press/faa-grants-initial-approval-allowing-bvlos-drone-flights-on-north-dakota-drone-system/>)

- uAvionix — Statement on Garmin patent lawsuit resolution (<https://uavionix.com/uavionix-statement-patent/>)
- uAvionix — tailBeaconX patent announcement (<https://uavionix.com/uavionix-receives-patent-for-tailbeaconx-transponder/>)
- uAvionix — ADS-B Inert & Alert for UAS patent announcement (<https://uavionix.com/uavionix-receives-patent-for-ads-b-inert-alert-capability-for-uas/>)
- uAvionix Announces Acquisition by DC Capital Partners (<https://uavionix.com/press/uavionix-announces-acquisition-by-dc-capital-partners/>)
- PitchBook — uAvionix company profile (<https://pitchbook.com/profiles/company/157321-54>)
- PitchBook — uAvionix valued at \$28M with \$5M financing (<https://pitchbook.com/newsletter/uavionix-valued-at-28m-with-5m-financing>)
- PitchBook — AeroVionics company profile (<https://pitchbook.com/profiles/company/471853-00>)
- Crunchbase — uAvionix (<https://www.crunchbase.com/organization/uavionix>)
- Crunchbase — UVIONIX (<https://www.crunchbase.com/organization/uvionix>)
- CB Insights — uAvionix financials (<https://www.cbinsights.com/company/uavionix/financials>)
- Tracxn — uAvionix company profile (https://tracxn.com/d/companies/uavionix/_RBr6QpDot0pvtwyNGdAX6xZ1I30nllG4meV5mUZQuXw)
- Tracxn — Uvionix Aerospace Corporation (https://tracxn.com/d/companies/uvionix-aerospace-corporation/_weRRSuTLZ7_73pwPQRW7JeTXwxqLaU2tb7r7-tApbQ)
- Owler — uAvionix (<https://www.owler.com/company/uavionix>)
- Growjo — uAvionix (<https://growjo.com/company/uAvionix>)
- RocketReach — uAvionix Corporation (https://rocketreach.co/uavionix-corporation-profile_b5580c82f6acb7fc)
- LeadIQ — uAvionix Corporation (<https://leadiq.com/c/uavionix-corporation/5a1d885d240000240062166d>)
- ZoomInfo — uAvionix (<https://www.zoominfo.com/c/uavionix-corp/374992928>)
- UVIONIX official site (<https://www.uvionix.com/about>)
- AOPA — uAvionix acquired by DC Capital Partners (<https://www.aopa.org/news-and-media/all-news/2022/march/14/uavionix-acquired-by-dc-capital-partners>)
- AVweb — DC Capital Partners Purchases uAvionix (<https://avweb.com/aviation-news/dc-capital-partners-purchases-uavionix/>)
- GovConWire — DC Capital Partners Buys uAvionix (<https://www.govconwire.com/articles/dc-capital-partners-buys-uas-communications-tech-developer-uavionix>)
- Washington Technology — DC Capital Partners acquires unmanned avionics provider (<https://www.washingtontechnology.com/companies/2022/03/dc-capital-partners-acquires-unmanned-avionics-provider/362972/>)
- PRNewswire — DC Capital Partners Makes Strategic Investment in uAvionix (<https://www.prnewswire.com/news-releases/dc-capital-partners-makes-strategic-investment-in-uavionix-301499363.html>)
- DC Capital Partners — official announcement (<https://www.dccp.com/2022/03/09/dc-capital-partners-makes-strategic-investment-in-uavionix/>)
- sUAS News — uAvionix Announces Acquisition by DC Capital Partners (<https://www.suasnews.com/2022/03/uavionix-announces-acquisition-by-dc-capital-partners/>)
- Aero-News Network — uAvionix Acquires Iris Automation (<https://www.aero-news.net/index.cfm?do=main.textpost&id=358874ee-1469-42fd-ae8-1234435a7a66>)
- Vertical Mag — uAvionix acquires Iris Automation (<https://verticalmag.com/press-releases/uavionix-acquires-iris-automation/>)
- DroneLife — uAvionix Acquires Iris Aviation (<https://dronelife.com/2023/10/24/uavionix-acquires-iris-aviation/>)
- DroneDJ — uAvionix acquires Iris Automation (<https://dronedj.com/2023/10/27/uavionix-acquires-iris-automation-reinforcing-drone-integration-tech/>)

- Iris Onboard — acquisition announcement (<https://www.irisonboard.com/a-new-era-for-aviation-safety-and-efficiency-uavionix-acquires-iris-automation/>)
- Commercial UAV News — interview with Jon Damush (<https://www.commercialuavnews.com/regulations/uavionix-acquires-iris-automation-an-interview-with-jon-damush>)
- DroneLife — uAvionix wins FAA BVLOS approval, Vantis (<https://dronelife.com/2023/01/09/uavionix-wins-faa-approval-for-bvlos-flights-in-north-dakota-drone-network-a-first-for-vantis/>)
- Unmanned Airspace — Reliable Robotics and Vantis partnership (<https://www.unmannedairspace.info/latest-news-and-information/reliable-robotics-and-vantis-partner-to-advance-automated-aviation-in-north-dakota/>)
- Unmanned Airspace — uAvionix BVLOS C2 for Vantis (<https://www.unmannedairspace.info/latest-news-and-information/uavionix-to-provide-bvlos-c2-infrastructure-and-service-to-vantis-north-dakotas-uas-network/>)
- Inside Unmanned Systems — FAA Grants BVLOS Approval in North Dakota (<https://insideunmannedsystems.com/faa-grants-approval-allowing-bvlos-drone-flights-in-north-dakota/>)
- Vantis UAS — News (<https://www.vantisuas.com/news/article/faa-grants-initial-approval-allowing-bvlos-drone-flights-on-north-dakota-drone-system>)
- AOPA — uAvionix transcoder patent approved (<https://www.aopa.org/news-and-media/all-news/2018/december/06/uavionix-transcoder-patent-approved>)
- FlyingMag — uAvionix Granted Power Transcoder Patent (<https://www.flyingmag.com/uavionix-granted-power-transcoder-patent/>)
- USPTO — Patent 10.890.657 (PDF) (<https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10890657>)
- Aviation Consumer — Why The Latest uAvionix Patent Is Significant (<http://aviationconsumer.com/aircraftreviews/why-the-latest-uavionix-patent-is-significant/>)
- Sporty's — tailBeacon product listing (<https://www.sportys.com/tailbeacon-ads-b-out-transmitter-by-uavionix.html>)
- AOPA — uAvionix sued for patent infringement (<https://www.aopa.org/news-and-media/all-news/2018/august/21/uavionix-sued-for-patent-infringement>)
- FlyingMag — Garmin and uAvionix Prepare to Tangle in Court (<https://www.flyingmag.com/garmin-uavionix-patent-lawsuit/>)
- Plane & Pilot — Garmin, uAvionix Patent Lawsuit Settled (<https://www.planeandpilotmag.com/article/garmin-uavionix-patent-lawsuit-settled/>)
- KITPLANES — Garmin, uAvionix Resolve Patent Lawsuit (<https://www.kitplanes.com/garmin-uavionix-resolve-patent-lawsuit/>)
- Aviation Week — Garmin, uAvionix Resolve Patent Lawsuit (<https://aviationweek.com/business-aviation/garmin-uavionix-resolve-patent-lawsuit>)
- AOPA — Garmin, uAvionix settle lawsuit (<https://www.aopa.org/news-and-media/all-news/2019/october/17/uavionix-garmin-settle-suit>)
- AVweb — Garmin, uAvionix Resolve Patent Lawsuit (<http://avweb.com/avionics/adsb/garmin-uavionix-resolve-patent-lawsuit/>)
- AVweb — FAA Adds 55 Towers to uAvionix SAI Program (<https://avweb.com/aviation-news/faa-adds-55-towers-to-uavionix-sai-program/>)
- Airports International — uAvionix wins large FAA SAI contract (<https://www.airportsinternational.com/article/uavionix-wins-large-faa-sai-contract>)
- Market Forecast — Innovative Avionics Leader uAvionix Lands Major FAA Certification (<https://www.marketforecast.com/industrynews/innovative-avionics-leader-uavionix-lands-major-faa-certification-64341>)
- Avionics News (AEA) — FAA Approves uAvionix (Jan 2026) (<https://digital.avionicsnews.net/articles/faa-approves-uavionix>)
- AVweb — uAvionix Adds ForeFlight Integration, Autopilot Coupling To AV-30-C (<https://avweb.com/avionics/uavionix-adds-foreflight-certified-av-30-c/>)

- AOPA — AV-30-C gains ForeFlight integration, advanced autopilot (<https://www.aopa.org/news-and-media/all-news/2026/february/26/av-30-c-gains-foreflight-integration-advanced-autopilot>)
- Airwise Solutions — uAvionix partnership announcement (https://airwisesolutions.com/company/airwise_solutions_uavionix_partner/)
- AUVSI — uAvionix Aims to "Connect Everything That Flies" (<https://www.auvsi.org/uavionix-aims-connect-everything-flies>)
- Unmanned Systems Technology — uAvionix Corporation profile (<https://www.unmannedsystemstechnology.com/company/uavionix-corporation/>)

RESEARCH REPORT: "ZERODRAG" (drone/UAS)

0. DISAMBIGUATION — read first

Three distinct things carry the "Zerodrag" / "zero drag" name. Only one is a drone/UAS company:

1. ZeroDrag Technologies Private Limited (India) — the subject of this report. An Indian deep-tech UAV avionics and drone-electronics manufacturer, founded 2022, registered in Nagpur, Maharashtra, operating from IIT Delhi's Research & Innovation Park and a Noida office. Web: zerodrag.in (<https://zerodrag.in/>). This is the only entity plausibly a competitor to Vector Technics.
2. ZeroDrag™ (inline skate bearings, US) — zerodrag.com (<https://zerodrag.com/>) is a completely unrelated sports-bearing brand that says it has "been in the sports bearing business for over 20 years," selling bearings for fitness skating, roller hockey and marathon racing. No aerospace connection whatsoever. Note the trap: the .com is NOT the drone company; the drone company is the .in.
3. "zero drag" as an aerodynamics marketing phrase — appears in UAV copy from unrelated firms (e.g. PteroDynamics' Transwing VTOL marketing around low-drag configurations, pterodynamics.com (<https://pterodynamics.com/>)). Not a company name.

No US or European defence contractor named "Zerodrag" was found in any source. Everything below concerns entity #1.

Note also the company's former name/brand: "DroneAdda." CB Insights still files the company under that name ([cbinsights.com/company/droneadda](https://www.cbinsights.com/company/droneadda) (<https://www.cbinsights.com/company/droneadda>)), and TheCompanyCheck lists two associated brands for the same CIN: "DroneAdda" and "Zerodrag." If you are deduplicating a database, DroneAdda and Zerodrag are the same legal entity.

1. COMPANY OVERVIEW

Legal entity: ZERODRAG TECHNOLOGIES PRIVATE LIMITED CIN: U29100MH2022PTC388953 Incorporated: 22 August 2022 Registrar of Companies: Nagpur (Maharashtra) Company type: Private, non-government, limited by shares Status: Active / "Active Compliant" DPIIT status: Recognised Startup (Source: [thecompanycheck.com](https://www.thecompanycheck.com/company/zerodrag-technologies-private-limited/U29100MH2022PTC388953) (<https://www.thecompanycheck.com/company/zerodrag-technologies-private-limited/U29100MH2022PTC388953>), [tofler.in](https://www.tofler.in/zerodrag-technologies-private-limited/company/U29100MH2022PTC388953) (<https://www.tofler.in/zerodrag-technologies-private-limited/company/U29100MH2022PTC388953>), tracxn.com (https://tracxn.com/d/companies/zerodrag/_COdrEKeVnxdEHzo5bHdARINM9Mn6wwR69RSWDJVq4W4))

Addresses (three, and they conflict across sources — all appear legitimate):

- Registered office (MCA of record): H. No. 3150-A, Diamond Nagar, Nagpur, Maharashtra 440022
- Operations/office listed on company website: A-27, 2nd Floor, Sector-3, Gautam Buddha Nagar, Noida, Uttar Pradesh 201301 ([zerodrag.in about page](https://zerodrag.in/pages/about-us-made-in-india-flight-controller-esc-els-vtx) (<https://zerodrag.in/pages/about-us-made-in-india-flight-controller-esc-els-vtx>))
- Incubator address / LinkedIn HQ: Research and Innovation Park, Block-A, Hauz Khas, IIT Delhi, New Delhi 110016 (in.linkedin.com/company/zerodrag (<https://in.linkedin.com/company/zerodrag>), CB Insights (<https://www.cbinsights.com/company/droneadda>))

This explains why press coverage variously calls it "Nagpur-based" (YourStory, Drone Federation India, Entrepreneur India) and "Delhi-based" (Tracxn, CB Insights). The correct framing: incorporated and registered in Nagpur, incubated and originally headquartered at IIT Delhi's R&I Park, with an NCR/Noida commercial address. YourStory's July 2026 feature states plainly that the company "began at the IIT Delhi research park, where it was incubated, and now operates out of Nagpur" (yourstory.com/2026/07/zerodrag-indian-drone-hardware (<https://yourstory.com/2026/07/zerodrag-indian-drone-hardware>)).

Industry classification (MCA): Main Activity Group C (Manufacturing); Business Activity Code C10 — "Electrical equipment, General Purpose and Special purpose Machinery & equipment, Transport equipment"; described as "Electronic Components & Equipment Manufacturing."

Incubation and government support:

- Incubated at FITT, IIT Delhi (Foundation for Innovation and Technology Transfer)
- Pre-incubated at NSRCEL, IIM Bangalore (Launchpad program)
- Support cited via Startup India, NIDHI, and MeitY-supported programs

(Sources: [zerodrag.in about](https://zerodrag.in/about)

(<https://zerodrag.in/pages/about-us-made-in-india-flight-controller-esc-els-vtx>), CB Insights

(<https://www.cbinsights.com/company/droneadda>), startupsamadhan.com

(<https://startupsamadhan.com/zerodrag-raises-6-5-crore-seed-transition-vc-drone-avionics-india/>))

Headcount (conflicting, all small):

- Inc42: 15 employees, +25% over the trailing 90 days (inc42.com/company/zerodrag (<https://inc42.com/company/zerodrag/>))
- Tracxn: 16 employees as of 31 July 2026
- LinkedIn: self-declared size band "2–10 employees," but 17 people listed as employees on the platform; 5,649 followers
- EPFO (via TheCompanyCheck): 1 active EPFO establishment, "0 employees as of Apr 08, 2024" — clearly a stale/underreported filing

Mission and positioning. ZeroDrag does not build drones. It builds the avionics and core electronics layer that drones run on — flight controllers/autopilots, ESCs (motor controllers), GNSS/navigation, telemetry and radio links, and payload-integration hardware — and sells them B2B to drone OEMs on an explicitly OEM-agnostic, cross-platform basis. The founder's formulation, quoted repeatedly in coverage: "We're not building drones. We're building what every drone needs." (startupsamadhan.com

(<https://startupsamadhan.com/zerodrag-raises-6-5-crore-seed-transition-vc-drone-avionics-india/>))

The strategic pitch is import substitution and supply-chain sovereignty: India has scaled domestic airframe manufacturing under PLI and related schemes, but the avionics inside those airframes remain overwhelmingly Chinese-sourced (DJI, Holybro, SpeedyBee and similar). ZeroDrag positions itself as the indigenous alternative for that layer, serving defence, agriculture, logistics, industrial inspection, and the FPV/racing community. Coverage consistently notes rising inbound interest from US and European manufacturers and governments seeking non-Chinese component sources (yourstory.com (<https://yourstory.com/2026/07/zerodrag-indian-drone-hardware>), [dronefederation.in](https://www.dronefederation.in) (<https://www.dronefederation.in/media-hub/zero-drag-raise-funds-to-build-drone-avionics-layer>)).

Go-to-market. Direct D2C via a Shopify storefront at zerodrag.in, plus six named Authorised Channel Partners in India: ZBOTIC, Vega Drones, TujoRC, Robu.in, drkstore.in, and Quadkart. Also resold by UAVGarage (uavgarage.com/product-brand/zerodrag (<https://uavgarage.com/product-brand/zerodrag/>)). B2B/OEM sales are direct, with custom design and white-labelling offered for bulk orders. Warranty is 6–12 months depending on product line (12 months on the core boards).

Contact: support@thezerodrag.com; technical support/WhatsApp +91 98217 34544; sales +91 92112 25855; MCA-registered email zerodragtech@gmail.com.

2. FINANCIALS

ZeroDrag is a private Indian company; financials come from MCA/ROC filings surfaced by Tofler and TheCompanyCheck. It is small and currently loss-making.

Revenue:

- FY2025 (year ended 31 March 2025): 27.61 lakh (~USD 33K) — TheCompanyCheck
- FY2024: ~42.1 lakh (implied)
- YoY change: –34.4% (Inc42 states "decrease of 34.4% from FY24"; Tofler shows –34.44%; TheCompanyCheck shows "▼ 34.00% YoY")
- Tofler revenue bracket: "< 1 crore"
- Tracxn revenue band: 0–10 Cr as of 31 Mar 2025

(Sources: inc42.com/company/zerodrag/ (<https://inc42.com/company/zerodrag/>), [thecompanycheck.com](https://www.thecompanycheck.com/company/zerodrag-technologies-private-limited/U29100MH2022PTC388953) (<https://www.thecompanycheck.com/company/zerodrag-technologies-private-limited/U29100MH2022PTC388953>), [tofler.in](https://www.tofler.in/zerodrag-technologies-private-limited/company/U29100MH2022PTC388953) (<https://www.tofler.in/zerodrag-technologies-private-limited/company/U29100MH2022PTC388953>))

A revenue *decline* in FY25 alongside heavy product launches is worth flagging — it most plausibly reflects a pivot from the earlier DroneAdda reseller/service model toward own-manufactured hardware, but no source states this explicitly.

Profitability and balance-sheet ratios (FY2025, Tofler; absolute rupee figures are paywalled):

- Net Profit Margin: -133.96
- Return on Equity: -84.78
- ROCE: -43.29
- Debt-to-Equity: 0.17
- Networth change YoY: -58.19%
- Total Assets change YoY: -52.22%
- Borrowings change YoY: 0.0%

Read together: the company burned roughly half its net worth and assets in FY25 while running a loss materially larger than its revenue — normal for a pre-scale hardware startup between rounds, and consistent with the 6.5 Cr raise that followed in May 2026.

Capital structure (note the source conflict, which is a filing-lag artefact):

- TheCompanyCheck (more recent): Authorised capital 66 lakh; paid-up (issued & subscribed) 51.2 lakh
- Tofler (stale snapshot): Authorised 15.0 lakh; paid-up 1.2 lakh

Debt / charges: None. Both Tofler and TheCompanyCheck state the company has no registered charges with the RoC — i.e., no secured borrowing. Equity-funded only.

Compliance:

- Balance sheet date: 31 March 2025
- Last AGM: 10 September 2025 (TheCompanyCheck); Tofler's stale record shows 27 December 2023
- Auditor: J.S. Uberoi & Co (FY2023 appointment)

Valuation: Not disclosed. Tracxn redacts it ("*\$") as of the 4 May 2026 round. No press report gives a post-money. The only adjacent datapoint: Tracxn estimates the founders' collective net worth at 28.9 Cr as of 4 May 2026, which — given a founder stake of 61.11% — implies an equity valuation loosely in the 45–50 Cr (~USD 5–6M) range if taken at face value. Treat that as inference, not a reported figure.

Cap table (Tracxn):

Holder class	%
Founders	61.11%
Funds (institutional)	21.72%
ESOP pool	9.92%
Angels	6.01%
Enterprise	1.24%

No public filings beyond MCA. No IPO, no SEC/Companies House presence, no bond or debt issuance.

3. FUNDING HISTORY

Totals differ by database because of how sub-rounds and grants are counted:

- Inc42: \$828.92K across 2 rounds, last funding 4 May 2026 (page also renders it as 03/05/2026)
- CB Insights: \$830K total, last raise \$680K (Seed, led by Transition VC)
- Tracxn: \$1.23M across 7 rounds, with 12 institutional + 16 angel investors (28 total)

Round-by-round:

Date	Round	Amount	Lead / investors	Source
17 Sep 2022	Seed (first, likely incubator/grant)	undisclosed	—	Tracxn
21 Nov 2023	Seed	undisclosed	Spring Star Capital (lead), "NikkenHotelmgt +19 more"	Tracxn
Sep 2023	Pre-seed	1.2 crore (~USD 145K)	Enrission India Capital + angels	Entrepreneur India (https://india.entrepreneur.com/business-news/zerodrag-veloce-aif-suncharge-motors-report-funding-and-investment-activity), startupsamadhan (https://startupsamadhan.com/zerodrag-raises-6-5-crore-seed-transition-vc-drone-avionics-india/)
11 Dec 2025	Seed	undisclosed/redacted	—	Tracxn
4 May 2026 (announced 27–28 May 2026)	Seed	6.5 crore (~USD 680K per CB Insights/startupsamadhan; ~USD 780K per Entrepreneur India)	Transition VC (lead); 2 investors participating	BW Disrupt (https://www.bwdisrupt.com/article/nagpur-based-drone-startup-zerodrag-technologies-raises-rs-6-5-cr-from-transition-vc-608569), Entrepreneur India (https://india.entrepreneur.com/business-news/zerodrag-veloce-aif-suncharge-motors-report-funding-and-investment-activity)

Total raised to date: ≈ 7.7 crore (~USD 0.83M–1.23M depending on database).

Named investors across all sources: Transition VC, Enrission India Capital, Spring Star Capital, Pilani Innovation & Entrepreneurship Development Society (PIEDS, BITS Pilani), iCreate (Gujarat), NSRCEL IIM Bangalore Launchpad, Startup India, Ministry of Electronics and Information Technology (MeitY), plus undisclosed angels.

Use of proceeds from the May 2026 seed (consistent across all outlets): expanding R&D; scaling manufacturing capacity; building testing and quality infrastructure including EMI/EMC certification capability; stabilising and localising the domestic supply chain; and funding entry into US and European markets.

About the lead investor — Transition VC. Bengaluru-based, co-founded by Raiyaan Shingati (Managing Partner) and Mohammed Shoeb Ali, with Shantanu Chaturvedi as Partner; describes itself as "India's first energy-transition VC fund, investing in engineering-led climate tech and deep-tech startups" ([transitionventurecapital.com/about](https://www.transitionventurecapital.com/about) (<https://www.transitionventurecapital.com/about>)). Fund I closed at 700 crore. In July 2026 the firm announced Fund II at ~1,500 crore / USD 150–155M, writing \$2M–\$5M cheques into roughly 20–23 companies. Shingati's framing of the ZeroDrag thesis, as reported: "supply chains are weapons," India needs "SovereignTech," and avionics is "the intelligence layer of every drone" (startupsamadhan.com (<https://startupsamadhan.com/zerodrag-raises-6-5-crore-seed-transition-vc-drone-avionics-india/>)).

The round was widely bucketed into a defence-adjacent deep-tech funding wave in Indian startup activity in May 2026 ([news.ayen.in](https://news.ayen.in/zerodrag-drone-avionics-firm-closes-rs-65-crore/) (<https://news.ayen.in/zerodrag-drone-avionics-firm-closes-rs-65-crore/>), citing StartupTalky's 28 May 2026 roundup).

M&A: "Zerodrag has made no investments or acquisitions yet" (Tracxn). No acquisition offers or exit activity reported.

4. LEADERSHIP / PEOPLE

Two founders, both directors since incorporation (22 Aug 2022), both still on the board at ~4 years tenure.

Shantanu Shivdas Bhede — Co-Founder & CEO

- DIN: 09710694
- Director since 22 August 2022
- Background: mechanical engineer; self-taught in embedded systems. His LinkedIn headline reads "Enabling intelligent drones with Zerodrag | Incubated @ IIT-Delhi," and his profile references prior drone-services work with clients including Adani Power, ACC Cement, and Mahindra (in.linkedin.com/in/shantanu-bhede-380bab17a (<https://in.linkedin.com/in/shantanu-bhede-380bab17a>))
- Also maintains an FPV-pilot public persona: Instagram @shan_fpv — bio "Building @zerodrag.in I build things that make cool things fly!"
- Tracxn notes he has founded one other company besides ZeroDrag
- Public appearances: featured in the NSRCEL Launchpad video series ([youtube.com/watch?v=0WvouxexDRU](https://www.youtube.com/watch?v=0WvouxexDRU) (<https://www.youtube.com/watch?v=0WvouxexDRU>)) and the "Made-in-India Drone Avionics | ZeroDRAG | E4" episode ([youtube.com/watch?v=OXI-Q7enQDE](https://www.youtube.com/watch?v=OXI-Q7enQDE) (<https://www.youtube.com/watch?v=OXI-Q7enQDE>))
- Profiled on Inc42 at inc42.com/person/shantanu-bhede (<https://inc42.com/person/shantanu-bhede/>)
- Signature quotes: "We're not building drones. We're building what every drone needs." and framing ZeroDrag's output as the "essential building block" every drone requires, arguing India's UAV sector needs a domestically sourced, dependable electronics foundation ([yourstory.com](https://yourstory.com/2026/07/zerodrag-indian-drone-hardware) (<https://yourstory.com/2026/07/zerodrag-indian-drone-hardware>))

Amit Ramesh Nimje — Co-Founder & COO

- DIN: 08104465
- Director since 22 August 2022
- His F6S profile describes the technical work directly: "At Zerodrag, we built a modular UAV avionics system featuring our proprietary flight controller," with emphasis on secure, power-efficient drone systems ([f6s.com/member/amit-nimje](https://www.f6s.com/member/amit-nimje) (<https://www.f6s.com/member/amit-nimje>))
- Tracxn notes he has also founded one other company
- Note: his DIN (08104465) predates the company by several years, indicating prior directorship experience elsewhere

Board: Only the two founders are listed as directors on MCA records. No investor-nominated director is on record, and no independent directors — normal at this round size.

Named executives beyond the founders: none disclosed in any public source. No CTO, CFO, VP Engineering, or advisory board is publicly named — a genuine gap in the record rather than an omission in this research.

Hiring signal (as of mid/late 2026, via LinkedIn): actively recruiting a Senior Accounts Executive and a Senior Hardware Design Engineer — consistent with the stated plan to grow the engineering team post-seed.

Key external person: Raiyaan Shingati, Managing Partner, Transition VC — the lead investor voice in all May 2026 coverage (see §3).

5. PRODUCTS

ZeroDrag's catalogue is 26 live SKUs across roughly 10 categories, all branded ZeroDrag except the motors (QPT) and a handful of sourced accessories. Core boards are designed and manufactured in India ("Country of origin: India" in product metadata); passive accessories are sourced from China and flagged as such. Firmware compatibility spans ArduPilot, Betaflight, INAV, and Pixhawk, with AM32 on the ESCs. The recurring design claim is "fully solder-less" modular construction. Warranty 6–12 months, 12 months on the core avionics.

(Full catalogue extracted from zerodrag.in/collections/all (<https://zerodrag.in/collections/all>), zerodrag.in/products.json (<https://zerodrag.in/products.json?limit=250>), and the product sitemap (https://zerodrag.in/sitemap_products_1.xml). Prices are INR, sale/list.)

5.1 Flight Controllers

Product	Price (sale / list)	First published
Zerodrag Nova F4 Flight Controller — fully solder-less design, 12-mo warranty	4,499 / 8,499	30 Jun 2025
Zerodrag Warp F7 Flight Controller — fully solder-less design, 12-mo warranty	4,999 / 10,999	10 Jul 2024
Zerodrag Strix H7 Flight Controller — fully solder-less, 12-mo warranty; flagship H7-class board	7,999 / 12,999	30 Sep 2025

UAVGarage lists the Nova F4 at 4,799 excl. GST (SKU FC-109-3586). A separate reseller listing references the Nova as "F4 FC – ICM42688P," indicating the InvenSense/TDK ICM-42688-P IMU on at least the Nova line. Detailed MCU/gyro/UART specs live in per-product PDF manuals linked from each page (e.g. "Zerodrag STRIX H7 Flight Controller User Manual") rather than in HTML, so they are not reproduced here.

5.2 FPV Stacks (FC + ESC bundles)

Product	Price (sale / list)	First published
Zerodrag Nova-Boost F4 FPV Stack (Nova F4 + Boost AM32 55A ESC)	7,499 / 12,049	30 Jun 2025
Zerodrag Warp-Boost F7 FPV Stack (Warp F7 + AM32 55A ESC)	7,999 / 14,999	31 Dec 2024
Zerodrag Strix-Boost H7 FPV Stack (Strix H7 + Boost AM32 55A ESC)	11,599 / 17,999	9 Jun 2026

UAVGarage lists "Zerodrag NOVA – Boost Stack (F4 FC – ICM42688P + 55A 4-in-1 AM32 ESC)" at 7,399 excl. GST (SKU FC-109-3558).

5.3 ESCs / Motor Controllers

Product	Price	First published
Zerodrag Boost AM32 55A 4-in-1 ESC, 12-mo warranty, Made in India	5,499 / 8,499	31 Dec 2024
Finch 55A AM32 Electronic Speed Controller — a dedicated product page exists at zerodrag.in/pages/finch-55a-am32-electronic-speed-controller (https://zerodrag.in/pages/finch-55a-am32-electronic-speed-controller) but the SKU is not in the live storefront; likely a rebrand of, or successor to, the Boost line	n/a	—

5.4 GNSS / Navigation — the "LOC8" family (launched 30 April 2026)

Product	Price	Notes
Zerodrag LOC8 DroneCAN RTK GNSS System	25,999 / 35,999	Flagship; RTK-capable, DroneCAN bus — targets survey/industrial and ArduPilot-class platforms, not FPV. Highest-ticket SKU in the catalogue.
Zerodrag LOC8 Lite GNSS System	20,999 / 29,999	Mid-tier
Zerodrag LOC8-M10 GPS Module	2,099 / 4,999	Entry module (naming implies a u-blox M10-class receiver); ships with user manual and 3D files

The LOC8 launch in April 2026 is strategically the most significant product event: it moves ZeroDrag out of the hobby/FPV price band (500–8,000) into enterprise/industrial avionics (21,000–36,000) and into DroneCAN/ArduPilot enterprise territory, which is where the seed-round roadmap (heavy-lift, eVTOL) points.

5.5 Radio / Receivers

Product	Price	Notes
Zerodrag Nexus 1 Express LRS 2.4 GHz Receiver, 12-mo warranty, Made in India	1,549 / 2,599	Homepage claims "Up to 15 km range"
Zerodrag Nexus Duo	not listed	Product page exists (zerodrag.in/pages/zerodrag-nexus-duo) (https://zerodrag.in/pages/zerodrag-nexus-duo) with a 7-image gallery but no price or storefront SKU — a diversity/dual-receiver variant appearing to be pre-launch or newly launched
Zerodrag 2.4G U.FL T Antenna for Receivers (SKU RD-109-3457)	299 / 499	Country of origin: China

VTX note: LinkedIn specialties and the site's own page slugs both list VTX (video transmitters) and GCS (ground control station) as product areas, and the about page URL literally reads "flight-controller-esc-els-vtx" — but no VTX or GCS SKU is currently in the storefront. Either a discontinued line or a not-yet-shipped one.

5.6 Propulsion — QPT motors (OEM-partnered)

Six brushless motors, all badged QPT, manufactured by Quanteon Powertrain (QPT) Pvt. Ltd. (quanteonworld.com (<https://quanteonworld.com>)) and resold by Zerodrag — i.e. a partner-manufactured line, not in-house silicon:

Motor	Price	Published
QPT 1404-4500KV	1,649 / 1,899	Jan 2026
QPT 2807-1300KV	1,999 / 2,599	Jan 2026
QPT 3115-900KV	3,149 / 3,999	Jan 2026
QPT 4008-380KV	4,599 / 5,499	Jun 2026
QPT 4214-450KV	3,899 / 4,499	Jun 2026
QPT 4720-380KV	4,499 / 5,499	Jun 2026

The June 2026 additions (4008/4214/4720) are large-diameter, low-KV motors — heavy-lift and long-endurance class, corroborating the stated heavy-lift roadmap.

5.7 Lighting — "Aurora" LED line

Product	Price	Published
Zerodrag Aurora Race-X LED	699 / 999	7 Mar 2024
Zerodrag Aurora 20×20 LED (20×20mm mounting)	499 / 799	3 Oct 2024
Zerodrag Aurora Freestyle LED	599 / 899	18 Oct 2024

5.8 Wiring & accessories

- Zerodrag Catalyst Race Wires — 349 / 799 (12 Mar 2024), Made in India
- AMASS XT-60 12AWG 10 cm pigtail, male — 129 / 199 (China)
- Rubycon 1000 µF 35 V 105 ° C capacitor — 49 / 99 (China)
- M3 silicone grommets 6.5 mm and 8 mm, packs of 10 — 199 each (China)

5.9 Product-line observations

- Naming architecture is coherent and deliberate: Nova (F4/entry) → Warp (F7/mid) → Strix (H7/flagship) for flight controllers; Boost for ESCs; LOC8 for GNSS; Nexus for receivers; Aurora for LEDs; Catalyst for wiring. This is a company thinking in product *lines*, not one-off SKUs.
- Near-total stockout at time of research. Virtually every SKU on the storefront was flagged "Sold out" — including all flight controllers, stacks, motors, LEDs and two of three GNSS units. Read either as demand outstripping a constrained supply chain (consistent with "supply chain stability" being an explicit use of the seed proceeds) or as a production transition. Either way it is a material operational datapoint for a competitive assessment.
- Services: custom design and white-labelling for bulk/OEM orders; firmware hosted and distributed via the site with ArduPilot/Betaflight/INAV builds.
- Roadmap products (stated, unlaunched): enterprise-grade UAV avionics, heavy-lift drone systems, and eVTOL platform hardware, over a 12–24 month horizon from May 2026.

6. PATENTS AND INTELLECTUAL PROPERTY

Bottom line: no granted patents or published patent applications attributable to ZeroDrag Technologies Pvt. Ltd., Shantanu Bhede, or Amit Nimje could be located in any accessible public source.

Methodology and what was checked:

- Google Patents keyword and XHR query endpoints for "zerodrag" — returned no retrievable results (the endpoint served 503/anti-automation responses on repeated attempts, so this is an inconclusive rather

than confirmed-zero check).

- General web searches explicitly combining "zerodrag" with patent / trademark / design registration returned zero organic results — the query surfaced only an unrelated LegalZoom advertisement, with the engine reporting "No more results found."
- Tracxn's company profile lists no patent or IP section for ZeroDrag, and no IP is referenced in any of the ~15 news articles and database profiles reviewed.
- MCA filings show no IP-related charges or licensing disclosures.

What can be said about the IP posture:

- The company describes its flight controller as "proprietary" (Amit Nimje, F&S profile) and markets a distinctive "fully solder-less" modular mechanical/electrical interface across the Nova, Warp and Strix boards. That solder-less interconnect is the most obviously patentable/design-registrable element of the portfolio, and an Indian design registration for it would be plausible — but nothing is findable in accessible sources.
- The firmware layer is explicitly open-source-derived (Betaflight, INAV, ArduPilot, AM32), so the defensible IP is necessarily hardware/design and manufacturing know-how, not software.
- DPIIT "Recognised Startup" status is confirmed, which in India confers patent-fee rebates and expedited examination — a mild indicator that filings may exist in the unpublished 18-month window, particularly around the April 2026 LOC8 GNSS launch and the H7 controller.
- Trademark: the brand is used with a claim mark in commerce ("Zerodrag," "Nova," "Warp," "Strix," "LOC8," "Nexus," "Aurora," "Catalyst"). Indian trademark registry records were not retrievable through accessible sources; TrademarkElite's India search blocked automated access. Note the collision risk with the unrelated US ZeroDrag™ skate-bearing brand, which is in a completely different Nice class, so likely non-conflicting.

Caveat for the record: absence of evidence here is weak evidence of absence. Indian Patent Office records (ipindiaservices.gov.in public search) and WIPO Patentscope are not accessible to automated retrieval and were not checked directly; a manual IPO search against applicant name "ZERODRAG TECHNOLOGIES PRIVATE LIMITED" and inventor names Bhede/Nimje would be the correct follow-up. For a four-year-old, 7.7 Cr-funded hardware startup, having zero granted patents would nonetheless be entirely unremarkable.

7. LATEST NEWS AND DEVELOPMENTS (2024 – Q3 2026)

Q3 2026

- 22 July 2026 — YourStory feature: "ZeroDrag is building the made-in-India electronics every drone needs" (yourstory.com/2026/07/zerodrag-indian-drone-hardware (<https://yourstory.com/2026/07/zerodrag-indian-drone-hardware>)). The most substantive profile to date: confirms the IIT Delhi → Nagpur operational move, the B2B/OEM model, defence-agriculture-logistics-industrial customer mix, growing US/Europe inbound, and the 7.7 Cr cumulative raise. Syndicated to DailyHunt ([m.dailyhunt.in](https://m.dailyhunt.in/news/india/english/yourstory-epaper-yourstory/zerodrag-is-building-the-madeinindia-electronics-every-drone-needs-newsid-n720544851) (<https://m.dailyhunt.in/news/india/english/yourstory-epaper-yourstory/zerodrag-is-building-the-madeinindia-electronics-every-drone-needs-newsid-n720544851>)).
- 22 July 2026 — lead investor scales up: Transition VC announced Fund II at 1,500 Cr / ~\$150–155M for energy and industrial deep tech, with \$2–5M cheques across 20–23 companies — meaningfully increasing the capital available to follow on into ZeroDrag ([thefinancialanalyst.net](https://thefinancialanalyst.net/2026/07/22/transition-vc-raises-155m-for-fund-ii-to-scale-deep-tech/) (<https://thefinancialanalyst.net/2026/07/22/transition-vc-raises-155m-for-fund-ii-to-scale-deep-tech/>), [esg-investing.com](https://esg-investing.com/2026/07/22/transition-vc-launches-150-million-venture-fund/) (<https://esg-investing.com/2026/07/22/transition-vc-launches-150-million-venture-fund/>)).
- 31 July 2026 — Tracxn headcount snapshot: 16 employees.

Q2 2026 — the pivotal quarter

- 30 April 2026 — LOC8 GNSS family launched: LOC8 DroneCAN RTK GNSS System (25,999), LOC8 Lite (20,999), LOC8-M10 module (2,099). First move into enterprise-priced avionics.

- 4 May 2026 — round closes (Tracxn round date).
- 27–28 May 2026 — 6.5 crore seed round announced, led by Transition VC. Covered by BW Disrupt ([bwdisrupt.com \(https://www.bwdisrupt.com/article/nagpur-based-drone-startup-zero-drag-technologies-raises-rs-6-5-cr-from-transition-vc-608569\)](https://www.bwdisrupt.com/article/nagpur-based-drone-startup-zero-drag-technologies-raises-rs-6-5-cr-from-transition-vc-608569)), Entrepreneur India ([india.entrepreneur.com \(https://india.entrepreneur.com/business-news/zerodrag-veloce-aif-suncharge-motors-report-funding-and-investment-activity\)](https://india.entrepreneur.com/business-news/zerodrag-veloce-aif-suncharge-motors-report-funding-and-investment-activity)), Startup Samadhan ([startupsamadhan.com \(https://startupsamadhan.com/zerodrag-raises-6-5-crore-seed-transition-vc-drone-avionics-india/\)](https://startupsamadhan.com/zerodrag-raises-6-5-crore-seed-transition-vc-drone-avionics-india/)), Startup Magazine India ([startupmagazine.in \(https://startupmagazine.in/zerodrag-seed-funding-defencetech-drone-avionics-india/\)](https://startupmagazine.in/zerodrag-seed-funding-defencetech-drone-avionics-india/)), Newskart ([newskart.com \(https://www.newskart.com/zerodrag-raises-rs-6-5-crore-why-indigenous-drone-electronics-matter-for-india/\)](https://www.newskart.com/zerodrag-raises-rs-6-5-crore-why-indigenous-drone-electronics-matter-for-india/)), World Startup News, IPO Platform ([ipoplatform.com \(https://www.ipoplatform.com/Funding-news/zerodrag-raises-funding\)](https://www.ipoplatform.com/Funding-news/zerodrag-raises-funding), 29 May), Indian Startup News (28 May), and aggregated by news.ayen.in (29 May) as part of a defence-adjacent deep-tech funding wave dominating Indian startup activity in May 2026. Underlying sources cited include ET Entrepreneur (Economic Times) and Dealroom.co.
- 9 June 2026 — Strix-Boost H7 FPV Stack launched (11,599), the flagship stack.
- June 2026 — heavy-lift motor line added: QPT 4008-380KV, 4214-450KV, 4720-380KV.
- 16 June 2026 — Drone Federation of India media-hub coverage: "ZeroDrag Raises Rs 6.5 Crore to Build India's Drone Avionics Layer" ([dronefederation.in \(https://www.dronefederation.in/media-hub/zero-drag-raise-funds-to-build-drone-avionics-layer\)](https://www.dronefederation.in/media-hub/zero-drag-raise-funds-to-build-drone-avionics-layer)). Notable because DFI is the Indian industry body — the coverage frames ZeroDrag as filling the avionics gap left by India's frame-focused manufacturing push, and situates the raise in a trend of Indian investors funding "the less visible components of the drone industry — sensors, autopilot systems, and communications — rather than concentrating solely on airframe manufacturers."

Q1 2026

- January 2026 — QPT motor line launched (1404-4500KV, 2807-1300KV, 3115-900KV), adding propulsion to the portfolio via the Quanteon Powertrain partnership.
- 11 December 2025 — an additional seed tranche recorded by Tracxn.

Undated 2026 (via LinkedIn company updates)

- "Interceptor Drone" flight tests reportedly exceeding 320 km/h using ZeroDrag's indigenous flight controller. This is the single most defence-relevant technical claim the company has made publicly — counter-UAS interceptor work is a distinct market from FPV components — and it appears only on the company's own LinkedIn, not in third-party press ([in.linkedin.com/company/zerodrag \(https://in.linkedin.com/company/zerodrag\)](https://in.linkedin.com/company/zerodrag)). Worth independent verification.
- Active hiring: Senior Hardware Design Engineer, Senior Accounts Executive.
- Video coverage: "Made-in-India Drone Avionics | ZeroDRAG | E4" ([youtube.com/watch?v=OXI-Q7enQDE \(https://www.youtube.com/watch?v=OXI-Q7enQDE\)](https://www.youtube.com/watch?v=OXI-Q7enQDE)) and Transition VC's "Inside The Investments E1: ZeroDrag" ([youtube.com/watch?v=NxR7ALx5LSI \(https://www.youtube.com/watch?v=NxR7ALx5LSI\)](https://www.youtube.com/watch?v=NxR7ALx5LSI)).

2025

- 30 June 2025 — Nova F4 flight controller and Nova-Boost F4 stack launched.
- 30 September 2025 — Strix H7 flight controller launched — the H7-class flagship, the technical step-change from F4/F7 hobby-grade to industrial-grade compute.
- 10 September 2025 — AGM held; FY25 accounts filed (revenue 27.61 lakh, down 34%).

2024

- 7 March 2024 — Aurora Race-X LED; 12 March 2024 — Catalyst Race Wires.
- 10 July 2024 — Warp F7 flight controller launched.
- 3–18 October 2024 — Aurora 20×20 and Aurora Freestyle LEDs, plus sourced accessories.
- 31 December 2024 — Boost AM32 55A 4-in-1 ESC and Warp-Boost F7 stack launched — completing the first full FC+ESC stack offering.

8. COMPETITIVE CONTEXT (relevant for the Vector Technics comparison)

Tracxn ranks ZeroDrag 2nd of 71 active competitors in its UAV-avionics/components category by Tracxn Score (46/100), and 4th by total funding raised:

Rank	Company	Score	Funding / status
1	uAvionix	57	\$10M, acquired
2	ZeroDrag	46	\$1.23M, Seed
3	Sagetech Avionics	42	\$12M, Series B
4	ARK Electronics (Cottonwood Heights, UT)	41	Unfunded
5	Defendron Technologies	37	\$422K, Seed
6	SpeedyBee	36	Unfunded
7	Starlogic Intelligent Technology	36	\$1.49M, Series A
8	Team BlackSheep	34	Unfunded
9	Skynetics	33	Unfunded
10	Falcon Skyworks	33	Unfunded

CB Insights' comparison set differs, listing ASATRobo, ARA Robotics, Berry Avionics, Argo Aerospace (Prato, Italy), Hdrones (Paris), Zerosum Technologies (Delhi — also founded 2022, UAV subsystem integration), Mayan Robotics (Chula Vista, CA), and Vecros (New Delhi).

The named incumbents ZeroDrag positions *against* in its own messaging are DJI and Holybro — i.e. the Chinese avionics supply chain, not Western avionics firms.

9. SUMMARY ASSESSMENT / DATA-QUALITY NOTES

- ZeroDrag is a very early-stage, sub-1 Cr-revenue, ~15–17 person Indian hardware startup with ~7.7 Cr (~\$0.83–1.23M) raised to date and no disclosed valuation. It is not a peer to a scaled avionics OEM on financial metrics; it is a peer on *product category* — flight controllers, ESCs, GNSS, radio links.
- Its differentiation is geopolitical and supply-chain, not technological: Made-in-India, non-Chinese, cross-platform, open-firmware-compatible, white-labellable. The technology stack is largely open-source firmware on in-house-designed hardware.
- Trajectory is clearly upward-market: FPV hobby components (2024) → prosumer H7 flight controllers (2025) → enterprise RTK GNSS and heavy-lift propulsion (2026) → stated intent toward heavy-lift and eVTOL avionics (2026–2028).
- Red flags for diligence: FY25 revenue declined 34% to 27.6 lakh; ROE –84.78 and net margin –134 indicate meaningful losses against a tiny top line; the entire storefront was showing "Sold out" at the time of research; no executives beyond the two founders are publicly named; no verifiable patent estate; and the flagship defence claim (320+ km/h interceptor drone) appears only on the company's own LinkedIn.
- Source conflicts to be aware of if this feeds a RAG index: HQ is variously Nagpur / New Delhi / Noida (all real, different functions); total funding is \$828.92K (Inc42) vs \$830K (CB Insights) vs \$1.23M (Tracxn); round count is 2 vs 7; the 6.5 Cr seed converts to both "\$680K" and "\$780K" across outlets; authorised capital reads 66 lakh (current) vs 15 lakh (stale Tofler); and the company appears under the legacy name DroneAdda in CB Insights.